

MySimulation_3

Simulation File: Z:\Greenfeild Conventional Ammonia\Sadiq Conventional Ammonia Plant.dwxmz
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PURGE-GAS-1201 (Material Stream)

Property	Value	Unit
Temperature	30.0015	C
Pressure	145	bar
Mass Flow	1133.21	kg/h
Molar Flow	117.308	kmol/h
Volumetric Flow	19.885	m3/h
Mixture Molar Fraction		
Methane	0.118325	
Ethane	2.77702E-17	
Propane	6.06032E-19	
N-butane	0	
Isobutane	0	
Nitrogen	0.138891	
Carbon dioxide	0.00126157	
Water	4.02081E-06	
Oxygen	7.47993E-14	
Argon	0.00294551	
Hydrogen sulfide	0	
Hydrogen	0.591454	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0.14712	
Density (Mixture)	56.9881	kg/m3
Molecular Weight (Mixture)	9.66011	kg/kmol
Specific Enthalpy (Mixture)	-61.1639	kJ/kg
Specific Entropy (Mixture)	-2.78941	kJ/[kg.K]
Molar Enthalpy (Mixture)	-590.85	kJ/kmol
Molar Entropy (Mixture)	-26.946	kJ/[kmol.K]
Thermal Conductivity (Mixture)	0.130784	W/[m.K]
Molar Fraction (Vapor)		
Methane	0.118325	
Ethane	2.77702E-17	
Propane	6.06032E-19	
N-butane	0	
Isobutane	0	
Nitrogen	0.138891	
Carbon dioxide	0.00126157	
Water	3.9785E-06	
Oxygen	7.47995E-14	
Argon	0.00294552	
Hydrogen sulfide	0	
Hydrogen	0.591455	

Property	Value	Unit
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0.147117	
Density (Vapor)	56.9878	kg/m ³
Molecular Weight (Vapor)	9.66008	kg/kmol
Specific Enthalpy (Vapor)	-61.1573	kJ/kg
Specific Entropy (Vapor)	-2.7894	kJ/[kg.K]
Molar Enthalpy (Vapor)	-590.785	kJ/kmol
Molar Entropy (Vapor)	-26.9459	kJ/[kmol.K]
Thermal Conductivity (Vapor)	0.130783	W/[m.K]
Kinematic Viscosity (Vapor)	2.17623E-07	m ² /s
Dynamic Viscosity (Vapor)	1.24018E-05	Pa.s
Heat Capacity (Vapor)	3.6305	kJ/[kg.K]
Heat Capacity Ratio (Vapor)	1.58876	
Mass Flow (Vapor)	1133.2	kg/h
Molar Flow (Vapor)	117.308	kmol/h
Volumetric Flow (Vapor)	19.885	m ³ /h
Compressibility Factor (Vapor)	0.975208	
Molar Fraction (Vapor)	0.999997	
Mass Fraction (Vapor)	0.999994	
Volumetric Fraction (Vapor)	0.999999	
Molar Fraction (Overall Liquid)		
Methane	0.0144063	
Ethane	1.47788E-17	
Propane	6.41408E-19	
N-butane	0	
Isobutane	0	
Nitrogen	0.00234315	
Carbon dioxide	0.00121146	
Water	0.0127787	
Oxygen	7.02883E-15	
Argon	0.000534263	
Hydrogen sulfide	0	
Hydrogen	0.0144582	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0.954268	
Density (Overall Liquid)	589.604	kg/m ³
Molecular Weight (Overall Liquid)	16.8824	kg/kmol
Specific Enthalpy (Overall Liquid)	-1193.75	kJ/kg
Specific Entropy (Overall Liquid)	-4.8817	kJ/[kg.K]
Molar Enthalpy (Overall Liquid)	-20153.5	kJ/kmol

Property	Value	Unit
Molar Entropy (Overall Liquid)	-82.4151	kJ/[kmol.K]
Thermal Conductivity (Overall Liquid)	0.437291	W/[m.K]
Kinematic Viscosity (Overall Liquid)	2.34273E-07	m ² /s
Dynamic Viscosity (Overall Liquid)	0.000138128	Pa.s
Heat Capacity (Overall Liquid)	4.48079	kJ/[kg.K]
Heat Capacity Ratio (Overall Liquid)	1.71934	
Mass Flow (Overall Liquid)	0.00656138	kg/h
Molar Flow (Overall Liquid)	0.000388651	kmol/h
Volumetric Flow (Overall Liquid)	1.11285E-05	m ³ /h
Compressibility Factor (Overall Liquid)	0.18111	
Molar Fraction (Overall Liquid)	3.31308E-06	
Mass Fraction (Overall Liquid)	5.7901E-06	
Volumetric Fraction (Overall Liquid)	5.59641E-07	
Molar Fraction (Liquid 1)		
Methane	0.0144063	
Ethane	1.47788E-17	
Propane	6.41408E-19	
N-butane	0	
Isobutane	0	
Nitrogen	0.00234315	
Carbon dioxide	0.00121146	
Water	0.0127787	
Oxygen	7.02883E-15	
Argon	0.000534263	
Hydrogen sulfide	0	
Hydrogen	0.0144582	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0.954268	
Molar Fraction (Liquid 2)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	

Property	Value	Unit
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Molar Fraction (Aqueous)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Molar Fraction (Solid)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	

NH3-LIQ-1201 (Material Stream)

Property	Value	Unit
Temperature	30	C
Pressure	145	bar
Mass Flow	5453.12	kg/h
Molar Flow	322.996	kmol/h
Volumetric Flow	9.24768	m3/h
Mixture Molar Fraction		
Methane	0.0143698	
Ethane	1.47319E-17	
Propane	6.388E-19	
N-butane	0	
Isobutane	0	
Nitrogen	0.00233802	
Carbon dioxide	0.00121013	
Water	0.0129247	
Oxygen	7.01703E-15	
Argon	0.000533384	
Hydrogen sulfide	0	
Hydrogen	0.0144305	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0.954193	
Density (Mixture)	589.674	kg/m3
Molecular Weight (Mixture)	16.8829	kg/kmol
Specific Enthalpy (Mixture)	-1194.15	kJ/kg
Specific Entropy (Mixture)	-4.8825	kJ/[kg.K]
Molar Enthalpy (Mixture)	-20160.7	kJ/kmol
Molar Entropy (Mixture)	-82.4309	kJ/[kmol.K]
Thermal Conductivity (Mixture)	0.437377	W/[m.K]
Molar Fraction (Vapor)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	

Property	Value	Unit
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Density (Vapor)	0	kg/m3
Molecular Weight (Vapor)	0	kg/kmol
Specific Enthalpy (Vapor)	0	kJ/kg
Specific Entropy (Vapor)	0	kJ/[kg.K]
Molar Enthalpy (Vapor)	0	kJ/kmol
Molar Entropy (Vapor)	0	kJ/[kmol.K]
Thermal Conductivity (Vapor)	0	W/[m.K]
Kinematic Viscosity (Vapor)	0	m2/s
Dynamic Viscosity (Vapor)	0	Pa.s
Heat Capacity (Vapor)	0	kJ/[kg.K]
Heat Capacity Ratio (Vapor)	NaN	
Mass Flow (Vapor)	0	kg/h
Molar Flow (Vapor)	0	kmol/h
Volumetric Flow (Vapor)	0	m3/h
Compressibility Factor (Vapor)	0	
Molar Fraction (Vapor)	0	
Mass Fraction (Vapor)	0	
Volumetric Fraction (Vapor)	0	
Molar Fraction (Overall Liquid)		
Methane	0.0143698	
Ethane	1.47319E-17	
Propane	6.388E-19	
N-butane	0	
Isobutane	0	
Nitrogen	0.00233802	
Carbon dioxide	0.00121013	
Water	0.0129247	
Oxygen	7.01703E-15	
Argon	0.000533384	
Hydrogen sulfide	0	
Hydrogen	0.0144305	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0.954193	
Density (Overall Liquid)	589.674	kg/m3
Molecular Weight (Overall Liquid)	16.8829	kg/kmol
Specific Enthalpy (Overall Liquid)	-1194.15	kJ/kg
Specific Entropy (Overall Liquid)	-4.8825	kJ/[kg.K]
Molar Enthalpy (Overall Liquid)	-20160.7	kJ/kmol

Property	Value	Unit
Molar Entropy (Overall Liquid)	-82.4309	kJ/[kmol.K]
Thermal Conductivity (Overall Liquid)	0.437377	W/[m.K]
Kinematic Viscosity (Overall Liquid)	2.34434E-07	m ² /s
Dynamic Viscosity (Overall Liquid)	0.00013824	Pa.s
Heat Capacity (Overall Liquid)	4.48026	kJ/[kg.K]
Heat Capacity Ratio (Overall Liquid)	1.71898	
Mass Flow (Overall Liquid)	5453.12	kg/h
Molar Flow (Overall Liquid)	322.996	kmol/h
Volumetric Flow (Overall Liquid)	9.24768	m ³ /h
Compressibility Factor (Overall Liquid)	0.18108	
Molar Fraction (Overall Liquid)	1	
Mass Fraction (Overall Liquid)	1	
Volumetric Fraction (Overall Liquid)	1	
Molar Fraction (Liquid 1)		
Methane	0.0143698	
Ethane	1.47319E-17	
Propane	6.388E-19	
N-butane	0	
Isobutane	0	
Nitrogen	0.00233802	
Carbon dioxide	0.00121013	
Water	0.0129247	
Oxygen	7.01703E-15	
Argon	0.000533384	
Hydrogen sulfide	0	
Hydrogen	0.0144305	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0.954193	
Molar Fraction (Liquid 2)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	

Property	Value	Unit
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Molar Fraction (Aqueous)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Molar Fraction (Solid)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	

SYNGAS-SEP-1201 (Material Stream)

Property	Value	Unit
Temperature	30	C
Pressure	145	bar
Mass Flow	37773.6	kg/h
Molar Flow	3910.27	kmol/h
Volumetric Flow	662.83	m3/h
Mixture Molar Fraction		
Methane	0.118325	
Ethane	2.77702E-17	
Propane	6.06032E-19	
N-butane	0	
Isobutane	0	
Nitrogen	0.138891	
Carbon dioxide	0.00126157	
Water	4.02081E-06	
Oxygen	7.47993E-14	
Argon	0.00294551	
Hydrogen sulfide	0	
Hydrogen	0.591454	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0.14712	
Density (Mixture)	56.9884	kg/m3
Molecular Weight (Mixture)	9.66011	kg/kmol
Specific Enthalpy (Mixture)	-61.1639	kJ/kg
Specific Entropy (Mixture)	-2.78941	kJ/[kg.K]
Molar Enthalpy (Mixture)	-590.85	kJ/kmol
Molar Entropy (Mixture)	-26.946	kJ/[kmol.K]
Thermal Conductivity (Mixture)	0.130782	W/[m.K]
Molar Fraction (Vapor)		
Methane	0.118325	
Ethane	2.77702E-17	
Propane	6.06032E-19	
N-butane	0	
Isobutane	0	
Nitrogen	0.138891	
Carbon dioxide	0.00126157	
Water	4.02081E-06	
Oxygen	7.47993E-14	
Argon	0.00294551	
Hydrogen sulfide	0	
Hydrogen	0.591454	

Property	Value	Unit
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0.14712	
Density (Vapor)	56.9884	kg/m3
Molecular Weight (Vapor)	9.66011	kg/kmol
Specific Enthalpy (Vapor)	-61.1639	kJ/kg
Specific Entropy (Vapor)	-2.78941	kJ/[kg.K]
Molar Enthalpy (Vapor)	-590.85	kJ/kmol
Molar Entropy (Vapor)	-26.946	kJ/[kmol.K]
Thermal Conductivity (Vapor)	0.130782	W/[m.K]
Kinematic Viscosity (Vapor)	2.1762E-07	m2/s
Dynamic Viscosity (Vapor)	1.24018E-05	Pa.s
Heat Capacity (Vapor)	3.6305	kJ/[kg.K]
Heat Capacity Ratio (Vapor)	1.58877	
Mass Flow (Vapor)	37773.6	kg/h
Molar Flow (Vapor)	3910.27	kmol/h
Volumetric Flow (Vapor)	662.83	m3/h
Compressibility Factor (Vapor)	0.975206	
Molar Fraction (Vapor)	1	
Mass Fraction (Vapor)	1	
Volumetric Fraction (Vapor)	1	
Molar Fraction (Overall Liquid)		
Methane	NaN	
Ethane	NaN	
Propane	NaN	
N-butane	NaN	
Isobutane	NaN	
Nitrogen	NaN	
Carbon dioxide	NaN	
Water	NaN	
Oxygen	NaN	
Argon	NaN	
Hydrogen sulfide	NaN	
Hydrogen	NaN	
Carbonyl sulfide	NaN	
Carbon monoxide	NaN	
Ammonia	NaN	
Density (Overall Liquid)	0	kg/m3
Molecular Weight (Overall Liquid)	NaN	kg/kmol
Specific Enthalpy (Overall Liquid)	0	kJ/kg
Specific Entropy (Overall Liquid)	0	kJ/[kg.K]
Molar Enthalpy (Overall Liquid)	NaN	kJ/kmol

Property	Value	Unit
Molar Entropy (Overall Liquid)	NaN	kJ/[kmol.K]
Thermal Conductivity (Overall Liquid)	0	W/[m.K]
Kinematic Viscosity (Overall Liquid)	0	m2/s
Dynamic Viscosity (Overall Liquid)	0	Pa.s
Heat Capacity (Overall Liquid)	0	kJ/[kg.K]
Heat Capacity Ratio (Overall Liquid)	NaN	
Mass Flow (Overall Liquid)	0	kg/h
Molar Flow (Overall Liquid)	0	kmol/h
Volumetric Flow (Overall Liquid)	0	m3/h
Compressibility Factor (Overall Liquid)	0	
Molar Fraction (Overall Liquid)	0	
Mass Fraction (Overall Liquid)	0	
Volumetric Fraction (Overall Liquid)	0	
Molar Fraction (Liquid 1)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Molar Fraction (Liquid 2)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	

Property	Value	Unit
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Molar Fraction (Aqueous)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Molar Fraction (Solid)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	

NH3-COLD-1201 (Material Stream)

Property	Value	Unit
Temperature	30	C
Pressure	145	bar
Mass Flow	43226.5	kg/h
Molar Flow	4233.26	kmol/h
Volumetric Flow	672.075	m3/h
Mixture Molar Fraction		
Methane	0.110394	
Ethane	2.67754E-17	
Propane	6.08532E-19	
N-butane	0	
Isobutane	0	
Nitrogen	0.128472	
Carbon dioxide	0.00125765	
Water	0.000989796	
Oxygen	6.96279E-14	
Argon	0.00276148	
Hydrogen sulfide	0	
Hydrogen	0.54743	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0.208695	
Density (Mixture)	64.318	kg/m3
Molecular Weight (Mixture)	10.2112	kg/kmol
Specific Enthalpy (Mixture)	-204.092	kJ/kg
Specific Entropy (Mixture)	-3.05345	kJ/[kg.K]
Molar Enthalpy (Mixture)	-2084.02	kJ/kmol
Molar Entropy (Mixture)	-31.1793	kJ/[kmol.K]
Thermal Conductivity (Mixture)	0.154175	W/[m.K]
Molar Fraction (Vapor)		
Methane	0.118325	
Ethane	2.77702E-17	
Propane	6.06032E-19	
N-butane	0	
Isobutane	0	
Nitrogen	0.138891	
Carbon dioxide	0.00126157	
Water	4.02081E-06	
Oxygen	7.47993E-14	
Argon	0.00294551	
Hydrogen sulfide	0	
Hydrogen	0.591454	

Property	Value	Unit
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0.14712	
Density (Vapor)	56.9884	kg/m3
Molecular Weight (Vapor)	9.66011	kg/kmol
Specific Enthalpy (Vapor)	-61.1639	kJ/kg
Specific Entropy (Vapor)	-2.78941	kJ/[kg.K]
Molar Enthalpy (Vapor)	-590.85	kJ/kmol
Molar Entropy (Vapor)	-26.946	kJ/[kmol.K]
Thermal Conductivity (Vapor)	0.130782	W/[m.K]
Kinematic Viscosity (Vapor)	2.1762E-07	m2/s
Dynamic Viscosity (Vapor)	1.24018E-05	Pa.s
Heat Capacity (Vapor)	3.6305	kJ/[kg.K]
Heat Capacity Ratio (Vapor)	1.58877	
Mass Flow (Vapor)	37773.6	kg/h
Molar Flow (Vapor)	3910.27	kmol/h
Volumetric Flow (Vapor)	662.83	m3/h
Compressibility Factor (Vapor)	0.975206	
Molar Fraction (Vapor)	0.9237	
Mass Fraction (Vapor)	0.873848	
Volumetric Fraction (Vapor)	0.986244	
Molar Fraction (Overall Liquid)		
Methane	0.0143698	
Ethane	1.47319E-17	
Propane	6.388E-19	
N-butane	0	
Isobutane	0	
Nitrogen	0.00233802	
Carbon dioxide	0.00121013	
Water	0.0129247	
Oxygen	7.01703E-15	
Argon	0.000533384	
Hydrogen sulfide	0	
Hydrogen	0.0144305	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0.954193	
Density (Overall Liquid)	589.674	kg/m3
Molecular Weight (Overall Liquid)	16.8829	kg/kmol
Specific Enthalpy (Overall Liquid)	-1194.15	kJ/kg
Specific Entropy (Overall Liquid)	-4.8825	kJ/[kg.K]
Molar Enthalpy (Overall Liquid)	-20160.7	kJ/kmol

Property	Value	Unit
Molar Entropy (Overall Liquid)	-82.4309	kJ/[kmol.K]
Thermal Conductivity (Overall Liquid)	0.437377	W/[m.K]
Kinematic Viscosity (Overall Liquid)	2.34434E-07	m ² /s
Dynamic Viscosity (Overall Liquid)	0.00013824	Pa.s
Heat Capacity (Overall Liquid)	4.48026	kJ/[kg.K]
Heat Capacity Ratio (Overall Liquid)	1.71898	
Mass Flow (Overall Liquid)	5453.12	kg/h
Molar Flow (Overall Liquid)	322.996	kmol/h
Volumetric Flow (Overall Liquid)	9.24768	m ³ /h
Compressibility Factor (Overall Liquid)	0.18108	
Molar Fraction (Overall Liquid)	0.0762996	
Mass Fraction (Overall Liquid)	0.126152	
Volumetric Fraction (Overall Liquid)	0.0137599	
Molar Fraction (Liquid 1)		
Methane	0.0143698	
Ethane	1.47319E-17	
Propane	6.388E-19	
N-butane	0	
Isobutane	0	
Nitrogen	0.00233802	
Carbon dioxide	0.00121013	
Water	0.0129247	
Oxygen	7.01703E-15	
Argon	0.000533384	
Hydrogen sulfide	0	
Hydrogen	0.0144305	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0.954193	
Molar Fraction (Liquid 2)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	

Property	Value	Unit
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Molar Fraction (Aqueous)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Molar Fraction (Solid)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	

NH3-HOT-RAW-1101 (Material Stream)

Property	Value	Unit
Temperature	430	C
Pressure	147	bar
Mass Flow	43227.4	kg/h
Molar Flow	4233.14	kmol/h
Volumetric Flow	1736.47	m3/h
Mixture Molar Fraction		
Methane	0.110414	
Ethane	2.67764E-17	
Propane	6.08556E-19	
N-butane	0	
Isobutane	0	
Nitrogen	0.128484	
Carbon dioxide	0.00125769	
Water	0.000989796	
Oxygen	3.4435E-08	
Argon	0.00276159	
Hydrogen sulfide	0	
Hydrogen	0.547403	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0.20869	
Density (Mixture)	24.8938	kg/m3
Molecular Weight (Mixture)	10.2117	kg/kmol
Specific Enthalpy (Mixture)	1335.44	kJ/kg
Specific Entropy (Mixture)	0.459074	kJ/[kg.K]
Molar Enthalpy (Mixture)	13637.1	kJ/kmol
Molar Entropy (Mixture)	4.68791	kJ/[kmol.K]
Thermal Conductivity (Mixture)	0.229675	W/[m.K]
Molar Fraction (Vapor)		
Methane	0.110414	
Ethane	2.67764E-17	
Propane	6.08556E-19	
N-butane	0	
Isobutane	0	
Nitrogen	0.128484	
Carbon dioxide	0.00125769	
Water	0.000989796	
Oxygen	3.4435E-08	
Argon	0.00276159	
Hydrogen sulfide	0	
Hydrogen	0.547403	

Property	Value	Unit
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0.20869	
Density (Vapor)	24.8938	kg/m3
Molecular Weight (Vapor)	10.2117	kg/kmol
Specific Enthalpy (Vapor)	1335.44	kJ/kg
Specific Entropy (Vapor)	0.459074	kJ/[kg.K]
Molar Enthalpy (Vapor)	13637.1	kJ/kmol
Molar Entropy (Vapor)	4.68791	kJ/[kmol.K]
Thermal Conductivity (Vapor)	0.229675	W/[m.K]
Kinematic Viscosity (Vapor)	8.55799E-07	m2/s
Dynamic Viscosity (Vapor)	2.13041E-05	Pa.s
Heat Capacity (Vapor)	3.68857	kJ/[kg.K]
Heat Capacity Ratio (Vapor)	1.3134	
Mass Flow (Vapor)	43227.4	kg/h
Molar Flow (Vapor)	4233.14	kmol/h
Volumetric Flow (Vapor)	1736.47	m3/h
Compressibility Factor (Vapor)	1.03149	
Molar Fraction (Vapor)	1	
Mass Fraction (Vapor)	1	
Volumetric Fraction (Vapor)	1	
Molar Fraction (Overall Liquid)		
Methane	NaN	
Ethane	NaN	
Propane	NaN	
N-butane	NaN	
Isobutane	NaN	
Nitrogen	NaN	
Carbon dioxide	NaN	
Water	NaN	
Oxygen	NaN	
Argon	NaN	
Hydrogen sulfide	NaN	
Hydrogen	NaN	
Carbonyl sulfide	NaN	
Carbon monoxide	NaN	
Ammonia	NaN	
Density (Overall Liquid)	0	kg/m3
Molecular Weight (Overall Liquid)	NaN	kg/kmol
Specific Enthalpy (Overall Liquid)	0	kJ/kg
Specific Entropy (Overall Liquid)	0	kJ/[kg.K]
Molar Enthalpy (Overall Liquid)	NaN	kJ/kmol

Property	Value	Unit
Molar Entropy (Overall Liquid)	NaN	kJ/[kmol.K]
Thermal Conductivity (Overall Liquid)	0	W/[m.K]
Kinematic Viscosity (Overall Liquid)	0	m2/s
Dynamic Viscosity (Overall Liquid)	0	Pa.s
Heat Capacity (Overall Liquid)	0	kJ/[kg.K]
Heat Capacity Ratio (Overall Liquid)	NaN	
Mass Flow (Overall Liquid)	0	kg/h
Molar Flow (Overall Liquid)	0	kmol/h
Volumetric Flow (Overall Liquid)	0	m3/h
Compressibility Factor (Overall Liquid)	0	
Molar Fraction (Overall Liquid)	0	
Mass Fraction (Overall Liquid)	0	
Volumetric Fraction (Overall Liquid)	0	
Molar Fraction (Liquid 1)		
Methane	1.57989E-05	
Ethane	4.41062E-18	
Propane	6.10799E-18	
N-butane	0	
Isobutane	0	
Nitrogen	7.8882E-15	
Carbon dioxide	3.68309E-05	
Water	0.0603923	
Oxygen	2.16564E-15	
Argon	7.57462E-13	
Hydrogen sulfide	0	
Hydrogen	1.80432E-68	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0.939555	
Molar Fraction (Liquid 2)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	

Property	Value	Unit
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Molar Fraction (Aqueous)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Molar Fraction (Solid)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	

METH-OUT-901 (Material Stream)

Property	Value	Unit
Temperature	40	C
Pressure	34	bar
Mass Flow	50605.1	kg/h
Molar Flow	5526.05	kmol/h
Volumetric Flow	3958.15	m3/h
Mixture Molar Fraction		
Methane	0.08458	
Ethane	2.05115E-17	
Propane	4.6617E-19	
N-butane	0	
Isobutane	0	
Nitrogen	0.178353	
Carbon dioxide	0.000964567	
Water	0.0748666	
Oxygen	2.63784E-08	
Argon	0.00211547	
Hydrogen sulfide	0	
Hydrogen	0.65912	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Density (Mixture)	12.7851	kg/m3
Molecular Weight (Mixture)	9.15756	kg/kmol
Specific Enthalpy (Mixture)	-314.701	kJ/kg
Specific Entropy (Mixture)	-2.73789	kJ/[kg.K]
Molar Enthalpy (Mixture)	-2881.89	kJ/kmol
Molar Entropy (Mixture)	-25.0724	kJ/[kmol.K]
Thermal Conductivity (Mixture)	0.177368	W/[m.K]
Molar Fraction (Vapor)		
Methane	0.0912208	
Ethane	2.2122E-17	
Propane	5.02772E-19	
N-butane	0	
Isobutane	0	
Nitrogen	0.192357	
Carbon dioxide	0.00103911	
Water	0.00223153	
Oxygen	2.84492E-08	
Argon	0.00228155	
Hydrogen sulfide	0	
Hydrogen	0.71087	

Property	Value	Unit
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Density (Vapor)	10.9781	kg/m3
Molecular Weight (Vapor)	8.46208	kg/kmol
Specific Enthalpy (Vapor)	44.9108	kJ/kg
Specific Entropy (Vapor)	-2.08992	kJ/[kg.K]
Molar Enthalpy (Vapor)	380.039	kJ/kmol
Molar Entropy (Vapor)	-17.6851	kJ/[kmol.K]
Thermal Conductivity (Vapor)	0.141764	W/[m.K]
Kinematic Viscosity (Vapor)	1.04633E-06	m2/s
Dynamic Viscosity (Vapor)	1.14867E-05	Pa.s
Heat Capacity (Vapor)	3.55293	kJ/[kg.K]
Heat Capacity Ratio (Vapor)	1.42364	
Mass Flow (Vapor)	43357.9	kg/h
Molar Flow (Vapor)	5123.79	kmol/h
Volumetric Flow (Vapor)	3949.5	m3/h
Compressibility Factor (Vapor)	1.00662	
Molar Fraction (Vapor)	0.927207	
Mass Fraction (Vapor)	0.856795	
Volumetric Fraction (Vapor)	0.997814	
Molar Fraction (Overall Liquid)		
Methane	2.91568E-08	
Ethane	1.86454E-25	
Propane	2.36866E-29	
N-butane	0	
Isobutane	0	
Nitrogen	1.24194E-06	
Carbon dioxide	1.51454E-05	
Water	0.999971	
Oxygen	3.31659E-12	
Argon	2.55208E-07	
Hydrogen sulfide	0	
Hydrogen	1.26365E-05	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Density (Overall Liquid)	840.395	kg/m3
Molecular Weight (Overall Liquid)	18.0155	kg/kmol
Specific Enthalpy (Overall Liquid)	-2466.25	kJ/kg
Specific Entropy (Overall Liquid)	-6.61468	kJ/[kg.K]
Molar Enthalpy (Overall Liquid)	-44430.7	kJ/kmol

Property	Value	Unit
Molar Entropy (Overall Liquid)	-119.167	kJ/[kmol.K]
Thermal Conductivity (Overall Liquid)	0.63088	W/[m.K]
Kinematic Viscosity (Overall Liquid)	7.81696E-07	m ² /s
Dynamic Viscosity (Overall Liquid)	0.000656933	Pa.s
Heat Capacity (Overall Liquid)	3.91689	kJ/[kg.K]
Heat Capacity Ratio (Overall Liquid)	1.30788	
Mass Flow (Overall Liquid)	7246.88	kg/h
Molar Flow (Overall Liquid)	402.258	kmol/h
Volumetric Flow (Overall Liquid)	8.62317	m ³ /h
Compressibility Factor (Overall Liquid)	0.0279949	
Molar Fraction (Overall Liquid)	0.072793	
Mass Fraction (Overall Liquid)	0.143205	
Volumetric Fraction (Overall Liquid)	0.00217859	
Molar Fraction (Liquid 1)		
Methane	2.91568E-08	
Ethane	1.86454E-25	
Propane	2.36866E-29	
N-butane	0	
Isobutane	0	
Nitrogen	1.24194E-06	
Carbon dioxide	1.51454E-05	
Water	0.999971	
Oxygen	3.31659E-12	
Argon	2.55208E-07	
Hydrogen sulfide	0	
Hydrogen	1.26365E-05	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Molar Fraction (Liquid 2)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	

Property	Value	Unit
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Molar Fraction (Aqueous)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Molar Fraction (Solid)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	

33 (Material Stream)

Property	Value	Unit
Temperature	852.093	C
Pressure	34	bar
Mass Flow	0	kg/h
Molar Flow	0	kmol/h
Volumetric Flow	0	m3/h
Mixture Molar Fraction		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Density (Mixture)	0	kg/m3
Molecular Weight (Mixture)	0	kg/kmol
Specific Enthalpy (Mixture)	NaN	kJ/kg
Specific Entropy (Mixture)	0	kJ/[kg.K]
Molar Enthalpy (Mixture)	0	kJ/kmol
Molar Entropy (Mixture)	0	kJ/[kmol.K]
Thermal Conductivity (Mixture)	0	W/[m.K]
Molar Fraction (Vapor)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	

Property	Value	Unit
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Density (Vapor)	0	kg/m3
Molecular Weight (Vapor)	0	kg/kmol
Specific Enthalpy (Vapor)	0	kJ/kg
Specific Entropy (Vapor)	0	kJ/[kg.K]
Molar Enthalpy (Vapor)	0	kJ/kmol
Molar Entropy (Vapor)	0	kJ/[kmol.K]
Thermal Conductivity (Vapor)	0	W/[m.K]
Kinematic Viscosity (Vapor)	0	m2/s
Dynamic Viscosity (Vapor)	0	Pa.s
Heat Capacity (Vapor)	0	kJ/[kg.K]
Heat Capacity Ratio (Vapor)	NaN	
Mass Flow (Vapor)	0	kg/h
Molar Flow (Vapor)	0	kmol/h
Volumetric Flow (Vapor)	0	m3/h
Compressibility Factor (Vapor)	0	
Molar Fraction (Vapor)	0	
Mass Fraction (Vapor)	0	
Volumetric Fraction (Vapor)	NaN	
Molar Fraction (Overall Liquid)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Density (Overall Liquid)	0	kg/m3
Molecular Weight (Overall Liquid)	0	kg/kmol
Specific Enthalpy (Overall Liquid)	0	kJ/kg
Specific Entropy (Overall Liquid)	0	kJ/[kg.K]
Molar Enthalpy (Overall Liquid)	0	kJ/kmol

Property	Value	Unit
Molar Entropy (Overall Liquid)	0	kJ/[kmol.K]
Thermal Conductivity (Overall Liquid)	0	W/[m.K]
Kinematic Viscosity (Overall Liquid)	0	m2/s
Dynamic Viscosity (Overall Liquid)	0	Pa.s
Heat Capacity (Overall Liquid)	0	kJ/[kg.K]
Heat Capacity Ratio (Overall Liquid)	NaN	
Mass Flow (Overall Liquid)	0	kg/h
Molar Flow (Overall Liquid)	0	kmol/h
Volumetric Flow (Overall Liquid)	0	m3/h
Compressibility Factor (Overall Liquid)	0	
Molar Fraction (Overall Liquid)	0	
Mass Fraction (Overall Liquid)	0	
Volumetric Fraction (Overall Liquid)	NaN	
Molar Fraction (Liquid 1)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Molar Fraction (Liquid 2)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	

Property	Value	Unit
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Molar Fraction (Aqueous)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Molar Fraction (Solid)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	

METH-902 (Material Stream)

Property	Value	Unit
Temperature	852.093	C
Pressure	34	bar
Mass Flow	50605.1	kg/h
Molar Flow	5526.05	kmol/h
Volumetric Flow	15301	m3/h
Mixture Molar Fraction		
Methane	0.08458	
Ethane	2.05115E-17	
Propane	4.6617E-19	
N-butane	0	
Isobutane	0	
Nitrogen	0.178353	
Carbon dioxide	0.000964567	
Water	0.0748666	
Oxygen	2.63784E-08	
Argon	0.00211547	
Hydrogen sulfide	0	
Hydrogen	0.65912	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Density (Mixture)	3.30731	kg/m3
Molecular Weight (Mixture)	9.15756	kg/kmol
Specific Enthalpy (Mixture)	2971.97	kJ/kg
Specific Entropy (Mixture)	2.90534	kJ/[kg.K]
Molar Enthalpy (Mixture)	27216	kJ/kmol
Molar Entropy (Mixture)	26.6058	kJ/[kmol.K]
Thermal Conductivity (Mixture)	0.364468	W/[m.K]
Molar Fraction (Vapor)		
Methane	0.08458	
Ethane	2.05115E-17	
Propane	4.6617E-19	
N-butane	0	
Isobutane	0	
Nitrogen	0.178353	
Carbon dioxide	0.000964567	
Water	0.0748666	
Oxygen	2.63784E-08	
Argon	0.00211547	
Hydrogen sulfide	0	
Hydrogen	0.65912	

Property	Value	Unit
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Density (Vapor)	3.30731	kg/m3
Molecular Weight (Vapor)	9.15756	kg/kmol
Specific Enthalpy (Vapor)	2971.97	kJ/kg
Specific Entropy (Vapor)	2.90534	kJ/[kg.K]
Molar Enthalpy (Vapor)	27216	kJ/kmol
Molar Entropy (Vapor)	26.6058	kJ/[kmol.K]
Thermal Conductivity (Vapor)	0.364468	W/[m.K]
Kinematic Viscosity (Vapor)	8.6415E-06	m2/s
Dynamic Viscosity (Vapor)	2.85801E-05	Pa.s
Heat Capacity (Vapor)	3.92409	kJ/[kg.K]
Heat Capacity Ratio (Vapor)	1.30251	
Mass Flow (Vapor)	50605.1	kg/h
Molar Flow (Vapor)	5526.05	kmol/h
Volumetric Flow (Vapor)	15301	m3/h
Compressibility Factor (Vapor)	1.0063	
Molar Fraction (Vapor)	1	
Mass Fraction (Vapor)	1	
Volumetric Fraction (Vapor)	1	
Molar Fraction (Overall Liquid)		
Methane	NaN	
Ethane	NaN	
Propane	NaN	
N-butane	NaN	
Isobutane	NaN	
Nitrogen	NaN	
Carbon dioxide	NaN	
Water	NaN	
Oxygen	NaN	
Argon	NaN	
Hydrogen sulfide	NaN	
Hydrogen	NaN	
Carbonyl sulfide	NaN	
Carbon monoxide	NaN	
Ammonia	NaN	
Density (Overall Liquid)	0	kg/m3
Molecular Weight (Overall Liquid)	NaN	kg/kmol
Specific Enthalpy (Overall Liquid)	0	kJ/kg
Specific Entropy (Overall Liquid)	0	kJ/[kg.K]
Molar Enthalpy (Overall Liquid)	NaN	kJ/kmol

Property	Value	Unit
Molar Entropy (Overall Liquid)	NaN	kJ/[kmol.K]
Thermal Conductivity (Overall Liquid)	0	W/[m.K]
Kinematic Viscosity (Overall Liquid)	0	m2/s
Dynamic Viscosity (Overall Liquid)	0	Pa.s
Heat Capacity (Overall Liquid)	0	kJ/[kg.K]
Heat Capacity Ratio (Overall Liquid)	NaN	
Mass Flow (Overall Liquid)	0	kg/h
Molar Flow (Overall Liquid)	0	kmol/h
Volumetric Flow (Overall Liquid)	0	m3/h
Compressibility Factor (Overall Liquid)	0	
Molar Fraction (Overall Liquid)	0	
Mass Fraction (Overall Liquid)	0	
Volumetric Fraction (Overall Liquid)	0	
Molar Fraction (Liquid 1)		
Methane	5.05091E-15	
Ethane	1.38864E-23	
Propane	8.66202E-20	
N-butane	0	
Isobutane	0	
Nitrogen	8.39881E-40	
Carbon dioxide	1.07552E-12	
Water	1	
Oxygen	2.5834E-30	
Argon	3.08807E-32	
Hydrogen sulfide	0	
Hydrogen	2.59449E-174	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Molar Fraction (Liquid 2)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	

Property	Value	Unit
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Molar Fraction (Aqueous)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Molar Fraction (Solid)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	

METH-901 (Material Stream)

Property	Value	Unit
Temperature	420	C
Pressure	34.5	bar
Mass Flow	50605.1	kg/h
Molar Flow	7397.36	kmol/h
Volumetric Flow	12466.2	m3/h
Mixture Molar Fraction		
Methane	0.0125779	
Ethane	1.90186E-17	
Propane	4.32241E-19	
N-butane	0	
Isobutane	0	
Nitrogen	0.165372	
Carbon dioxide	0.00226842	
Water	0.00219743	
Oxygen	2.44585E-08	
Argon	0.0019615	
Hydrogen sulfide	0	
Hydrogen	0.81006	
Carbonyl sulfide	0	
Carbon monoxide	0.00556211	
Ammonia	0	
Density (Mixture)	4.05939	kg/m3
Molecular Weight (Mixture)	6.84098	kg/kmol
Specific Enthalpy (Mixture)	1715.59	kJ/kg
Specific Entropy (Mixture)	0.193758	kJ/[kg.K]
Molar Enthalpy (Mixture)	11736.3	kJ/kmol
Molar Entropy (Mixture)	1.3255	kJ/[kmol.K]
Thermal Conductivity (Mixture)	0.287582	W/[m.K]
Molar Fraction (Vapor)		
Methane	0.0125779	
Ethane	1.90186E-17	
Propane	4.32241E-19	
N-butane	0	
Isobutane	0	
Nitrogen	0.165372	
Carbon dioxide	0.00226842	
Water	0.00219743	
Oxygen	2.44585E-08	
Argon	0.0019615	
Hydrogen sulfide	0	
Hydrogen	0.81006	

Property	Value	Unit
Carbonyl sulfide	0	
Carbon monoxide	0.00556211	
Ammonia	0	
Density (Vapor)	4.05939	kg/m3
Molecular Weight (Vapor)	6.84098	kg/kmol
Specific Enthalpy (Vapor)	1715.59	kJ/kg
Specific Entropy (Vapor)	0.193758	kJ/[kg.K]
Molar Enthalpy (Vapor)	11736.3	kJ/kmol
Molar Entropy (Vapor)	1.3255	kJ/[kmol.K]
Thermal Conductivity (Vapor)	0.287582	W/[m.K]
Kinematic Viscosity (Vapor)	4.66741E-06	m2/s
Dynamic Viscosity (Vapor)	1.89468E-05	Pa.s
Heat Capacity (Vapor)	4.42169	kJ/[kg.K]
Heat Capacity Ratio (Vapor)	1.38375	
Mass Flow (Vapor)	50605.1	kg/h
Molar Flow (Vapor)	7397.36	kmol/h
Volumetric Flow (Vapor)	12466.2	m3/h
Compressibility Factor (Vapor)	1.00888	
Molar Fraction (Vapor)	1	
Mass Fraction (Vapor)	1	
Volumetric Fraction (Vapor)	1	
Molar Fraction (Overall Liquid)		
Methane	NaN	
Ethane	NaN	
Propane	NaN	
N-butane	NaN	
Isobutane	NaN	
Nitrogen	NaN	
Carbon dioxide	NaN	
Water	NaN	
Oxygen	NaN	
Argon	NaN	
Hydrogen sulfide	NaN	
Hydrogen	NaN	
Carbonyl sulfide	NaN	
Carbon monoxide	NaN	
Ammonia	NaN	
Density (Overall Liquid)	0	kg/m3
Molecular Weight (Overall Liquid)	NaN	kg/kmol
Specific Enthalpy (Overall Liquid)	0	kJ/kg
Specific Entropy (Overall Liquid)	0	kJ/[kg.K]
Molar Enthalpy (Overall Liquid)	NaN	kJ/kmol

Property	Value	Unit
Molar Entropy (Overall Liquid)	NaN	kJ/[kmol.K]
Thermal Conductivity (Overall Liquid)	0	W/[m.K]
Kinematic Viscosity (Overall Liquid)	0	m ² /s
Dynamic Viscosity (Overall Liquid)	0	Pa.s
Heat Capacity (Overall Liquid)	0	kJ/[kg.K]
Heat Capacity Ratio (Overall Liquid)	NaN	
Mass Flow (Overall Liquid)	0	kg/h
Molar Flow (Overall Liquid)	0	kmol/h
Volumetric Flow (Overall Liquid)	0	m ³ /h
Compressibility Factor (Overall Liquid)	0	
Molar Fraction (Overall Liquid)	0	
Mass Fraction (Overall Liquid)	0	
Volumetric Fraction (Overall Liquid)	0	
Molar Fraction (Liquid 1)		
Methane	1.84921E-05	
Ethane	2.71859E-17	
Propane	3.27952E-17	
N-butane	0	
Isobutane	0	
Nitrogen	1.99271E-13	
Carbon dioxide	0.00061677	
Water	0.999365	
Oxygen	1.97463E-14	
Argon	8.28113E-12	
Hydrogen sulfide	0	
Hydrogen	1.55179E-65	
Carbonyl sulfide	0	
Carbon monoxide	1.35576E-11	
Ammonia	0	
Molar Fraction (Liquid 2)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	

Property	Value	Unit
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Molar Fraction (Aqueous)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Molar Fraction (Solid)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	

WATER-OUT-801 (Material Stream)

Property	Value	Unit
Temperature	40	C
Pressure	34.5	bar
Mass Flow	38251.1	kg/h
Molar Flow	2123.18	kmol/h
Volumetric Flow	45.5147	m3/h
Mixture Molar Fraction		
Methane	4.10344E-09	
Ethane	1.64628E-25	
Propane	2.10015E-29	
N-butane	0	
Isobutane	0	
Nitrogen	1.08588E-06	
Carbon dioxide	3.37618E-05	
Water	0.99995	
Oxygen	2.90174E-12	
Argon	2.23225E-07	
Hydrogen sulfide	0	
Hydrogen	1.45967E-05	
Carbonyl sulfide	0	
Carbon monoxide	3.80598E-08	
Ammonia	0	
Density (Mixture)	840.412	kg/m3
Molecular Weight (Mixture)	18.0159	kg/kmol
Specific Enthalpy (Mixture)	-2466.09	kJ/kg
Specific Entropy (Mixture)	-6.61437	kJ/[kg.K]
Molar Enthalpy (Mixture)	-44428.9	kJ/kmol
Molar Entropy (Mixture)	-119.164	kJ/[kmol.K]
Thermal Conductivity (Mixture)	0.630845	W/[m.K]
Molar Fraction (Vapor)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	

Property	Value	Unit
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Density (Vapor)	0	kg/m3
Molecular Weight (Vapor)	0	kg/kmol
Specific Enthalpy (Vapor)	0	kJ/kg
Specific Entropy (Vapor)	0	kJ/[kg.K]
Molar Enthalpy (Vapor)	0	kJ/kmol
Molar Entropy (Vapor)	0	kJ/[kmol.K]
Thermal Conductivity (Vapor)	0	W/[m.K]
Kinematic Viscosity (Vapor)	0	m2/s
Dynamic Viscosity (Vapor)	0	Pa.s
Heat Capacity (Vapor)	0	kJ/[kg.K]
Heat Capacity Ratio (Vapor)	NaN	
Mass Flow (Vapor)	0	kg/h
Molar Flow (Vapor)	0	kmol/h
Volumetric Flow (Vapor)	0	m3/h
Compressibility Factor (Vapor)	0	
Molar Fraction (Vapor)	0	
Mass Fraction (Vapor)	0	
Volumetric Fraction (Vapor)	0	
Molar Fraction (Overall Liquid)		
Methane	4.10344E-09	
Ethane	1.64628E-25	
Propane	2.10015E-29	
N-butane	0	
Isobutane	0	
Nitrogen	1.08588E-06	
Carbon dioxide	3.37618E-05	
Water	0.99995	
Oxygen	2.90174E-12	
Argon	2.23225E-07	
Hydrogen sulfide	0	
Hydrogen	1.45967E-05	
Carbonyl sulfide	0	
Carbon monoxide	3.80598E-08	
Ammonia	0	
Density (Overall Liquid)	840.412	kg/m3
Molecular Weight (Overall Liquid)	18.0159	kg/kmol
Specific Enthalpy (Overall Liquid)	-2466.09	kJ/kg
Specific Entropy (Overall Liquid)	-6.61437	kJ/[kg.K]
Molar Enthalpy (Overall Liquid)	-44428.9	kJ/kmol

Property	Value	Unit
Molar Entropy (Overall Liquid)	-119.164	kJ/[kmol.K]
Thermal Conductivity (Overall Liquid)	0.630845	W/[m.K]
Kinematic Viscosity (Overall Liquid)	7.81747E-07	m ² /s
Dynamic Viscosity (Overall Liquid)	0.00065699	Pa.s
Heat Capacity (Overall Liquid)	3.91672	kJ/[kg.K]
Heat Capacity Ratio (Overall Liquid)	1.30786	
Mass Flow (Overall Liquid)	38251.1	kg/h
Molar Flow (Overall Liquid)	2123.18	kmol/h
Volumetric Flow (Overall Liquid)	45.5147	m ³ /h
Compressibility Factor (Overall Liquid)	0.0284067	
Molar Fraction (Overall Liquid)	1	
Mass Fraction (Overall Liquid)	1	
Volumetric Fraction (Overall Liquid)	1	
Molar Fraction (Liquid 1)		
Methane	4.10344E-09	
Ethane	1.64628E-25	
Propane	2.10015E-29	
N-butane	0	
Isobutane	0	
Nitrogen	1.08588E-06	
Carbon dioxide	3.37618E-05	
Water	0.99995	
Oxygen	2.90174E-12	
Argon	2.23225E-07	
Hydrogen sulfide	0	
Hydrogen	1.45967E-05	
Carbonyl sulfide	0	
Carbon monoxide	3.80598E-08	
Ammonia	0	
Molar Fraction (Liquid 2)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	

Property	Value	Unit
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Molar Fraction (Aqueous)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Molar Fraction (Solid)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	

SYNGAS-802 (Material Stream)

Property	Value	Unit
Temperature	40	C
Pressure	34.5	bar
Mass Flow	50605.1	kg/h
Molar Flow	7397.36	kmol/h
Volumetric Flow	5634.34	m3/h
Mixture Molar Fraction		
Methane	0.0125779	
Ethane	1.90186E-17	
Propane	4.32241E-19	
N-butane	0	
Isobutane	0	
Nitrogen	0.165372	
Carbon dioxide	0.00226842	
Water	0.00219743	
Oxygen	2.44585E-08	
Argon	0.0019615	
Hydrogen sulfide	0	
Hydrogen	0.81006	
Carbonyl sulfide	0	
Carbon monoxide	0.00556211	
Ammonia	0	
Density (Mixture)	8.98156	kg/m3
Molecular Weight (Mixture)	6.84098	kg/kmol
Specific Enthalpy (Mixture)	58.3602	kJ/kg
Specific Entropy (Mixture)	-3.26476	kJ/[kg.K]
Molar Enthalpy (Mixture)	399.241	kJ/kmol
Molar Entropy (Mixture)	-22.3341	kJ/[kmol.K]
Thermal Conductivity (Mixture)	0.156607	W/[m.K]
Molar Fraction (Vapor)		
Methane	0.0125779	
Ethane	1.90186E-17	
Propane	4.32241E-19	
N-butane	0	
Isobutane	0	
Nitrogen	0.165372	
Carbon dioxide	0.00226842	
Water	0.00219743	
Oxygen	2.44585E-08	
Argon	0.0019615	
Hydrogen sulfide	0	
Hydrogen	0.81006	

Property	Value	Unit
Carbonyl sulfide	0	
Carbon monoxide	0.00556211	
Ammonia	0	
Density (Vapor)	8.98156	kg/m3
Molecular Weight (Vapor)	6.84098	kg/kmol
Specific Enthalpy (Vapor)	58.3602	kJ/kg
Specific Entropy (Vapor)	-3.26476	kJ/[kg.K]
Molar Enthalpy (Vapor)	399.241	kJ/kmol
Molar Entropy (Vapor)	-22.3341	kJ/[kmol.K]
Thermal Conductivity (Vapor)	0.156607	W/[m.K]
Kinematic Viscosity (Vapor)	1.23156E-06	m2/s
Dynamic Viscosity (Vapor)	1.10614E-05	Pa.s
Heat Capacity (Vapor)	4.28919	kJ/[kg.K]
Heat Capacity Ratio (Vapor)	1.42903	
Mass Flow (Vapor)	50605.1	kg/h
Molar Flow (Vapor)	7397.36	kmol/h
Volumetric Flow (Vapor)	5634.34	m3/h
Compressibility Factor (Vapor)	1.00931	
Molar Fraction (Vapor)	1	
Mass Fraction (Vapor)	1	
Volumetric Fraction (Vapor)	1	
Molar Fraction (Overall Liquid)		
Methane	NaN	
Ethane	NaN	
Propane	NaN	
N-butane	NaN	
Isobutane	NaN	
Nitrogen	NaN	
Carbon dioxide	NaN	
Water	NaN	
Oxygen	NaN	
Argon	NaN	
Hydrogen sulfide	NaN	
Hydrogen	NaN	
Carbonyl sulfide	NaN	
Carbon monoxide	NaN	
Ammonia	NaN	
Density (Overall Liquid)	0	kg/m3
Molecular Weight (Overall Liquid)	NaN	kg/kmol
Specific Enthalpy (Overall Liquid)	0	kJ/kg
Specific Entropy (Overall Liquid)	0	kJ/[kg.K]
Molar Enthalpy (Overall Liquid)	NaN	kJ/kmol

Property	Value	Unit
Molar Entropy (Overall Liquid)	NaN	kJ/[kmol.K]
Thermal Conductivity (Overall Liquid)	0	W/[m.K]
Kinematic Viscosity (Overall Liquid)	0	m2/s
Dynamic Viscosity (Overall Liquid)	0	Pa.s
Heat Capacity (Overall Liquid)	0	kJ/[kg.K]
Heat Capacity Ratio (Overall Liquid)	NaN	
Mass Flow (Overall Liquid)	0	kg/h
Molar Flow (Overall Liquid)	0	kmol/h
Volumetric Flow (Overall Liquid)	0	m3/h
Compressibility Factor (Overall Liquid)	0	
Molar Fraction (Overall Liquid)	0	
Mass Fraction (Overall Liquid)	0	
Volumetric Fraction (Overall Liquid)	0	
Molar Fraction (Liquid 1)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Molar Fraction (Liquid 2)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	

Property	Value	Unit
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Molar Fraction (Aqueous)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Molar Fraction (Solid)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	

SYNGAS-801 (Material Stream)

Property	Value	Unit
Temperature	40	C
Pressure	34.5	bar
Mass Flow	88856.2	kg/h
Molar Flow	9520.54	kmol/h
Volumetric Flow	5679.85	m3/h
Mixture Molar Fraction		
Methane	0.00977286	
Ethane	1.47773E-17	
Propane	3.35847E-19	
N-butane	0	
Isobutane	0	
Nitrogen	0.128493	
Carbon dioxide	0.00177007	
Water	0.224707	
Oxygen	1.90046E-08	
Argon	0.00152412	
Hydrogen sulfide	0	
Hydrogen	0.629412	
Carbonyl sulfide	0	
Carbon monoxide	0.00432171	
Ammonia	0	
Density (Mixture)	15.6441	kg/m3
Molecular Weight (Mixture)	9.33311	kg/kmol
Specific Enthalpy (Mixture)	-1028.37	kJ/kg
Specific Entropy (Mixture)	-4.70671	kJ/[kg.K]
Molar Enthalpy (Mixture)	-9597.92	kJ/kmol
Molar Entropy (Mixture)	-43.9282	kJ/[kmol.K]
Thermal Conductivity (Mixture)	0.262367	W/[m.K]
Molar Fraction (Vapor)		
Methane	0.0125779	
Ethane	1.90186E-17	
Propane	4.32241E-19	
N-butane	0	
Isobutane	0	
Nitrogen	0.165372	
Carbon dioxide	0.00226842	
Water	0.00219743	
Oxygen	2.44585E-08	
Argon	0.0019615	
Hydrogen sulfide	0	
Hydrogen	0.81006	

Property	Value	Unit
Carbonyl sulfide	0	
Carbon monoxide	0.00556211	
Ammonia	0	
Density (Vapor)	8.98156	kg/m3
Molecular Weight (Vapor)	6.84098	kg/kmol
Specific Enthalpy (Vapor)	58.3602	kJ/kg
Specific Entropy (Vapor)	-3.26476	kJ/[kg.K]
Molar Enthalpy (Vapor)	399.241	kJ/kmol
Molar Entropy (Vapor)	-22.3341	kJ/[kmol.K]
Thermal Conductivity (Vapor)	0.156607	W/[m.K]
Kinematic Viscosity (Vapor)	1.23156E-06	m2/s
Dynamic Viscosity (Vapor)	1.10614E-05	Pa.s
Heat Capacity (Vapor)	4.28919	kJ/[kg.K]
Heat Capacity Ratio (Vapor)	1.42903	
Mass Flow (Vapor)	50605.1	kg/h
Molar Flow (Vapor)	7397.36	kmol/h
Volumetric Flow (Vapor)	5634.34	m3/h
Compressibility Factor (Vapor)	1.00931	
Molar Fraction (Vapor)	0.776989	
Mass Fraction (Vapor)	0.569517	
Volumetric Fraction (Vapor)	0.991987	
Molar Fraction (Overall Liquid)		
Methane	4.10344E-09	
Ethane	1.64628E-25	
Propane	2.10015E-29	
N-butane	0	
Isobutane	0	
Nitrogen	1.08588E-06	
Carbon dioxide	3.37618E-05	
Water	0.99995	
Oxygen	2.90174E-12	
Argon	2.23225E-07	
Hydrogen sulfide	0	
Hydrogen	1.45967E-05	
Carbonyl sulfide	0	
Carbon monoxide	3.80598E-08	
Ammonia	0	
Density (Overall Liquid)	840.412	kg/m3
Molecular Weight (Overall Liquid)	18.0159	kg/kmol
Specific Enthalpy (Overall Liquid)	-2466.09	kJ/kg
Specific Entropy (Overall Liquid)	-6.61437	kJ/[kg.K]
Molar Enthalpy (Overall Liquid)	-44428.9	kJ/kmol

Property	Value	Unit
Molar Entropy (Overall Liquid)	-119.164	kJ/[kmol.K]
Thermal Conductivity (Overall Liquid)	0.630845	W/[m.K]
Kinematic Viscosity (Overall Liquid)	7.81747E-07	m ² /s
Dynamic Viscosity (Overall Liquid)	0.00065699	Pa.s
Heat Capacity (Overall Liquid)	3.91672	kJ/[kg.K]
Heat Capacity Ratio (Overall Liquid)	1.30786	
Mass Flow (Overall Liquid)	38251.1	kg/h
Molar Flow (Overall Liquid)	2123.18	kmol/h
Volumetric Flow (Overall Liquid)	45.5147	m ³ /h
Compressibility Factor (Overall Liquid)	0.0284067	
Molar Fraction (Overall Liquid)	0.223011	
Mass Fraction (Overall Liquid)	0.430483	
Volumetric Fraction (Overall Liquid)	0.00801336	
Molar Fraction (Liquid 1)		
Methane	4.10344E-09	
Ethane	1.64628E-25	
Propane	2.10015E-29	
N-butane	0	
Isobutane	0	
Nitrogen	1.08588E-06	
Carbon dioxide	3.37618E-05	
Water	0.99995	
Oxygen	2.90174E-12	
Argon	2.23225E-07	
Hydrogen sulfide	0	
Hydrogen	1.45967E-05	
Carbonyl sulfide	0	
Carbon monoxide	3.80598E-08	
Ammonia	0	
Molar Fraction (Liquid 2)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	

Property	Value	Unit
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Molar Fraction (Aqueous)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Molar Fraction (Solid)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	

CO2-OUT-801 (Material Stream)

Property	Value	Unit
Temperature	40	C
Pressure	34.5	bar
Mass Flow	73423.2	kg/h
Molar Flow	1668.35	kmol/h
Volumetric Flow	1040.5	m3/h
Mixture Molar Fraction		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	1	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Density (Mixture)	70.5655	kg/m3
Molecular Weight (Mixture)	44.0095	kg/kmol
Specific Enthalpy (Mixture)	-21.1999	kJ/kg
Specific Entropy (Mixture)	-0.701275	kJ/[kg.K]
Molar Enthalpy (Mixture)	-932.997	kJ/kmol
Molar Entropy (Mixture)	-30.8628	kJ/[kmol.K]
Thermal Conductivity (Mixture)	0.0209487	W/[m.K]
Molar Fraction (Vapor)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	1	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	

Property	Value	Unit
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Density (Vapor)	70.5655	kg/m3
Molecular Weight (Vapor)	44.0095	kg/kmol
Specific Enthalpy (Vapor)	-21.1999	kJ/kg
Specific Entropy (Vapor)	-0.701275	kJ/[kg.K]
Molar Enthalpy (Vapor)	-932.997	kJ/kmol
Molar Entropy (Vapor)	-30.8628	kJ/[kmol.K]
Thermal Conductivity (Vapor)	0.0209487	W/[m.K]
Kinematic Viscosity (Vapor)	2.4253E-07	m2/s
Dynamic Viscosity (Vapor)	1.71143E-05	Pa.s
Heat Capacity (Vapor)	1.0939	kJ/[kg.K]
Heat Capacity Ratio (Vapor)	1.6234	
Mass Flow (Vapor)	73423.2	kg/h
Molar Flow (Vapor)	1668.35	kmol/h
Volumetric Flow (Vapor)	1040.5	m3/h
Compressibility Factor (Vapor)	0.826438	
Molar Fraction (Vapor)	1	
Mass Fraction (Vapor)	1	
Volumetric Fraction (Vapor)	1	
Molar Fraction (Overall Liquid)		
Methane	NaN	
Ethane	NaN	
Propane	NaN	
N-butane	NaN	
Isobutane	NaN	
Nitrogen	NaN	
Carbon dioxide	NaN	
Water	NaN	
Oxygen	NaN	
Argon	NaN	
Hydrogen sulfide	NaN	
Hydrogen	NaN	
Carbonyl sulfide	NaN	
Carbon monoxide	NaN	
Ammonia	NaN	
Density (Overall Liquid)	0	kg/m3
Molecular Weight (Overall Liquid)	NaN	kg/kmol
Specific Enthalpy (Overall Liquid)	0	kJ/kg
Specific Entropy (Overall Liquid)	0	kJ/[kg.K]
Molar Enthalpy (Overall Liquid)	NaN	kJ/kmol

Property	Value	Unit
Molar Entropy (Overall Liquid)	NaN	kJ/[kmol.K]
Thermal Conductivity (Overall Liquid)	0	W/[m.K]
Kinematic Viscosity (Overall Liquid)	0	m2/s
Dynamic Viscosity (Overall Liquid)	0	Pa.s
Heat Capacity (Overall Liquid)	0	kJ/[kg.K]
Heat Capacity Ratio (Overall Liquid)	NaN	
Mass Flow (Overall Liquid)	0	kg/h
Molar Flow (Overall Liquid)	0	kmol/h
Volumetric Flow (Overall Liquid)	0	m3/h
Compressibility Factor (Overall Liquid)	0	
Molar Fraction (Overall Liquid)	0	
Mass Fraction (Overall Liquid)	0	
Volumetric Fraction (Overall Liquid)	0	
Molar Fraction (Liquid 1)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	1	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Molar Fraction (Liquid 2)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	1	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	

Property	Value	Unit
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Molar Fraction (Aqueous)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Molar Fraction (Solid)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	1	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	

CO2-801 (Material Stream)

Property	Value	Unit
Temperature	40	C
Pressure	34.5	bar
Mass Flow	162279	kg/h
Molar Flow	11188.9	kmol/h
Volumetric Flow	6834.04	m3/h
Mixture Molar Fraction		
Methane	0.00831565	
Ethane	1.25739E-17	
Propane	2.85769E-19	
N-butane	0	
Isobutane	0	
Nitrogen	0.109333	
Carbon dioxide	0.150614	
Water	0.191201	
Oxygen	1.61709E-08	
Argon	0.00129686	
Hydrogen sulfide	0	
Hydrogen	0.535562	
Carbonyl sulfide	0	
Carbon monoxide	0.00367731	
Ammonia	0	
Density (Mixture)	23.7458	kg/m3
Molecular Weight (Mixture)	14.5036	kg/kmol
Specific Enthalpy (Mixture)	-563.814	kJ/kg
Specific Entropy (Mixture)	-2.13837	kJ/[kg.K]
Molar Enthalpy (Mixture)	-8177.36	kJ/kmol
Molar Entropy (Mixture)	-31.0142	kJ/[kmol.K]
Thermal Conductivity (Mixture)	0.225455	W/[m.K]
Molar Fraction (Vapor)		
Methane	0.0102638	
Ethane	1.55197E-17	
Propane	3.5272E-19	
N-butane	0	
Isobutane	0	
Nitrogen	0.134948	
Carbon dioxide	0.185256	
Water	0.002363	
Oxygen	1.99588E-08	
Argon	0.00160064	
Hydrogen sulfide	0	
Hydrogen	0.66103	

Property	Value	Unit
Carbonyl sulfide	0	
Carbon monoxide	0.00453882	
Ammonia	0	
Density (Vapor)	18.2466	kg/m3
Molecular Weight (Vapor)	13.6642	kg/kmol
Specific Enthalpy (Vapor)	21.223	kJ/kg
Specific Entropy (Vapor)	-0.761679	kJ/[kg.K]
Molar Enthalpy (Vapor)	289.996	kJ/kmol
Molar Entropy (Vapor)	-10.4077	kJ/[kmol.K]
Thermal Conductivity (Vapor)	0.131569	W/[m.K]
Kinematic Viscosity (Vapor)	6.63634E-07	m2/s
Dynamic Viscosity (Vapor)	1.2109E-05	Pa.s
Heat Capacity (Vapor)	2.30892	kJ/[kg.K]
Heat Capacity Ratio (Vapor)	1.42823	
Mass Flow (Vapor)	123866	kg/h
Molar Flow (Vapor)	9064.98	kmol/h
Volumetric Flow (Vapor)	6788.45	m3/h
Compressibility Factor (Vapor)	0.992341	
Molar Fraction (Vapor)	0.810177	
Mass Fraction (Vapor)	0.763284	
Volumetric Fraction (Vapor)	0.99333	
Molar Fraction (Overall Liquid)		
Methane	3.52154E-09	
Ethane	1.42597E-25	
Propane	1.84765E-29	
N-butane	0	
Isobutane	0	
Nitrogen	9.23225E-07	
Carbon dioxide	0.00274902	
Water	0.997237	
Oxygen	2.43441E-12	
Argon	1.87299E-07	
Hydrogen sulfide	0	
Hydrogen	1.24042E-05	
Carbonyl sulfide	0	
Carbon monoxide	3.23423E-08	
Ammonia	0	
Density (Overall Liquid)	842.219	kg/m3
Molecular Weight (Overall Liquid)	18.0866	kg/kmol
Specific Enthalpy (Overall Liquid)	-2450.26	kJ/kg
Specific Entropy (Overall Liquid)	-6.5775	kJ/[kg.K]
Molar Enthalpy (Overall Liquid)	-44316.7	kJ/kmol

Property	Value	Unit
Molar Entropy (Overall Liquid)	-118.964	kJ/[kmol.K]
Thermal Conductivity (Overall Liquid)	0.626167	W/[m.K]
Kinematic Viscosity (Overall Liquid)	7.79012E-07	m ² /s
Dynamic Viscosity (Overall Liquid)	0.000656099	Pa.s
Heat Capacity (Overall Liquid)	3.90144	kJ/[kg.K]
Heat Capacity Ratio (Overall Liquid)	1.30862	
Mass Flow (Overall Liquid)	38414.2	kg/h
Molar Flow (Overall Liquid)	2123.91	kmol/h
Volumetric Flow (Overall Liquid)	45.6107	m ³ /h
Compressibility Factor (Overall Liquid)	0.0284569	
Molar Fraction (Overall Liquid)	0.189823	
Mass Fraction (Overall Liquid)	0.236716	
Volumetric Fraction (Overall Liquid)	0.00667405	
Molar Fraction (Liquid 1)		
Methane	3.52154E-09	
Ethane	1.42597E-25	
Propane	1.84765E-29	
N-butane	0	
Isobutane	0	
Nitrogen	9.23225E-07	
Carbon dioxide	0.00274902	
Water	0.997237	
Oxygen	2.43441E-12	
Argon	1.87299E-07	
Hydrogen sulfide	0	
Hydrogen	1.24042E-05	
Carbonyl sulfide	0	
Carbon monoxide	3.23423E-08	
Ammonia	0	
Molar Fraction (Liquid 2)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	

Property	Value	Unit
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Molar Fraction (Aqueous)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Molar Fraction (Solid)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	

25 (Material Stream)

Property	Value	Unit
Temperature	235.019	C
Pressure	34.5	bar
Mass Flow	0	kg/h
Molar Flow	0	kmol/h
Volumetric Flow	0	m3/h
Mixture Molar Fraction		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Density (Mixture)	0	kg/m3
Molecular Weight (Mixture)	0	kg/kmol
Specific Enthalpy (Mixture)	NaN	kJ/kg
Specific Entropy (Mixture)	0	kJ/[kg.K]
Molar Enthalpy (Mixture)	0	kJ/kmol
Molar Entropy (Mixture)	0	kJ/[kmol.K]
Thermal Conductivity (Mixture)	0	W/[m.K]
Molar Fraction (Vapor)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	

Property	Value	Unit
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Density (Vapor)	0	kg/m3
Molecular Weight (Vapor)	0	kg/kmol
Specific Enthalpy (Vapor)	0	kJ/kg
Specific Entropy (Vapor)	0	kJ/[kg.K]
Molar Enthalpy (Vapor)	0	kJ/kmol
Molar Entropy (Vapor)	0	kJ/[kmol.K]
Thermal Conductivity (Vapor)	0	W/[m.K]
Kinematic Viscosity (Vapor)	0	m2/s
Dynamic Viscosity (Vapor)	0	Pa.s
Heat Capacity (Vapor)	0	kJ/[kg.K]
Heat Capacity Ratio (Vapor)	NaN	
Mass Flow (Vapor)	0	kg/h
Molar Flow (Vapor)	0	kmol/h
Volumetric Flow (Vapor)	0	m3/h
Compressibility Factor (Vapor)	0	
Molar Fraction (Vapor)	0	
Mass Fraction (Vapor)	0	
Volumetric Fraction (Vapor)	NaN	
Molar Fraction (Overall Liquid)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Density (Overall Liquid)	0	kg/m3
Molecular Weight (Overall Liquid)	0	kg/kmol
Specific Enthalpy (Overall Liquid)	0	kJ/kg
Specific Entropy (Overall Liquid)	0	kJ/[kg.K]
Molar Enthalpy (Overall Liquid)	0	kJ/kmol

Property	Value	Unit
Molar Entropy (Overall Liquid)	0	kJ/[kmol.K]
Thermal Conductivity (Overall Liquid)	0	W/[m.K]
Kinematic Viscosity (Overall Liquid)	0	m2/s
Dynamic Viscosity (Overall Liquid)	0	Pa.s
Heat Capacity (Overall Liquid)	0	kJ/[kg.K]
Heat Capacity Ratio (Overall Liquid)	NaN	
Mass Flow (Overall Liquid)	0	kg/h
Molar Flow (Overall Liquid)	0	kmol/h
Volumetric Flow (Overall Liquid)	0	m3/h
Compressibility Factor (Overall Liquid)	0	
Molar Fraction (Overall Liquid)	0	
Mass Fraction (Overall Liquid)	0	
Volumetric Fraction (Overall Liquid)	NaN	
Molar Fraction (Liquid 1)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Molar Fraction (Liquid 2)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	

Property	Value	Unit
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Molar Fraction (Aqueous)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Molar Fraction (Solid)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	

LTS-OUT-701 (Material Stream)

Property	Value	Unit
Temperature	235.019	C
Pressure	34.5	bar
Mass Flow	162279	kg/h
Molar Flow	11188.9	kmol/h
Volumetric Flow	13613.2	m3/h
Mixture Molar Fraction		
Methane	0.00831565	
Ethane	1.25739E-17	
Propane	2.85769E-19	
N-butane	0	
Isobutane	0	
Nitrogen	0.109333	
Carbon dioxide	0.150614	
Water	0.191201	
Oxygen	1.61709E-08	
Argon	0.00129686	
Hydrogen sulfide	0	
Hydrogen	0.535562	
Carbonyl sulfide	0	
Carbon monoxide	0.00367731	
Ammonia	0	
Density (Mixture)	11.9207	kg/m3
Molecular Weight (Mixture)	14.5036	kg/kmol
Specific Enthalpy (Mixture)	450.304	kJ/kg
Specific Entropy (Mixture)	0.809586	kJ/[kg.K]
Molar Enthalpy (Mixture)	6531.05	kJ/kmol
Molar Entropy (Mixture)	11.7419	kJ/[kmol.K]
Thermal Conductivity (Mixture)	0.163192	W/[m.K]
Molar Fraction (Vapor)		
Methane	0.00831565	
Ethane	1.25739E-17	
Propane	2.85769E-19	
N-butane	0	
Isobutane	0	
Nitrogen	0.109333	
Carbon dioxide	0.150614	
Water	0.191201	
Oxygen	1.61709E-08	
Argon	0.00129686	
Hydrogen sulfide	0	
Hydrogen	0.535562	

Property	Value	Unit
Carbonyl sulfide	0	
Carbon monoxide	0.00367731	
Ammonia	0	
Density (Vapor)	11.9207	kg/m3
Molecular Weight (Vapor)	14.5036	kg/kmol
Specific Enthalpy (Vapor)	450.304	kJ/kg
Specific Entropy (Vapor)	0.809586	kJ/[kg.K]
Molar Enthalpy (Vapor)	6531.05	kJ/kmol
Molar Entropy (Vapor)	11.7419	kJ/[kmol.K]
Thermal Conductivity (Vapor)	0.163192	W/[m.K]
Kinematic Viscosity (Vapor)	1.46258E-06	m2/s
Dynamic Viscosity (Vapor)	1.7435E-05	Pa.s
Heat Capacity (Vapor)	2.32791	kJ/[kg.K]
Heat Capacity Ratio (Vapor)	1.36798	
Mass Flow (Vapor)	162279	kg/h
Molar Flow (Vapor)	11188.9	kmol/h
Volumetric Flow (Vapor)	13613.2	m3/h
Compressibility Factor (Vapor)	0.993515	
Molar Fraction (Vapor)	1	
Mass Fraction (Vapor)	1	
Volumetric Fraction (Vapor)	1	
Molar Fraction (Overall Liquid)		
Methane	NaN	
Ethane	NaN	
Propane	NaN	
N-butane	NaN	
Isobutane	NaN	
Nitrogen	NaN	
Carbon dioxide	NaN	
Water	NaN	
Oxygen	NaN	
Argon	NaN	
Hydrogen sulfide	NaN	
Hydrogen	NaN	
Carbonyl sulfide	NaN	
Carbon monoxide	NaN	
Ammonia	NaN	
Density (Overall Liquid)	0	kg/m3
Molecular Weight (Overall Liquid)	NaN	kg/kmol
Specific Enthalpy (Overall Liquid)	0	kJ/kg
Specific Entropy (Overall Liquid)	0	kJ/[kg.K]
Molar Enthalpy (Overall Liquid)	NaN	kJ/kmol

Property	Value	Unit
Molar Entropy (Overall Liquid)	NaN	kJ/[kmol.K]
Thermal Conductivity (Overall Liquid)	0	W/[m.K]
Kinematic Viscosity (Overall Liquid)	0	m2/s
Dynamic Viscosity (Overall Liquid)	0	Pa.s
Heat Capacity (Overall Liquid)	0	kJ/[kg.K]
Heat Capacity Ratio (Overall Liquid)	NaN	
Mass Flow (Overall Liquid)	0	kg/h
Molar Flow (Overall Liquid)	0	kmol/h
Volumetric Flow (Overall Liquid)	0	m3/h
Compressibility Factor (Overall Liquid)	0	
Molar Fraction (Overall Liquid)	0	
Mass Fraction (Overall Liquid)	0	
Volumetric Fraction (Overall Liquid)	0	
Molar Fraction (Liquid 1)		
Methane	2.04102E-05	
Ethane	1.28095E-20	
Propane	7.19988E-23	
N-butane	0	
Isobutane	0	
Nitrogen	0.000560785	
Carbon dioxide	0.0133824	
Water	0.981028	
Oxygen	2.42675E-10	
Argon	2.02606E-05	
Hydrogen sulfide	0	
Hydrogen	0.0049686	
Carbonyl sulfide	0	
Carbon monoxide	1.93591E-05	
Ammonia	0	
Molar Fraction (Liquid 2)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	

Property	Value	Unit
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Molar Fraction (Aqueous)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Molar Fraction (Solid)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	

LTS-701 (Material Stream)

Property	Value	Unit
Temperature	220	C
Pressure	35	bar
Mass Flow	162279	kg/h
Molar Flow	11188.9	kmol/h
Volumetric Flow	14292.9	m3/h
Mixture Molar Fraction		
Methane	0.00831565	
Ethane	1.25739E-17	
Propane	2.85769E-19	
N-butane	0	
Isobutane	0	
Nitrogen	0.109333	
Carbon dioxide	0.137611	
Water	0.204204	
Oxygen	1.61709E-08	
Argon	0.00129686	
Hydrogen sulfide	0	
Hydrogen	0.522559	
Carbonyl sulfide	0	
Carbon monoxide	0.0166799	
Ammonia	0	
Density (Mixture)	11.3539	kg/m3
Molecular Weight (Mixture)	14.5036	kg/kmol
Specific Enthalpy (Mixture)	413.399	kJ/kg
Specific Entropy (Mixture)	0.762619	kJ/[kg.K]
Molar Enthalpy (Mixture)	5995.78	kJ/kmol
Molar Entropy (Mixture)	11.0607	kJ/[kmol.K]
Thermal Conductivity (Mixture)	0.156367	W/[m.K]
Molar Fraction (Vapor)		
Methane	0.00831565	
Ethane	1.25739E-17	
Propane	2.85769E-19	
N-butane	0	
Isobutane	0	
Nitrogen	0.109333	
Carbon dioxide	0.137611	
Water	0.204204	
Oxygen	1.61709E-08	
Argon	0.00129686	
Hydrogen sulfide	0	
Hydrogen	0.522559	

Property	Value	Unit
Carbonyl sulfide	0	
Carbon monoxide	0.0166799	
Ammonia	0	
Density (Vapor)	11.3539	kg/m3
Molecular Weight (Vapor)	14.5036	kg/kmol
Specific Enthalpy (Vapor)	413.399	kJ/kg
Specific Entropy (Vapor)	0.762619	kJ/[kg.K]
Molar Enthalpy (Vapor)	5995.78	kJ/kmol
Molar Entropy (Vapor)	11.0607	kJ/[kmol.K]
Thermal Conductivity (Vapor)	0.156367	W/[m.K]
Kinematic Viscosity (Vapor)	1.50688E-06	m2/s
Dynamic Viscosity (Vapor)	1.71089E-05	Pa.s
Heat Capacity (Vapor)	2.31803	kJ/[kg.K]
Heat Capacity Ratio (Vapor)	1.37597	
Mass Flow (Vapor)	162279	kg/h
Molar Flow (Vapor)	11188.9	kmol/h
Volumetric Flow (Vapor)	14292.9	m3/h
Compressibility Factor (Vapor)	1.09046	
Molar Fraction (Vapor)	1	
Mass Fraction (Vapor)	1	
Volumetric Fraction (Vapor)	1	
Molar Fraction (Overall Liquid)		
Methane	NaN	
Ethane	NaN	
Propane	NaN	
N-butane	NaN	
Isobutane	NaN	
Nitrogen	NaN	
Carbon dioxide	NaN	
Water	NaN	
Oxygen	NaN	
Argon	NaN	
Hydrogen sulfide	NaN	
Hydrogen	NaN	
Carbonyl sulfide	NaN	
Carbon monoxide	NaN	
Ammonia	NaN	
Density (Overall Liquid)	0	kg/m3
Molecular Weight (Overall Liquid)	NaN	kg/kmol
Specific Enthalpy (Overall Liquid)	0	kJ/kg
Specific Entropy (Overall Liquid)	0	kJ/[kg.K]
Molar Enthalpy (Overall Liquid)	NaN	kJ/kmol

Property	Value	Unit
Molar Entropy (Overall Liquid)	NaN	kJ/[kmol.K]
Thermal Conductivity (Overall Liquid)	0	W/[m.K]
Kinematic Viscosity (Overall Liquid)	0	m ² /s
Dynamic Viscosity (Overall Liquid)	0	Pa.s
Heat Capacity (Overall Liquid)	0	kJ/[kg.K]
Heat Capacity Ratio (Overall Liquid)	NaN	
Mass Flow (Overall Liquid)	0	kg/h
Molar Flow (Overall Liquid)	0	kmol/h
Volumetric Flow (Overall Liquid)	0	m ³ /h
Compressibility Factor (Overall Liquid)	0	
Molar Fraction (Overall Liquid)	0	
Mass Fraction (Overall Liquid)	0	
Volumetric Fraction (Overall Liquid)	0	
Molar Fraction (Liquid 1)		
Methane	9.17743E-06	
Ethane	4.87962E-21	
Propane	2.19077E-23	
N-butane	0	
Isobutane	0	
Nitrogen	0.000282471	
Carbon dioxide	0.00796169	
Water	0.989103	
Oxygen	1.36638E-10	
Argon	1.13914E-05	
Hydrogen sulfide	0	
Hydrogen	0.00258831	
Carbonyl sulfide	0	
Carbon monoxide	4.42244E-05	
Ammonia	0	
Molar Fraction (Liquid 2)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	

Property	Value	Unit
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Molar Fraction (Aqueous)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Molar Fraction (Solid)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	

22 (Material Stream)

Property	Value	Unit
Temperature	350	C
Pressure	35	bar
Mass Flow	0	kg/h
Molar Flow	0	kmol/h
Volumetric Flow	0	m3/h
Mixture Molar Fraction		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Density (Mixture)	0	kg/m3
Molecular Weight (Mixture)	0	kg/kmol
Specific Enthalpy (Mixture)	NaN	kJ/kg
Specific Entropy (Mixture)	0	kJ/[kg.K]
Molar Enthalpy (Mixture)	0	kJ/kmol
Molar Entropy (Mixture)	0	kJ/[kmol.K]
Thermal Conductivity (Mixture)	0	W/[m.K]
Molar Fraction (Vapor)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	

Property	Value	Unit
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Density (Vapor)	0	kg/m3
Molecular Weight (Vapor)	0	kg/kmol
Specific Enthalpy (Vapor)	0	kJ/kg
Specific Entropy (Vapor)	0	kJ/[kg.K]
Molar Enthalpy (Vapor)	0	kJ/kmol
Molar Entropy (Vapor)	0	kJ/[kmol.K]
Thermal Conductivity (Vapor)	0	W/[m.K]
Kinematic Viscosity (Vapor)	0	m2/s
Dynamic Viscosity (Vapor)	0	Pa.s
Heat Capacity (Vapor)	0	kJ/[kg.K]
Heat Capacity Ratio (Vapor)	NaN	
Mass Flow (Vapor)	0	kg/h
Molar Flow (Vapor)	0	kmol/h
Volumetric Flow (Vapor)	0	m3/h
Compressibility Factor (Vapor)	0	
Molar Fraction (Vapor)	0	
Mass Fraction (Vapor)	0	
Volumetric Fraction (Vapor)	NaN	
Molar Fraction (Overall Liquid)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Density (Overall Liquid)	0	kg/m3
Molecular Weight (Overall Liquid)	0	kg/kmol
Specific Enthalpy (Overall Liquid)	0	kJ/kg
Specific Entropy (Overall Liquid)	0	kJ/[kg.K]
Molar Enthalpy (Overall Liquid)	0	kJ/kmol

Property	Value	Unit
Molar Entropy (Overall Liquid)	0	kJ/[kmol.K]
Thermal Conductivity (Overall Liquid)	0	W/[m.K]
Kinematic Viscosity (Overall Liquid)	0	m2/s
Dynamic Viscosity (Overall Liquid)	0	Pa.s
Heat Capacity (Overall Liquid)	0	kJ/[kg.K]
Heat Capacity Ratio (Overall Liquid)	NaN	
Mass Flow (Overall Liquid)	0	kg/h
Molar Flow (Overall Liquid)	0	kmol/h
Volumetric Flow (Overall Liquid)	0	m3/h
Compressibility Factor (Overall Liquid)	0	
Molar Fraction (Overall Liquid)	0	
Mass Fraction (Overall Liquid)	0	
Volumetric Fraction (Overall Liquid)	NaN	
Molar Fraction (Liquid 1)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Molar Fraction (Liquid 2)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	

Property	Value	Unit
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Molar Fraction (Aqueous)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Molar Fraction (Solid)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	

HTS-OUT-601 (Material Stream)

Property	Value	Unit
Temperature	350	C
Pressure	35	bar
Mass Flow	162279	kg/h
Molar Flow	11188.9	kmol/h
Volumetric Flow	16563.6	m3/h
Mixture Molar Fraction		
Methane	0.00831565	
Ethane	1.25739E-17	
Propane	2.85769E-19	
N-butane	0	
Isobutane	0	
Nitrogen	0.109333	
Carbon dioxide	0.137611	
Water	0.204204	
Oxygen	1.61709E-08	
Argon	0.00129686	
Hydrogen sulfide	0	
Hydrogen	0.522559	
Carbonyl sulfide	0	
Carbon monoxide	0.0166799	
Ammonia	0	
Density (Mixture)	9.79736	kg/m3
Molecular Weight (Mixture)	14.5036	kg/kmol
Specific Enthalpy (Mixture)	717.774	kJ/kg
Specific Entropy (Mixture)	1.31023	kJ/[kg.K]
Molar Enthalpy (Mixture)	10410.3	kJ/kmol
Molar Entropy (Mixture)	19.003	kJ/[kmol.K]
Thermal Conductivity (Mixture)	0.188581	W/[m.K]
Molar Fraction (Vapor)		
Methane	0.00831565	
Ethane	1.25739E-17	
Propane	2.85769E-19	
N-butane	0	
Isobutane	0	
Nitrogen	0.109333	
Carbon dioxide	0.137611	
Water	0.204204	
Oxygen	1.61709E-08	
Argon	0.00129686	
Hydrogen sulfide	0	
Hydrogen	0.522559	

Property	Value	Unit
Carbonyl sulfide	0	
Carbon monoxide	0.0166799	
Ammonia	0	
Density (Vapor)	9.79736	kg/m3
Molecular Weight (Vapor)	14.5036	kg/kmol
Specific Enthalpy (Vapor)	717.774	kJ/kg
Specific Entropy (Vapor)	1.31023	kJ/[kg.K]
Molar Enthalpy (Vapor)	10410.3	kJ/kmol
Molar Entropy (Vapor)	19.003	kJ/[kmol.K]
Thermal Conductivity (Vapor)	0.188581	W/[m.K]
Kinematic Viscosity (Vapor)	2.09811E-06	m2/s
Dynamic Viscosity (Vapor)	2.0556E-05	Pa.s
Heat Capacity (Vapor)	2.36006	kJ/[kg.K]
Heat Capacity Ratio (Vapor)	1.3442	
Mass Flow (Vapor)	162279	kg/h
Molar Flow (Vapor)	11188.9	kmol/h
Volumetric Flow (Vapor)	16563.6	m3/h
Compressibility Factor (Vapor)	1.00008	
Molar Fraction (Vapor)	1	
Mass Fraction (Vapor)	1	
Volumetric Fraction (Vapor)	1	
Molar Fraction (Overall Liquid)		
Methane	NaN	
Ethane	NaN	
Propane	NaN	
N-butane	NaN	
Isobutane	NaN	
Nitrogen	NaN	
Carbon dioxide	NaN	
Water	NaN	
Oxygen	NaN	
Argon	NaN	
Hydrogen sulfide	NaN	
Hydrogen	NaN	
Carbonyl sulfide	NaN	
Carbon monoxide	NaN	
Ammonia	NaN	
Density (Overall Liquid)	0	kg/m3
Molecular Weight (Overall Liquid)	NaN	kg/kmol
Specific Enthalpy (Overall Liquid)	0	kJ/kg
Specific Entropy (Overall Liquid)	0	kJ/[kg.K]
Molar Enthalpy (Overall Liquid)	NaN	kJ/kmol

Property	Value	Unit
Molar Entropy (Overall Liquid)	NaN	kJ/[kmol.K]
Thermal Conductivity (Overall Liquid)	0	W/[m.K]
Kinematic Viscosity (Overall Liquid)	0	m2/s
Dynamic Viscosity (Overall Liquid)	0	Pa.s
Heat Capacity (Overall Liquid)	0	kJ/[kg.K]
Heat Capacity Ratio (Overall Liquid)	NaN	
Mass Flow (Overall Liquid)	0	kg/h
Molar Flow (Overall Liquid)	0	kmol/h
Volumetric Flow (Overall Liquid)	0	m3/h
Compressibility Factor (Overall Liquid)	0	
Molar Fraction (Overall Liquid)	0	
Mass Fraction (Overall Liquid)	0	
Volumetric Fraction (Overall Liquid)	0	
Molar Fraction (Liquid 1)		
Methane	9.7656E-07	
Ethane	4.71508E-19	
Propane	2.36177E-19	
N-butane	0	
Isobutane	0	
Nitrogen	7.39261E-13	
Carbon dioxide	0.00148814	
Water	0.998511	
Oxygen	4.49514E-15	
Argon	6.18725E-12	
Hydrogen sulfide	0	
Hydrogen	3.00118E-55	
Carbonyl sulfide	0	
Carbon monoxide	4.79861E-11	
Ammonia	0	
Molar Fraction (Liquid 2)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	

Property	Value	Unit
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Molar Fraction (Aqueous)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Molar Fraction (Solid)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	

HTS- 601 (Material Stream)

Property	Value	Unit
Temperature	350	C
Pressure	35.5	bar
Mass Flow	162279	kg/h
Molar Flow	11188.9	kmol/h
Volumetric Flow	16283.3	m3/h
Mixture Molar Fraction		
Methane	0.00831565	
Ethane	1.25739E-17	
Propane	2.85769E-19	
N-butane	0	
Isobutane	0	
Nitrogen	0.109333	
Carbon dioxide	0.0478962	
Water	0.293919	
Oxygen	1.61709E-08	
Argon	0.00129686	
Hydrogen sulfide	0	
Hydrogen	0.432844	
Carbonyl sulfide	0	
Carbon monoxide	0.106395	
Ammonia	0	
Density (Mixture)	9.96603	kg/m3
Molecular Weight (Mixture)	14.5036	kg/kmol
Specific Enthalpy (Mixture)	699.053	kJ/kg
Specific Entropy (Mixture)	1.40433	kJ/[kg.K]
Molar Enthalpy (Mixture)	10138.8	kJ/kmol
Molar Entropy (Mixture)	20.3679	kJ/[kmol.K]
Thermal Conductivity (Mixture)	0.165304	W/[m.K]
Molar Fraction (Vapor)		
Methane	0.00831565	
Ethane	1.25739E-17	
Propane	2.85769E-19	
N-butane	0	
Isobutane	0	
Nitrogen	0.109333	
Carbon dioxide	0.0478962	
Water	0.293919	
Oxygen	1.61709E-08	
Argon	0.00129686	
Hydrogen sulfide	0	
Hydrogen	0.432844	

Property	Value	Unit
Carbonyl sulfide	0	
Carbon monoxide	0.106395	
Ammonia	0	
Density (Vapor)	9.96603	kg/m3
Molecular Weight (Vapor)	14.5036	kg/kmol
Specific Enthalpy (Vapor)	699.053	kJ/kg
Specific Entropy (Vapor)	1.40433	kJ/[kg.K]
Molar Enthalpy (Vapor)	10138.8	kJ/kmol
Molar Entropy (Vapor)	20.3679	kJ/[kmol.K]
Thermal Conductivity (Vapor)	0.165304	W/[m.K]
Kinematic Viscosity (Vapor)	2.14808E-06	m2/s
Dynamic Viscosity (Vapor)	2.14078E-05	Pa.s
Heat Capacity (Vapor)	2.30303	kJ/[kg.K]
Heat Capacity Ratio (Vapor)	1.36138	
Mass Flow (Vapor)	162279	kg/h
Molar Flow (Vapor)	11188.9	kmol/h
Volumetric Flow (Vapor)	16283.3	m3/h
Compressibility Factor (Vapor)	0.997195	
Molar Fraction (Vapor)	1	
Mass Fraction (Vapor)	1	
Volumetric Fraction (Vapor)	1	
Molar Fraction (Overall Liquid)		
Methane	NaN	
Ethane	NaN	
Propane	NaN	
N-butane	NaN	
Isobutane	NaN	
Nitrogen	NaN	
Carbon dioxide	NaN	
Water	NaN	
Oxygen	NaN	
Argon	NaN	
Hydrogen sulfide	NaN	
Hydrogen	NaN	
Carbonyl sulfide	NaN	
Carbon monoxide	NaN	
Ammonia	NaN	
Density (Overall Liquid)	0	kg/m3
Molecular Weight (Overall Liquid)	NaN	kg/kmol
Specific Enthalpy (Overall Liquid)	0	kJ/kg
Specific Entropy (Overall Liquid)	0	kJ/[kg.K]
Molar Enthalpy (Overall Liquid)	NaN	kJ/kmol

Property	Value	Unit
Molar Entropy (Overall Liquid)	NaN	kJ/[kmol.K]
Thermal Conductivity (Overall Liquid)	0	W/[m.K]
Kinematic Viscosity (Overall Liquid)	0	m2/s
Dynamic Viscosity (Overall Liquid)	0	Pa.s
Heat Capacity (Overall Liquid)	0	kJ/[kg.K]
Heat Capacity Ratio (Overall Liquid)	NaN	
Mass Flow (Overall Liquid)	0	kg/h
Molar Flow (Overall Liquid)	0	kmol/h
Volumetric Flow (Overall Liquid)	0	m3/h
Compressibility Factor (Overall Liquid)	0	
Molar Fraction (Overall Liquid)	0	
Mass Fraction (Overall Liquid)	0	
Volumetric Fraction (Overall Liquid)	0	
Molar Fraction (Liquid 1)		
Methane	6.79244E-07	
Ethane	3.27956E-19	
Propane	1.64272E-19	
N-butane	0	
Isobutane	0	
Nitrogen	5.14191E-13	
Carbon dioxide	0.000360262	
Water	0.999639	
Oxygen	3.12658E-15	
Argon	4.30353E-12	
Hydrogen sulfide	0	
Hydrogen	1.72908E-55	
Carbonyl sulfide	0	
Carbon monoxide	2.12897E-10	
Ammonia	0	
Molar Fraction (Liquid 2)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	

Property	Value	Unit
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Molar Fraction (Aqueous)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Molar Fraction (Solid)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	

13 (Material Stream)

Property	Value	Unit
Temperature	850	C
Pressure	36.5	bar
Mass Flow	0	kg/h
Molar Flow	0	kmol/h
Volumetric Flow	0	m3/h
Mixture Molar Fraction		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Density (Mixture)	0	kg/m3
Molecular Weight (Mixture)	0	kg/kmol
Specific Enthalpy (Mixture)	NaN	kJ/kg
Specific Entropy (Mixture)	0	kJ/[kg.K]
Molar Enthalpy (Mixture)	0	kJ/kmol
Molar Entropy (Mixture)	0	kJ/[kmol.K]
Thermal Conductivity (Mixture)	0	W/[m.K]
Molar Fraction (Vapor)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	

Property	Value	Unit
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Density (Vapor)	0	kg/m3
Molecular Weight (Vapor)	0	kg/kmol
Specific Enthalpy (Vapor)	0	kJ/kg
Specific Entropy (Vapor)	0	kJ/[kg.K]
Molar Enthalpy (Vapor)	0	kJ/kmol
Molar Entropy (Vapor)	0	kJ/[kmol.K]
Thermal Conductivity (Vapor)	0	W/[m.K]
Kinematic Viscosity (Vapor)	0	m2/s
Dynamic Viscosity (Vapor)	0	Pa.s
Heat Capacity (Vapor)	0	kJ/[kg.K]
Heat Capacity Ratio (Vapor)	NaN	
Mass Flow (Vapor)	0	kg/h
Molar Flow (Vapor)	0	kmol/h
Volumetric Flow (Vapor)	0	m3/h
Compressibility Factor (Vapor)	0	
Molar Fraction (Vapor)	0	
Mass Fraction (Vapor)	0	
Volumetric Fraction (Vapor)	NaN	
Molar Fraction (Overall Liquid)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Density (Overall Liquid)	0	kg/m3
Molecular Weight (Overall Liquid)	0	kg/kmol
Specific Enthalpy (Overall Liquid)	0	kJ/kg
Specific Entropy (Overall Liquid)	0	kJ/[kg.K]
Molar Enthalpy (Overall Liquid)	0	kJ/kmol

Property	Value	Unit
Molar Entropy (Overall Liquid)	0	kJ/[kmol.K]
Thermal Conductivity (Overall Liquid)	0	W/[m.K]
Kinematic Viscosity (Overall Liquid)	0	m2/s
Dynamic Viscosity (Overall Liquid)	0	Pa.s
Heat Capacity (Overall Liquid)	0	kJ/[kg.K]
Heat Capacity Ratio (Overall Liquid)	NaN	
Mass Flow (Overall Liquid)	0	kg/h
Molar Flow (Overall Liquid)	0	kmol/h
Volumetric Flow (Overall Liquid)	0	m3/h
Compressibility Factor (Overall Liquid)	0	
Molar Fraction (Overall Liquid)	0	
Mass Fraction (Overall Liquid)	0	
Volumetric Fraction (Overall Liquid)	NaN	
Molar Fraction (Liquid 1)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Molar Fraction (Liquid 2)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	

Property	Value	Unit
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Molar Fraction (Aqueous)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Molar Fraction (Solid)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	

RFG-302 (Material Stream)

Property	Value	Unit
Temperature	850	C
Pressure	36.5	bar
Mass Flow	117207	kg/h
Molar Flow	9276.25	kmol/h
Volumetric Flow	23838.4	m3/h
Mixture Molar Fraction		
Methane	0.0469659	
Ethane	1.51664E-17	
Propane	3.44691E-19	
N-butane	0	
Isobutane	0	
Nitrogen	0.001113	
Carbon dioxide	0.0547663	
Water	0.323728	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0.479086	
Carbonyl sulfide	0	
Carbon monoxide	0.0943407	
Ammonia	0	
Density (Mixture)	4.91674	kg/m3
Molecular Weight (Mixture)	12.6352	kg/kmol
Specific Enthalpy (Mixture)	2267.44	kJ/kg
Specific Entropy (Mixture)	2.99949	kJ/[kg.K]
Molar Enthalpy (Mixture)	28649.6	kJ/kmol
Molar Entropy (Mixture)	37.8992	kJ/[kmol.K]
Thermal Conductivity (Mixture)	0.29412	W/[m.K]
Molar Fraction (Vapor)		
Methane	0.0469659	
Ethane	1.51664E-17	
Propane	3.44691E-19	
N-butane	0	
Isobutane	0	
Nitrogen	0.001113	
Carbon dioxide	0.0547663	
Water	0.323728	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0.479086	

Property	Value	Unit
Carbonyl sulfide	0	
Carbon monoxide	0.0943407	
Ammonia	0	
Density (Vapor)	4.91674	kg/m3
Molecular Weight (Vapor)	12.6352	kg/kmol
Specific Enthalpy (Vapor)	2267.44	kJ/kg
Specific Entropy (Vapor)	2.99949	kJ/[kg.K]
Molar Enthalpy (Vapor)	28649.6	kJ/kmol
Molar Entropy (Vapor)	37.8992	kJ/[kmol.K]
Thermal Conductivity (Vapor)	0.29412	W/[m.K]
Kinematic Viscosity (Vapor)	6.64665E-06	m2/s
Dynamic Viscosity (Vapor)	3.26799E-05	Pa.s
Heat Capacity (Vapor)	3.04873	kJ/[kg.K]
Heat Capacity Ratio (Vapor)	1.27953	
Mass Flow (Vapor)	117207	kg/h
Molar Flow (Vapor)	9276.25	kmol/h
Volumetric Flow (Vapor)	23838.4	m3/h
Compressibility Factor (Vapor)	1.0045	
Molar Fraction (Vapor)	1	
Mass Fraction (Vapor)	1	
Volumetric Fraction (Vapor)	1	
Molar Fraction (Overall Liquid)		
Methane	NaN	
Ethane	NaN	
Propane	NaN	
N-butane	NaN	
Isobutane	NaN	
Nitrogen	NaN	
Carbon dioxide	NaN	
Water	NaN	
Oxygen	NaN	
Argon	NaN	
Hydrogen sulfide	NaN	
Hydrogen	NaN	
Carbonyl sulfide	NaN	
Carbon monoxide	NaN	
Ammonia	NaN	
Density (Overall Liquid)	0	kg/m3
Molecular Weight (Overall Liquid)	NaN	kg/kmol
Specific Enthalpy (Overall Liquid)	0	kJ/kg
Specific Entropy (Overall Liquid)	0	kJ/[kg.K]
Molar Enthalpy (Overall Liquid)	NaN	kJ/kmol

Property	Value	Unit
Molar Entropy (Overall Liquid)	NaN	kJ/[kmol.K]
Thermal Conductivity (Overall Liquid)	0	W/[m.K]
Kinematic Viscosity (Overall Liquid)	0	m2/s
Dynamic Viscosity (Overall Liquid)	0	Pa.s
Heat Capacity (Overall Liquid)	0	kJ/[kg.K]
Heat Capacity Ratio (Overall Liquid)	NaN	
Mass Flow (Overall Liquid)	0	kg/h
Molar Flow (Overall Liquid)	0	kmol/h
Volumetric Flow (Overall Liquid)	0	m3/h
Compressibility Factor (Overall Liquid)	0	
Molar Fraction (Overall Liquid)	0	
Mass Fraction (Overall Liquid)	0	
Volumetric Fraction (Overall Liquid)	0	
Molar Fraction (Liquid 1)		
Methane	7.37241E-16	
Ethane	2.554E-24	
Propane	1.51E-20	
N-butane	0	
Isobutane	0	
Nitrogen	1.72063E-42	
Carbon dioxide	1.56813E-11	
Water	1	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	1.91763E-174	
Carbonyl sulfide	0	
Carbon monoxide	1.80145E-31	
Ammonia	0	
Molar Fraction (Liquid 2)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	

Property	Value	Unit
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Molar Fraction (Aqueous)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Molar Fraction (Solid)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	

MIX-302 (Material Stream)

Property	Value	Unit
Temperature	850	C
Pressure	37.5	bar
Mass Flow	117207	kg/h
Molar Flow	6561.56	kmol/h
Volumetric Flow	16348.9	m3/h
Mixture Molar Fraction		
Methane	0.243888	
Ethane	0.00917859	
Propane	0.00262245	
N-butane	0.000786736	
Isobutane	0	
Nitrogen	0.00157347	
Carbon dioxide	0.00393368	
Water	0.738017	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Density (Mixture)	7.16912	kg/m3
Molecular Weight (Mixture)	17.8627	kg/kmol
Specific Enthalpy (Mixture)	1999.26	kJ/kg
Specific Entropy (Mixture)	1.69569	kJ/[kg.K]
Molar Enthalpy (Mixture)	35712.2	kJ/kmol
Molar Entropy (Mixture)	30.2897	kJ/[kmol.K]
Thermal Conductivity (Mixture)	0.137607	W/[m.K]
Molar Fraction (Vapor)		
Methane	0.243888	
Ethane	0.00917859	
Propane	0.00262245	
N-butane	0.000786736	
Isobutane	0	
Nitrogen	0.00157347	
Carbon dioxide	0.00393368	
Water	0.738017	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	

Property	Value	Unit
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Density (Vapor)	7.16912	kg/m3
Molecular Weight (Vapor)	17.8627	kg/kmol
Specific Enthalpy (Vapor)	1999.26	kJ/kg
Specific Entropy (Vapor)	1.69569	kJ/[kg.K]
Molar Enthalpy (Vapor)	35712.2	kJ/kmol
Molar Entropy (Vapor)	30.2897	kJ/[kmol.K]
Thermal Conductivity (Vapor)	0.137607	W/[m.K]
Kinematic Viscosity (Vapor)	5.58495E-06	m2/s
Dynamic Viscosity (Vapor)	4.00392E-05	Pa.s
Heat Capacity (Vapor)	2.96918	kJ/[kg.K]
Heat Capacity Ratio (Vapor)	1.19341	
Mass Flow (Vapor)	117207	kg/h
Molar Flow (Vapor)	6561.56	kmol/h
Volumetric Flow (Vapor)	16348.9	m3/h
Compressibility Factor (Vapor)	1.00061	
Molar Fraction (Vapor)	1	
Mass Fraction (Vapor)	1	
Volumetric Fraction (Vapor)	1	
Molar Fraction (Overall Liquid)		
Methane	NaN	
Ethane	NaN	
Propane	NaN	
N-butane	NaN	
Isobutane	NaN	
Nitrogen	NaN	
Carbon dioxide	NaN	
Water	NaN	
Oxygen	NaN	
Argon	NaN	
Hydrogen sulfide	NaN	
Hydrogen	NaN	
Carbonyl sulfide	NaN	
Carbon monoxide	NaN	
Ammonia	NaN	
Density (Overall Liquid)	0	kg/m3
Molecular Weight (Overall Liquid)	NaN	kg/kmol
Specific Enthalpy (Overall Liquid)	0	kJ/kg
Specific Entropy (Overall Liquid)	0	kJ/[kg.K]
Molar Enthalpy (Overall Liquid)	NaN	kJ/kmol

Property	Value	Unit
Molar Entropy (Overall Liquid)	NaN	kJ/[kmol.K]
Thermal Conductivity (Overall Liquid)	0	W/[m.K]
Kinematic Viscosity (Overall Liquid)	0	m2/s
Dynamic Viscosity (Overall Liquid)	0	Pa.s
Heat Capacity (Overall Liquid)	0	kJ/[kg.K]
Heat Capacity Ratio (Overall Liquid)	NaN	
Mass Flow (Overall Liquid)	0	kg/h
Molar Flow (Overall Liquid)	0	kmol/h
Volumetric Flow (Overall Liquid)	0	m3/h
Compressibility Factor (Overall Liquid)	0	
Molar Fraction (Overall Liquid)	0	
Mass Fraction (Overall Liquid)	0	
Volumetric Fraction (Overall Liquid)	0	
Molar Fraction (Liquid 1)		
Methane	1.67919E-15	
Ethane	6.77949E-10	
Propane	5.03892E-05	
N-butane	2.12916E-05	
Isobutane	0	
Nitrogen	1.06693E-42	
Carbon dioxide	4.94028E-13	
Water	0.999928	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Molar Fraction (Liquid 2)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	

Property	Value	Unit
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Molar Fraction (Aqueous)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Molar Fraction (Solid)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	

9 (Material Stream)

Property	Value	Unit
Temperature	378.218	C
Pressure	39.5	bar
Mass Flow	117207	kg/h
Molar Flow	6561.56	kmol/h
Volumetric Flow	8679.17	m3/h
Mixture Molar Fraction		
Methane	0.243888	
Ethane	0.00917859	
Propane	0.00262245	
N-butane	0.000786736	
Isobutane	0	
Nitrogen	0.00157347	
Carbon dioxide	0.00393368	
Water	0.738017	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Density (Mixture)	13.5044	kg/m3
Molecular Weight (Mixture)	17.8627	kg/kmol
Specific Enthalpy (Mixture)	711.583	kJ/kg
Specific Entropy (Mixture)	0.196997	kJ/[kg.K]
Molar Enthalpy (Mixture)	12710.8	kJ/kmol
Molar Entropy (Mixture)	3.5189	kJ/[kmol.K]
Thermal Conductivity (Mixture)	0.067405	W/[m.K]
Molar Fraction (Vapor)		
Methane	0.243888	
Ethane	0.00917859	
Propane	0.00262245	
N-butane	0.000786736	
Isobutane	0	
Nitrogen	0.00157347	
Carbon dioxide	0.00393368	
Water	0.738017	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	

Property	Value	Unit
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Density (Vapor)	13.5044	kg/m ³
Molecular Weight (Vapor)	17.8627	kg/kmol
Specific Enthalpy (Vapor)	711.583	kJ/kg
Specific Entropy (Vapor)	0.196997	kJ/[kg.K]
Molar Enthalpy (Vapor)	12710.8	kJ/kmol
Molar Entropy (Vapor)	3.5189	kJ/[kmol.K]
Thermal Conductivity (Vapor)	0.067405	W/[m.K]
Kinematic Viscosity (Vapor)	1.73867E-06	m ² /s
Dynamic Viscosity (Vapor)	2.34798E-05	Pa.s
Heat Capacity (Vapor)	2.47431	kJ/[kg.K]
Heat Capacity Ratio (Vapor)	1.29811	
Mass Flow (Vapor)	117207	kg/h
Molar Flow (Vapor)	6561.56	kmol/h
Volumetric Flow (Vapor)	8679.17	m ³ /h
Compressibility Factor (Vapor)	0.964788	
Molar Fraction (Vapor)	1	
Mass Fraction (Vapor)	1	
Volumetric Fraction (Vapor)	1	
Molar Fraction (Overall Liquid)		
Methane	NaN	
Ethane	NaN	
Propane	NaN	
N-butane	NaN	
Isobutane	NaN	
Nitrogen	NaN	
Carbon dioxide	NaN	
Water	NaN	
Oxygen	NaN	
Argon	NaN	
Hydrogen sulfide	NaN	
Hydrogen	NaN	
Carbonyl sulfide	NaN	
Carbon monoxide	NaN	
Ammonia	NaN	
Density (Overall Liquid)	0	kg/m ³
Molecular Weight (Overall Liquid)	NaN	kg/kmol
Specific Enthalpy (Overall Liquid)	0	kJ/kg
Specific Entropy (Overall Liquid)	0	kJ/[kg.K]
Molar Enthalpy (Overall Liquid)	NaN	kJ/kmol

Property	Value	Unit
Molar Entropy (Overall Liquid)	NaN	kJ/[kmol.K]
Thermal Conductivity (Overall Liquid)	0	W/[m.K]
Kinematic Viscosity (Overall Liquid)	0	m2/s
Dynamic Viscosity (Overall Liquid)	0	Pa.s
Heat Capacity (Overall Liquid)	0	kJ/[kg.K]
Heat Capacity Ratio (Overall Liquid)	NaN	
Mass Flow (Overall Liquid)	0	kg/h
Molar Flow (Overall Liquid)	0	kmol/h
Volumetric Flow (Overall Liquid)	0	m3/h
Compressibility Factor (Overall Liquid)	0	
Molar Fraction (Overall Liquid)	0	
Mass Fraction (Overall Liquid)	0	
Volumetric Fraction (Overall Liquid)	0	
Molar Fraction (Liquid 1)		
Methane	3.7227E-06	
Ethane	6.90535E-05	
Propane	0.000608297	
N-butane	0.000367585	
Isobutane	0	
Nitrogen	2.64168E-16	
Carbon dioxide	7.30302E-06	
Water	0.998944	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Molar Fraction (Liquid 2)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	

Property	Value	Unit
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Molar Fraction (Aqueous)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Molar Fraction (Solid)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	

NG-101 (Material Stream)

Property	Value	Unit
Temperature	30	C
Pressure	40	bar
Mass Flow	30000	kg/h
Molar Flow	1720.74	kmol/h
Volumetric Flow	995.341	m3/h
Mixture Molar Fraction		
Methane	0.93	
Ethane	0.035	
Propane	0.01	
N-butane	0.003	
Isobutane	0	
Nitrogen	0.006	
Carbon dioxide	0.015	
Water	0.00096	
Oxygen	0	
Argon	0	
Hydrogen sulfide	3E-05	
Hydrogen	0	
Carbonyl sulfide	1E-05	
Carbon monoxide	0	
Ammonia	0	
Density (Mixture)	30.1404	kg/m3
Molecular Weight (Mixture)	17.4344	kg/kmol
Specific Enthalpy (Mixture)	-34.8748	kJ/kg
Specific Entropy (Mixture)	-1.7209	kJ/[kg.K]
Molar Enthalpy (Mixture)	-608.02	kJ/kmol
Molar Entropy (Mixture)	-30.0028	kJ/[kmol.K]
Thermal Conductivity (Mixture)	0.0377705	W/[m.K]
Molar Fraction (Vapor)		
Methane	0.93	
Ethane	0.035	
Propane	0.01	
N-butane	0.003	
Isobutane	0	
Nitrogen	0.006	
Carbon dioxide	0.015	
Water	0.00096	
Oxygen	0	
Argon	0	
Hydrogen sulfide	3E-05	
Hydrogen	0	

Property	Value	Unit
Carbonyl sulfide	1E-05	
Carbon monoxide	0	
Ammonia	0	
Density (Vapor)	30.1404	kg/m3
Molecular Weight (Vapor)	17.4344	kg/kmol
Specific Enthalpy (Vapor)	-34.8748	kJ/kg
Specific Entropy (Vapor)	-1.7209	kJ/[kg.K]
Molar Enthalpy (Vapor)	-608.02	kJ/kmol
Molar Entropy (Vapor)	-30.0028	kJ/[kmol.K]
Thermal Conductivity (Vapor)	0.0377705	W/[m.K]
Kinematic Viscosity (Vapor)	4.01383E-07	m2/s
Dynamic Viscosity (Vapor)	1.20978E-05	Pa.s
Heat Capacity (Vapor)	2.40027	kJ/[kg.K]
Heat Capacity Ratio (Vapor)	1.46263	
Mass Flow (Vapor)	30000	kg/h
Molar Flow (Vapor)	1720.74	kmol/h
Volumetric Flow (Vapor)	995.341	m3/h
Compressibility Factor (Vapor)	0.918014	
Molar Fraction (Vapor)	1	
Mass Fraction (Vapor)	1	
Volumetric Fraction (Vapor)	1	
Molar Fraction (Overall Liquid)		
Methane	NaN	
Ethane	NaN	
Propane	NaN	
N-butane	NaN	
Isobutane	NaN	
Nitrogen	NaN	
Carbon dioxide	NaN	
Water	NaN	
Oxygen	NaN	
Argon	NaN	
Hydrogen sulfide	NaN	
Hydrogen	NaN	
Carbonyl sulfide	NaN	
Carbon monoxide	NaN	
Ammonia	NaN	
Density (Overall Liquid)	0	kg/m3
Molecular Weight (Overall Liquid)	NaN	kg/kmol
Specific Enthalpy (Overall Liquid)	0	kJ/kg
Specific Entropy (Overall Liquid)	0	kJ/[kg.K]
Molar Enthalpy (Overall Liquid)	NaN	kJ/kmol

Property	Value	Unit
Molar Entropy (Overall Liquid)	NaN	kJ/[kmol.K]
Thermal Conductivity (Overall Liquid)	0	W/[m.K]
Kinematic Viscosity (Overall Liquid)	0	m2/s
Dynamic Viscosity (Overall Liquid)	0	Pa.s
Heat Capacity (Overall Liquid)	0	kJ/[kg.K]
Heat Capacity Ratio (Overall Liquid)	NaN	
Mass Flow (Overall Liquid)	0	kg/h
Molar Flow (Overall Liquid)	0	kmol/h
Volumetric Flow (Overall Liquid)	0	m3/h
Compressibility Factor (Overall Liquid)	0	
Molar Fraction (Overall Liquid)	0	
Mass Fraction (Overall Liquid)	0	
Volumetric Fraction (Overall Liquid)	0	
Molar Fraction (Liquid 1)		
Methane	0.924592	
Ethane	0.0377889	
Propane	0.0116093	
N-butane	0.00373925	
Isobutane	0	
Nitrogen	0.00572833	
Carbon dioxide	0.0154912	
Water	0.00100796	
Oxygen	0	
Argon	0	
Hydrogen sulfide	3.20792E-05	
Hydrogen	0	
Carbonyl sulfide	1.13434E-05	
Carbon monoxide	0	
Ammonia	0	
Molar Fraction (Liquid 2)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	

Property	Value	Unit
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Molar Fraction (Aqueous)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Molar Fraction (Solid)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	

STM - 101 (Material Stream)

Property	Value	Unit
Temperature	400	C
Pressure	40	bar
Mass Flow	87210	kg/h
Molar Flow	4840.89	kmol/h
Volumetric Flow	6353.21	m3/h
Mixture Molar Fraction		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	1	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Density (Mixture)	13.7269	kg/m3
Molecular Weight (Mixture)	18.0153	kg/kmol
Specific Enthalpy (Mixture)	664.997	kJ/kg
Specific Entropy (Mixture)	-0.189286	kJ/[kg.K]
Molar Enthalpy (Mixture)	11980.1	kJ/kmol
Molar Entropy (Mixture)	-3.41003	kJ/[kmol.K]
Thermal Conductivity (Mixture)	0.0612816	W/[m.K]
Molar Fraction (Vapor)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	1	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	

Property	Value	Unit
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Density (Vapor)	13.7269	kg/m3
Molecular Weight (Vapor)	18.0153	kg/kmol
Specific Enthalpy (Vapor)	664.997	kJ/kg
Specific Entropy (Vapor)	-0.189286	kJ/[kg.K]
Molar Enthalpy (Vapor)	11980.1	kJ/kmol
Molar Entropy (Vapor)	-3.41003	kJ/[kmol.K]
Thermal Conductivity (Vapor)	0.0612816	W/[m.K]
Kinematic Viscosity (Vapor)	1.83171E-06	m2/s
Dynamic Viscosity (Vapor)	2.51437E-05	Pa.s
Heat Capacity (Vapor)	2.22469	kJ/[kg.K]
Heat Capacity Ratio (Vapor)	1.38607	
Mass Flow (Vapor)	87210	kg/h
Molar Flow (Vapor)	4840.89	kmol/h
Volumetric Flow (Vapor)	6353.21	m3/h
Compressibility Factor (Vapor)	0.938007	
Molar Fraction (Vapor)	1	
Mass Fraction (Vapor)	1	
Volumetric Fraction (Vapor)	1	
Molar Fraction (Overall Liquid)		
Methane	NaN	
Ethane	NaN	
Propane	NaN	
N-butane	NaN	
Isobutane	NaN	
Nitrogen	NaN	
Carbon dioxide	NaN	
Water	NaN	
Oxygen	NaN	
Argon	NaN	
Hydrogen sulfide	NaN	
Hydrogen	NaN	
Carbonyl sulfide	NaN	
Carbon monoxide	NaN	
Ammonia	NaN	
Density (Overall Liquid)	0	kg/m3
Molecular Weight (Overall Liquid)	NaN	kg/kmol
Specific Enthalpy (Overall Liquid)	0	kJ/kg
Specific Entropy (Overall Liquid)	0	kJ/[kg.K]
Molar Enthalpy (Overall Liquid)	NaN	kJ/kmol

Property	Value	Unit
Molar Entropy (Overall Liquid)	NaN	kJ/[kmol.K]
Thermal Conductivity (Overall Liquid)	0	W/[m.K]
Kinematic Viscosity (Overall Liquid)	0	m2/s
Dynamic Viscosity (Overall Liquid)	0	Pa.s
Heat Capacity (Overall Liquid)	0	kJ/[kg.K]
Heat Capacity Ratio (Overall Liquid)	NaN	
Mass Flow (Overall Liquid)	0	kg/h
Molar Flow (Overall Liquid)	0	kmol/h
Volumetric Flow (Overall Liquid)	0	m3/h
Compressibility Factor (Overall Liquid)	0	
Molar Fraction (Overall Liquid)	0	
Mass Fraction (Overall Liquid)	0	
Volumetric Fraction (Overall Liquid)	0	
Molar Fraction (Liquid 1)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	1	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Molar Fraction (Liquid 2)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	

Property	Value	Unit
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Molar Fraction (Aqueous)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Molar Fraction (Solid)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	1	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	

AIR - 101 (Material Stream)

Property	Value	Unit
Temperature	25	C
Pressure	40	bar
Mass Flow	45000	kg/h
Molar Flow	1553.58	kmol/h
Volumetric Flow	951.78	m3/h
Mixture Molar Fraction		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0.78084	
Carbon dioxide	0.00036	
Water	0	
Oxygen	0.20946	
Argon	0.00934	
Hydrogen sulfide	0	
Hydrogen	0	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Density (Mixture)	47.2798	kg/m3
Molecular Weight (Mixture)	28.9654	kg/kmol
Specific Enthalpy (Mixture)	-10.6083	kJ/kg
Specific Entropy (Mixture)	-0.932892	kJ/[kg.K]
Molar Enthalpy (Mixture)	-307.274	kJ/kmol
Molar Entropy (Mixture)	-27.0216	kJ/[kmol.K]
Thermal Conductivity (Mixture)	0.02763	W/[m.K]
Molar Fraction (Vapor)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0.78084	
Carbon dioxide	0.00036	
Water	0	
Oxygen	0.20946	
Argon	0.00934	
Hydrogen sulfide	0	
Hydrogen	0	

Property	Value	Unit
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Density (Vapor)	47.2798	kg/m3
Molecular Weight (Vapor)	28.9654	kg/kmol
Specific Enthalpy (Vapor)	-10.6083	kJ/kg
Specific Entropy (Vapor)	-0.932892	kJ/[kg.K]
Molar Enthalpy (Vapor)	-307.274	kJ/kmol
Molar Entropy (Vapor)	-27.0216	kJ/[kmol.K]
Thermal Conductivity (Vapor)	0.02763	W/[m.K]
Kinematic Viscosity (Vapor)	4.07054E-07	m2/s
Dynamic Viscosity (Vapor)	1.92454E-05	Pa.s
Heat Capacity (Vapor)	1.07342	kJ/[kg.K]
Heat Capacity Ratio (Vapor)	1.48622	
Mass Flow (Vapor)	45000	kg/h
Molar Flow (Vapor)	1553.58	kmol/h
Volumetric Flow (Vapor)	951.78	m3/h
Compressibility Factor (Vapor)	0.988596	
Molar Fraction (Vapor)	1	
Mass Fraction (Vapor)	1	
Volumetric Fraction (Vapor)	1	
Molar Fraction (Overall Liquid)		
Methane	NaN	
Ethane	NaN	
Propane	NaN	
N-butane	NaN	
Isobutane	NaN	
Nitrogen	NaN	
Carbon dioxide	NaN	
Water	NaN	
Oxygen	NaN	
Argon	NaN	
Hydrogen sulfide	NaN	
Hydrogen	NaN	
Carbonyl sulfide	NaN	
Carbon monoxide	NaN	
Ammonia	NaN	
Density (Overall Liquid)	0	kg/m3
Molecular Weight (Overall Liquid)	NaN	kg/kmol
Specific Enthalpy (Overall Liquid)	0	kJ/kg
Specific Entropy (Overall Liquid)	0	kJ/[kg.K]
Molar Enthalpy (Overall Liquid)	NaN	kJ/kmol

Property	Value	Unit
Molar Entropy (Overall Liquid)	NaN	kJ/[kmol.K]
Thermal Conductivity (Overall Liquid)	0	W/[m.K]
Kinematic Viscosity (Overall Liquid)	0	m2/s
Dynamic Viscosity (Overall Liquid)	0	Pa.s
Heat Capacity (Overall Liquid)	0	kJ/[kg.K]
Heat Capacity Ratio (Overall Liquid)	NaN	
Mass Flow (Overall Liquid)	0	kg/h
Molar Flow (Overall Liquid)	0	kmol/h
Volumetric Flow (Overall Liquid)	0	m3/h
Compressibility Factor (Overall Liquid)	0	
Molar Fraction (Overall Liquid)	0	
Mass Fraction (Overall Liquid)	0	
Volumetric Fraction (Overall Liquid)	0	
Molar Fraction (Liquid 1)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0.746287	
Carbon dioxide	0.00450015	
Water	0	
Oxygen	0.238324	
Argon	0.0108892	
Hydrogen sulfide	0	
Hydrogen	0	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Molar Fraction (Liquid 2)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	

Property	Value	Unit
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Molar Fraction (Aqueous)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Molar Fraction (Solid)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	

NG-102 (Material Stream)

Property	Value	Unit
Temperature	350	C
Pressure	39.5	bar
Mass Flow	30000	kg/h
Molar Flow	1720.74	kmol/h
Volumetric Flow	2272.17	m3/h
Mixture Molar Fraction		
Methane	0.93	
Ethane	0.035	
Propane	0.01	
N-butane	0.003	
Isobutane	0	
Nitrogen	0.006	
Carbon dioxide	0.015	
Water	0.00096	
Oxygen	0	
Argon	0	
Hydrogen sulfide	3E-05	
Hydrogen	0	
Carbonyl sulfide	1E-05	
Carbon monoxide	0	
Ammonia	0	
Density (Mixture)	13.2032	kg/m3
Molecular Weight (Mixture)	17.4344	kg/kmol
Specific Enthalpy (Mixture)	846.972	kJ/kg
Specific Entropy (Mixture)	0.232771	kJ/[kg.K]
Molar Enthalpy (Mixture)	14766.4	kJ/kmol
Molar Entropy (Mixture)	4.05822	kJ/[kmol.K]
Thermal Conductivity (Mixture)	0.0885978	W/[m.K]
Molar Fraction (Vapor)		
Methane	0.93	
Ethane	0.035	
Propane	0.01	
N-butane	0.003	
Isobutane	0	
Nitrogen	0.006	
Carbon dioxide	0.015	
Water	0.00096	
Oxygen	0	
Argon	0	
Hydrogen sulfide	3E-05	
Hydrogen	0	

Property	Value	Unit
Carbonyl sulfide	1E-05	
Carbon monoxide	0	
Ammonia	0	
Density (Vapor)	13.2032	kg/m3
Molecular Weight (Vapor)	17.4344	kg/kmol
Specific Enthalpy (Vapor)	846.972	kJ/kg
Specific Entropy (Vapor)	0.232771	kJ/[kg.K]
Molar Enthalpy (Vapor)	14766.4	kJ/kmol
Molar Entropy (Vapor)	4.05822	kJ/[kmol.K]
Thermal Conductivity (Vapor)	0.0885978	W/[m.K]
Kinematic Viscosity (Vapor)	1.54736E-06	m2/s
Dynamic Viscosity (Vapor)	2.04301E-05	Pa.s
Heat Capacity (Vapor)	3.24699	kJ/[kg.K]
Heat Capacity Ratio (Vapor)	1.18648	
Mass Flow (Vapor)	30000	kg/h
Molar Flow (Vapor)	1720.74	kmol/h
Volumetric Flow (Vapor)	2272.17	m3/h
Compressibility Factor (Vapor)	1.00675	
Molar Fraction (Vapor)	1	
Mass Fraction (Vapor)	1	
Volumetric Fraction (Vapor)	1	
Molar Fraction (Overall Liquid)		
Methane	NaN	
Ethane	NaN	
Propane	NaN	
N-butane	NaN	
Isobutane	NaN	
Nitrogen	NaN	
Carbon dioxide	NaN	
Water	NaN	
Oxygen	NaN	
Argon	NaN	
Hydrogen sulfide	NaN	
Hydrogen	NaN	
Carbonyl sulfide	NaN	
Carbon monoxide	NaN	
Ammonia	NaN	
Density (Overall Liquid)	0	kg/m3
Molecular Weight (Overall Liquid)	NaN	kg/kmol
Specific Enthalpy (Overall Liquid)	0	kJ/kg
Specific Entropy (Overall Liquid)	0	kJ/[kg.K]
Molar Enthalpy (Overall Liquid)	NaN	kJ/kmol

Property	Value	Unit
Molar Entropy (Overall Liquid)	NaN	kJ/[kmol.K]
Thermal Conductivity (Overall Liquid)	0	W/[m.K]
Kinematic Viscosity (Overall Liquid)	0	m2/s
Dynamic Viscosity (Overall Liquid)	0	Pa.s
Heat Capacity (Overall Liquid)	0	kJ/[kg.K]
Heat Capacity Ratio (Overall Liquid)	NaN	
Mass Flow (Overall Liquid)	0	kg/h
Molar Flow (Overall Liquid)	0	kmol/h
Volumetric Flow (Overall Liquid)	0	m3/h
Compressibility Factor (Overall Liquid)	0	
Molar Fraction (Overall Liquid)	0	
Mass Fraction (Overall Liquid)	0	
Volumetric Fraction (Overall Liquid)	0	
Molar Fraction (Liquid 1)		
Methane	0.00555934	
Ethane	0.0668076	
Propane	0.420686	
N-butane	0.258832	
Isobutane	0	
Nitrogen	2.06506E-12	
Carbon dioxide	0.00825693	
Water	0.238945	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0.000804911	
Hydrogen	0	
Carbonyl sulfide	0.000108052	
Carbon monoxide	0	
Ammonia	0	
Molar Fraction (Liquid 2)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	

Property	Value	Unit
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Molar Fraction (Aqueous)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Molar Fraction (Solid)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	

SULF-OUT-101 (Material Stream)

Property	Value	Unit
Temperature	350	C
Pressure	39.5	bar
Mass Flow	2.79306	kg/h
Molar Flow	0.0688296	kmol/h
Volumetric Flow	0.0875704	m3/h
Mixture Molar Fraction		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0.75	
Hydrogen	0	
Carbonyl sulfide	0.25	
Carbon monoxide	0	
Ammonia	0	
Density (Mixture)	31.8951	kg/m3
Molecular Weight (Mixture)	40.5794	kg/kmol
Specific Enthalpy (Mixture)	297.953	kJ/kg
Specific Entropy (Mixture)	0.0369037	kJ/[kg.K]
Molar Enthalpy (Mixture)	12090.8	kJ/kmol
Molar Entropy (Mixture)	1.49753	kJ/[kmol.K]
Thermal Conductivity (Mixture)	0.0355567	W/[m.K]
Molar Fraction (Vapor)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0.75	
Hydrogen	0	

Property	Value	Unit
Carbonyl sulfide	0.25	
Carbon monoxide	0	
Ammonia	0	
Density (Vapor)	31.8951	kg/m3
Molecular Weight (Vapor)	40.5794	kg/kmol
Specific Enthalpy (Vapor)	297.953	kJ/kg
Specific Entropy (Vapor)	0.0369037	kJ/[kg.K]
Molar Enthalpy (Vapor)	12090.8	kJ/kmol
Molar Entropy (Vapor)	1.49753	kJ/[kmol.K]
Thermal Conductivity (Vapor)	0.0355567	W/[m.K]
Kinematic Viscosity (Vapor)	8.6801E-07	m2/s
Dynamic Viscosity (Vapor)	2.76853E-05	Pa.s
Heat Capacity (Vapor)	1.09845	kJ/[kg.K]
Heat Capacity Ratio (Vapor)	1.29789	
Mass Flow (Vapor)	2.79306	kg/h
Molar Flow (Vapor)	0.0688296	kmol/h
Volumetric Flow (Vapor)	0.0875704	m3/h
Compressibility Factor (Vapor)	0.970011	
Molar Fraction (Vapor)	1	
Mass Fraction (Vapor)	1	
Volumetric Fraction (Vapor)	1	
Molar Fraction (Overall Liquid)		
Methane	NaN	
Ethane	NaN	
Propane	NaN	
N-butane	NaN	
Isobutane	NaN	
Nitrogen	NaN	
Carbon dioxide	NaN	
Water	NaN	
Oxygen	NaN	
Argon	NaN	
Hydrogen sulfide	NaN	
Hydrogen	NaN	
Carbonyl sulfide	NaN	
Carbon monoxide	NaN	
Ammonia	NaN	
Density (Overall Liquid)	0	kg/m3
Molecular Weight (Overall Liquid)	NaN	kg/kmol
Specific Enthalpy (Overall Liquid)	0	kJ/kg
Specific Entropy (Overall Liquid)	0	kJ/[kg.K]
Molar Enthalpy (Overall Liquid)	NaN	kJ/kmol

Property	Value	Unit
Molar Entropy (Overall Liquid)	NaN	kJ/[kmol.K]
Thermal Conductivity (Overall Liquid)	0	W/[m.K]
Kinematic Viscosity (Overall Liquid)	0	m2/s
Dynamic Viscosity (Overall Liquid)	0	Pa.s
Heat Capacity (Overall Liquid)	0	kJ/[kg.K]
Heat Capacity Ratio (Overall Liquid)	NaN	
Mass Flow (Overall Liquid)	0	kg/h
Molar Flow (Overall Liquid)	0	kmol/h
Volumetric Flow (Overall Liquid)	0	m3/h
Compressibility Factor (Overall Liquid)	0	
Molar Fraction (Overall Liquid)	0	
Mass Fraction (Overall Liquid)	0	
Volumetric Fraction (Overall Liquid)	0	
Molar Fraction (Liquid 1)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0.881647	
Hydrogen	0	
Carbonyl sulfide	0.118353	
Carbon monoxide	0	
Ammonia	0	
Molar Fraction (Liquid 2)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	

Property	Value	Unit
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Molar Fraction (Aqueous)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Molar Fraction (Solid)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	

NG-103 (Material Stream)

Property	Value	Unit
Temperature	350	C
Pressure	39.5	bar
Mass Flow	29997.2	kg/h
Molar Flow	1720.67	kmol/h
Volumetric Flow	2272.08	m3/h
Mixture Molar Fraction		
Methane	0.930037	
Ethane	0.0350014	
Propane	0.0100004	
N-butane	0.00300012	
Isobutane	0	
Nitrogen	0.00600024	
Carbon dioxide	0.0150006	
Water	0.000960038	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Density (Mixture)	13.2025	kg/m3
Molecular Weight (Mixture)	17.4334	kg/kmol
Specific Enthalpy (Mixture)	847.023	kJ/kg
Specific Entropy (Mixture)	0.232681	kJ/[kg.K]
Molar Enthalpy (Mixture)	14766.5	kJ/kmol
Molar Entropy (Mixture)	4.05643	kJ/[kmol.K]
Thermal Conductivity (Mixture)	0.0885999	W/[m.K]
Molar Fraction (Vapor)		
Methane	0.930037	
Ethane	0.0350014	
Propane	0.0100004	
N-butane	0.00300012	
Isobutane	0	
Nitrogen	0.00600024	
Carbon dioxide	0.0150006	
Water	0.000960038	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	

Property	Value	Unit
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Density (Vapor)	13.2025	kg/m3
Molecular Weight (Vapor)	17.4334	kg/kmol
Specific Enthalpy (Vapor)	847.023	kJ/kg
Specific Entropy (Vapor)	0.232681	kJ/[kg.K]
Molar Enthalpy (Vapor)	14766.5	kJ/kmol
Molar Entropy (Vapor)	4.05643	kJ/[kmol.K]
Thermal Conductivity (Vapor)	0.0885999	W/[m.K]
Kinematic Viscosity (Vapor)	1.54742E-06	m2/s
Dynamic Viscosity (Vapor)	2.04298E-05	Pa.s
Heat Capacity (Vapor)	3.24719	kJ/[kg.K]
Heat Capacity Ratio (Vapor)	1.18648	
Mass Flow (Vapor)	29997.2	kg/h
Molar Flow (Vapor)	1720.67	kmol/h
Volumetric Flow (Vapor)	2272.08	m3/h
Compressibility Factor (Vapor)	1.00675	
Molar Fraction (Vapor)	1	
Mass Fraction (Vapor)	1	
Volumetric Fraction (Vapor)	1	
Molar Fraction (Overall Liquid)		
Methane	NaN	
Ethane	NaN	
Propane	NaN	
N-butane	NaN	
Isobutane	NaN	
Nitrogen	NaN	
Carbon dioxide	NaN	
Water	NaN	
Oxygen	NaN	
Argon	NaN	
Hydrogen sulfide	NaN	
Hydrogen	NaN	
Carbonyl sulfide	NaN	
Carbon monoxide	NaN	
Ammonia	NaN	
Density (Overall Liquid)	0	kg/m3
Molecular Weight (Overall Liquid)	NaN	kg/kmol
Specific Enthalpy (Overall Liquid)	0	kJ/kg
Specific Entropy (Overall Liquid)	0	kJ/[kg.K]
Molar Enthalpy (Overall Liquid)	NaN	kJ/kmol

Property	Value	Unit
Molar Entropy (Overall Liquid)	NaN	kJ/[kmol.K]
Thermal Conductivity (Overall Liquid)	0	W/[m.K]
Kinematic Viscosity (Overall Liquid)	0	m2/s
Dynamic Viscosity (Overall Liquid)	0	Pa.s
Heat Capacity (Overall Liquid)	0	kJ/[kg.K]
Heat Capacity Ratio (Overall Liquid)	NaN	
Mass Flow (Overall Liquid)	0	kg/h
Molar Flow (Overall Liquid)	0	kmol/h
Volumetric Flow (Overall Liquid)	0	m3/h
Compressibility Factor (Overall Liquid)	0	
Molar Fraction (Overall Liquid)	0	
Mass Fraction (Overall Liquid)	0	
Volumetric Fraction (Overall Liquid)	0	
Molar Fraction (Liquid 1)		
Methane	0.00556442	
Ethane	0.0668686	
Propane	0.421071	
N-butane	0.259069	
Isobutane	0	
Nitrogen	2.06695E-12	
Carbon dioxide	0.00826448	
Water	0.239163	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Molar Fraction (Liquid 2)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	

Property	Value	Unit
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Molar Fraction (Aqueous)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Molar Fraction (Solid)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	

RFG-301 (Material Stream)

Property	Value	Unit
Temperature	850	C
Pressure	37	bar
Mass Flow	117207	kg/h
Molar Flow	8890.8	kmol/h
Volumetric Flow	22533.9	m3/h
Mixture Molar Fraction		
Methane	0.0490021	
Ethane	0.00677395	
Propane	0.00193542	
N-butane	0.000580625	
Isobutane	0	
Nitrogen	0.00116125	
Carbon dioxide	0.0571406	
Water	0.35944	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0.447212	
Carbonyl sulfide	0	
Carbon monoxide	0.0767541	
Ammonia	0	
Density (Mixture)	5.20137	kg/m3
Molecular Weight (Mixture)	13.183	kg/kmol
Specific Enthalpy (Mixture)	2242.04	kJ/kg
Specific Entropy (Mixture)	3.07884	kJ/[kg.K]
Molar Enthalpy (Mixture)	29556.7	kJ/kmol
Molar Entropy (Mixture)	40.5882	kJ/[kmol.K]
Thermal Conductivity (Mixture)	0.283639	W/[m.K]
Molar Fraction (Vapor)		
Methane	0.0490021	
Ethane	0.00677395	
Propane	0.00193542	
N-butane	0.000580625	
Isobutane	0	
Nitrogen	0.00116125	
Carbon dioxide	0.0571406	
Water	0.35944	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0.447212	

Property	Value	Unit
Carbonyl sulfide	0	
Carbon monoxide	0.0767541	
Ammonia	0	
Density (Vapor)	5.20137	kg/m3
Molecular Weight (Vapor)	13.183	kg/kmol
Specific Enthalpy (Vapor)	2242.04	kJ/kg
Specific Entropy (Vapor)	3.07884	kJ/[kg.K]
Molar Enthalpy (Vapor)	29556.7	kJ/kmol
Molar Entropy (Vapor)	40.5882	kJ/[kmol.K]
Thermal Conductivity (Vapor)	0.283639	W/[m.K]
Kinematic Viscosity (Vapor)	6.38053E-06	m2/s
Dynamic Viscosity (Vapor)	3.31875E-05	Pa.s
Heat Capacity (Vapor)	3.04754	kJ/[kg.K]
Heat Capacity Ratio (Vapor)	1.26559	
Mass Flow (Vapor)	117207	kg/h
Molar Flow (Vapor)	8890.8	kmol/h
Volumetric Flow (Vapor)	22533.9	m3/h
Compressibility Factor (Vapor)	1.00427	
Molar Fraction (Vapor)	1	
Mass Fraction (Vapor)	1	
Volumetric Fraction (Vapor)	1	
Molar Fraction (Overall Liquid)		
Methane	NaN	
Ethane	NaN	
Propane	NaN	
N-butane	NaN	
Isobutane	NaN	
Nitrogen	NaN	
Carbon dioxide	NaN	
Water	NaN	
Oxygen	NaN	
Argon	NaN	
Hydrogen sulfide	NaN	
Hydrogen	NaN	
Carbonyl sulfide	NaN	
Carbon monoxide	NaN	
Ammonia	NaN	
Density (Overall Liquid)	0	kg/m3
Molecular Weight (Overall Liquid)	NaN	kg/kmol
Specific Enthalpy (Overall Liquid)	0	kJ/kg
Specific Entropy (Overall Liquid)	0	kJ/[kg.K]
Molar Enthalpy (Overall Liquid)	NaN	kJ/kmol

Property	Value	Unit
Molar Entropy (Overall Liquid)	NaN	kJ/[kmol.K]
Thermal Conductivity (Overall Liquid)	0	W/[m.K]
Kinematic Viscosity (Overall Liquid)	0	m2/s
Dynamic Viscosity (Overall Liquid)	0	Pa.s
Heat Capacity (Overall Liquid)	0	kJ/[kg.K]
Heat Capacity Ratio (Overall Liquid)	NaN	
Mass Flow (Overall Liquid)	0	kg/h
Molar Flow (Overall Liquid)	0	kmol/h
Volumetric Flow (Overall Liquid)	0	m3/h
Compressibility Factor (Overall Liquid)	0	
Molar Fraction (Overall Liquid)	0	
Mass Fraction (Overall Liquid)	0	
Volumetric Fraction (Overall Liquid)	0	
Molar Fraction (Liquid 1)		
Methane	6.92705E-16	
Ethane	1.02728E-09	
Propane	7.63533E-05	
N-butane	3.22626E-05	
Isobutane	0	
Nitrogen	1.61669E-42	
Carbon dioxide	1.4734E-11	
Water	0.999891	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	1.61203E-174	
Carbonyl sulfide	0	
Carbon monoxide	1.31987E-31	
Ammonia	0	
Molar Fraction (Liquid 2)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	

Property	Value	Unit
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Molar Fraction (Aqueous)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Molar Fraction (Solid)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	

Outlet stream 2 (Material Stream)

Property	Value	Unit
Temperature	850	C
Pressure	37	bar
Mass Flow	0	kg/h
Molar Flow	0	kmol/h
Volumetric Flow	0	m3/h
Mixture Molar Fraction		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Density (Mixture)	0	kg/m3
Molecular Weight (Mixture)	0	kg/kmol
Specific Enthalpy (Mixture)	NaN	kJ/kg
Specific Entropy (Mixture)	0	kJ/[kg.K]
Molar Enthalpy (Mixture)	0	kJ/kmol
Molar Entropy (Mixture)	0	kJ/[kmol.K]
Thermal Conductivity (Mixture)	0	W/[m.K]
Molar Fraction (Vapor)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	

Property	Value	Unit
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Density (Vapor)	0	kg/m3
Molecular Weight (Vapor)	0	kg/kmol
Specific Enthalpy (Vapor)	0	kJ/kg
Specific Entropy (Vapor)	0	kJ/[kg.K]
Molar Enthalpy (Vapor)	0	kJ/kmol
Molar Entropy (Vapor)	0	kJ/[kmol.K]
Thermal Conductivity (Vapor)	0	W/[m.K]
Kinematic Viscosity (Vapor)	0	m2/s
Dynamic Viscosity (Vapor)	0	Pa.s
Heat Capacity (Vapor)	0	kJ/[kg.K]
Heat Capacity Ratio (Vapor)	NaN	
Mass Flow (Vapor)	0	kg/h
Molar Flow (Vapor)	0	kmol/h
Volumetric Flow (Vapor)	0	m3/h
Compressibility Factor (Vapor)	0	
Molar Fraction (Vapor)	0	
Mass Fraction (Vapor)	0	
Volumetric Fraction (Vapor)	NaN	
Molar Fraction (Overall Liquid)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Density (Overall Liquid)	0	kg/m3
Molecular Weight (Overall Liquid)	0	kg/kmol
Specific Enthalpy (Overall Liquid)	0	kJ/kg
Specific Entropy (Overall Liquid)	0	kJ/[kg.K]
Molar Enthalpy (Overall Liquid)	0	kJ/kmol

Property	Value	Unit
Molar Entropy (Overall Liquid)	0	kJ/[kmol.K]
Thermal Conductivity (Overall Liquid)	0	W/[m.K]
Kinematic Viscosity (Overall Liquid)	0	m2/s
Dynamic Viscosity (Overall Liquid)	0	Pa.s
Heat Capacity (Overall Liquid)	0	kJ/[kg.K]
Heat Capacity Ratio (Overall Liquid)	NaN	
Mass Flow (Overall Liquid)	0	kg/h
Molar Flow (Overall Liquid)	0	kmol/h
Volumetric Flow (Overall Liquid)	0	m3/h
Compressibility Factor (Overall Liquid)	0	
Molar Fraction (Overall Liquid)	0	
Mass Fraction (Overall Liquid)	0	
Volumetric Fraction (Overall Liquid)	NaN	
Molar Fraction (Liquid 1)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Molar Fraction (Liquid 2)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	

Property	Value	Unit
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Molar Fraction (Aqueous)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Molar Fraction (Solid)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	

16 (Material Stream)

Property	Value	Unit
Temperature	749.819	C
Pressure	36.5	bar
Mass Flow	162207	kg/h
Molar Flow	10829.8	kmol/h
Volumetric Flow	25369.6	m3/h
Mixture Molar Fraction		
Methane	0.0402285	
Ethane	1.29907E-17	
Propane	2.95244E-19	
N-butane	0	
Isobutane	0	
Nitrogen	0.112968	
Carbon dioxide	0.0469615	
Water	0.277288	
Oxygen	0.0300478	
Argon	0.00133986	
Hydrogen sulfide	0	
Hydrogen	0.410359	
Carbonyl sulfide	0	
Carbon monoxide	0.0808072	
Ammonia	0	
Density (Mixture)	6.39376	kg/m3
Molecular Weight (Mixture)	14.9778	kg/kmol
Specific Enthalpy (Mixture)	1635.46	kJ/kg
Specific Entropy (Mixture)	2.66947	kJ/[kg.K]
Molar Enthalpy (Mixture)	24495.6	kJ/kmol
Molar Entropy (Mixture)	39.9828	kJ/[kmol.K]
Thermal Conductivity (Mixture)	0.242736	W/[m.K]
Molar Fraction (Vapor)		
Methane	0.0402285	
Ethane	1.29907E-17	
Propane	2.95244E-19	
N-butane	0	
Isobutane	0	
Nitrogen	0.112968	
Carbon dioxide	0.0469615	
Water	0.277288	
Oxygen	0.0300478	
Argon	0.00133986	
Hydrogen sulfide	0	
Hydrogen	0.410359	

Property	Value	Unit
Carbonyl sulfide	0	
Carbon monoxide	0.0808072	
Ammonia	0	
Density (Vapor)	6.39376	kg/m3
Molecular Weight (Vapor)	14.9778	kg/kmol
Specific Enthalpy (Vapor)	1635.46	kJ/kg
Specific Entropy (Vapor)	2.66947	kJ/[kg.K]
Molar Enthalpy (Vapor)	24495.6	kJ/kmol
Molar Entropy (Vapor)	39.9828	kJ/[kmol.K]
Thermal Conductivity (Vapor)	0.242736	W/[m.K]
Kinematic Viscosity (Vapor)	5.03521E-06	m2/s
Dynamic Viscosity (Vapor)	3.21939E-05	Pa.s
Heat Capacity (Vapor)	2.47366	kJ/[kg.K]
Heat Capacity Ratio (Vapor)	1.29442	
Mass Flow (Vapor)	162207	kg/h
Molar Flow (Vapor)	10829.8	kmol/h
Volumetric Flow (Vapor)	25369.6	m3/h
Compressibility Factor (Vapor)	1.00534	
Molar Fraction (Vapor)	1	
Mass Fraction (Vapor)	1	
Volumetric Fraction (Vapor)	1	
Molar Fraction (Overall Liquid)		
Methane	NaN	
Ethane	NaN	
Propane	NaN	
N-butane	NaN	
Isobutane	NaN	
Nitrogen	NaN	
Carbon dioxide	NaN	
Water	NaN	
Oxygen	NaN	
Argon	NaN	
Hydrogen sulfide	NaN	
Hydrogen	NaN	
Carbonyl sulfide	NaN	
Carbon monoxide	NaN	
Ammonia	NaN	
Density (Overall Liquid)	0	kg/m3
Molecular Weight (Overall Liquid)	NaN	kg/kmol
Specific Enthalpy (Overall Liquid)	0	kJ/kg
Specific Entropy (Overall Liquid)	0	kJ/[kg.K]
Molar Enthalpy (Overall Liquid)	NaN	kJ/kmol

Property	Value	Unit
Molar Entropy (Overall Liquid)	NaN	kJ/[kmol.K]
Thermal Conductivity (Overall Liquid)	0	W/[m.K]
Kinematic Viscosity (Overall Liquid)	0	m2/s
Dynamic Viscosity (Overall Liquid)	0	Pa.s
Heat Capacity (Overall Liquid)	0	kJ/[kg.K]
Heat Capacity Ratio (Overall Liquid)	NaN	
Mass Flow (Overall Liquid)	0	kg/h
Molar Flow (Overall Liquid)	0	kmol/h
Volumetric Flow (Overall Liquid)	0	m3/h
Compressibility Factor (Overall Liquid)	0	
Molar Fraction (Overall Liquid)	0	
Mass Fraction (Overall Liquid)	0	
Volumetric Fraction (Overall Liquid)	0	
Molar Fraction (Liquid 1)		
Methane	2.4704E-13	
Ethane	6.80219E-23	
Propane	3.53926E-20	
N-butane	0	
Isobutane	0	
Nitrogen	1.73905E-33	
Carbon dioxide	1.73994E-09	
Water	1	
Oxygen	1.08495E-20	
Argon	1.38866E-27	
Hydrogen sulfide	0	
Hydrogen	4.58568E-145	
Carbonyl sulfide	0	
Carbon monoxide	3.88858E-26	
Ammonia	0	
Molar Fraction (Liquid 2)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	

Property	Value	Unit
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Molar Fraction (Aqueous)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Molar Fraction (Solid)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	

BFW-501 (Material Stream)

Property	Value	Unit
Temperature	105	C
Pressure	42	bar
Mass Flow	50000	kg/h
Molar Flow	2775.42	kmol/h
Volumetric Flow	62.6819	m3/h
Mixture Molar Fraction		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	1	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Density (Mixture)	797.679	kg/m3
Molecular Weight (Mixture)	18.0153	kg/kmol
Specific Enthalpy (Mixture)	-2171.12	kJ/kg
Specific Entropy (Mixture)	-5.76189	kJ/[kg.K]
Molar Enthalpy (Mixture)	-39113.4	kJ/kmol
Molar Entropy (Mixture)	-103.802	kJ/[kmol.K]
Thermal Conductivity (Mixture)	0.682922	W/[m.K]
Molar Fraction (Vapor)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	1	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	

Property	Value	Unit
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Density (Vapor)	0	kg/m3
Molecular Weight (Vapor)	0	kg/kmol
Specific Enthalpy (Vapor)	0	kJ/kg
Specific Entropy (Vapor)	0	kJ/[kg.K]
Molar Enthalpy (Vapor)	0	kJ/kmol
Molar Entropy (Vapor)	0	kJ/[kmol.K]
Thermal Conductivity (Vapor)	0	W/[m.K]
Kinematic Viscosity (Vapor)	0	m2/s
Dynamic Viscosity (Vapor)	0	Pa.s
Heat Capacity (Vapor)	0	kJ/[kg.K]
Heat Capacity Ratio (Vapor)	NaN	
Mass Flow (Vapor)	0	kg/h
Molar Flow (Vapor)	0	kmol/h
Volumetric Flow (Vapor)	0	m3/h
Compressibility Factor (Vapor)	0	
Molar Fraction (Vapor)	0	
Mass Fraction (Vapor)	0	
Volumetric Fraction (Vapor)	0	
Molar Fraction (Overall Liquid)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	1	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Density (Overall Liquid)	797.679	kg/m3
Molecular Weight (Overall Liquid)	18.0153	kg/kmol
Specific Enthalpy (Overall Liquid)	-2171.12	kJ/kg
Specific Entropy (Overall Liquid)	-5.76189	kJ/[kg.K]
Molar Enthalpy (Overall Liquid)	-39113.4	kJ/kmol

Property	Value	Unit
Molar Entropy (Overall Liquid)	-103.802	kJ/[kmol.K]
Thermal Conductivity (Overall Liquid)	0.682922	W/[m.K]
Kinematic Viscosity (Overall Liquid)	3.34607E-07	m2/s
Dynamic Viscosity (Overall Liquid)	0.000266909	Pa.s
Heat Capacity (Overall Liquid)	4.02407	kJ/[kg.K]
Heat Capacity Ratio (Overall Liquid)	1.41686	
Mass Flow (Overall Liquid)	50000	kg/h
Molar Flow (Overall Liquid)	2775.42	kmol/h
Volumetric Flow (Overall Liquid)	62.6819	m3/h
Compressibility Factor (Overall Liquid)	0.0301709	
Molar Fraction (Overall Liquid)	1	
Mass Fraction (Overall Liquid)	1	
Volumetric Fraction (Overall Liquid)	1	
Molar Fraction (Liquid 1)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	1	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Molar Fraction (Liquid 2)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	

Property	Value	Unit
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Molar Fraction (Aqueous)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Molar Fraction (Solid)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	1	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	

RFG-401 (Material Stream)

Property	Value	Unit
Temperature	961.749	C
Pressure	36.5	bar
Mass Flow	162279	kg/h
Molar Flow	11188.9	kmol/h
Volumetric Flow	31641.2	m3/h
Mixture Molar Fraction		
Methane	0.00831565	
Ethane	1.25739E-17	
Propane	2.85769E-19	
N-butane	0	
Isobutane	0	
Nitrogen	0.109333	
Carbon dioxide	0.0478962	
Water	0.293919	
Oxygen	1.61709E-08	
Argon	0.00129686	
Hydrogen sulfide	0	
Hydrogen	0.432844	
Carbonyl sulfide	0	
Carbon monoxide	0.106395	
Ammonia	0	
Density (Mixture)	5.12874	kg/m3
Molecular Weight (Mixture)	14.5036	kg/kmol
Specific Enthalpy (Mixture)	2186.21	kJ/kg
Specific Entropy (Mixture)	3.04066	kJ/[kg.K]
Molar Enthalpy (Mixture)	31708	kJ/kmol
Molar Entropy (Mixture)	44.1006	kJ/[kmol.K]
Thermal Conductivity (Mixture)	0.290778	W/[m.K]
Molar Fraction (Vapor)		
Methane	0.00831565	
Ethane	1.25739E-17	
Propane	2.85769E-19	
N-butane	0	
Isobutane	0	
Nitrogen	0.109333	
Carbon dioxide	0.0478962	
Water	0.293919	
Oxygen	1.61709E-08	
Argon	0.00129686	
Hydrogen sulfide	0	
Hydrogen	0.432844	

Property	Value	Unit
Carbonyl sulfide	0	
Carbon monoxide	0.106395	
Ammonia	0	
Density (Vapor)	5.12874	kg/m3
Molecular Weight (Vapor)	14.5036	kg/kmol
Specific Enthalpy (Vapor)	2186.21	kJ/kg
Specific Entropy (Vapor)	3.04066	kJ/[kg.K]
Molar Enthalpy (Vapor)	31708	kJ/kmol
Molar Entropy (Vapor)	44.1006	kJ/[kmol.K]
Thermal Conductivity (Vapor)	0.290778	W/[m.K]
Kinematic Viscosity (Vapor)	7.21445E-06	m2/s
Dynamic Viscosity (Vapor)	3.7001E-05	Pa.s
Heat Capacity (Vapor)	2.56872	kJ/[kg.K]
Heat Capacity Ratio (Vapor)	1.28989	
Mass Flow (Vapor)	162279	kg/h
Molar Flow (Vapor)	11188.9	kmol/h
Volumetric Flow (Vapor)	31641.2	m3/h
Compressibility Factor (Vapor)	1.00535	
Molar Fraction (Vapor)	1	
Mass Fraction (Vapor)	1	
Volumetric Fraction (Vapor)	1	
Molar Fraction (Overall Liquid)		
Methane	NaN	
Ethane	NaN	
Propane	NaN	
N-butane	NaN	
Isobutane	NaN	
Nitrogen	NaN	
Carbon dioxide	NaN	
Water	NaN	
Oxygen	NaN	
Argon	NaN	
Hydrogen sulfide	NaN	
Hydrogen	NaN	
Carbonyl sulfide	NaN	
Carbon monoxide	NaN	
Ammonia	NaN	
Density (Overall Liquid)	0	kg/m3
Molecular Weight (Overall Liquid)	NaN	kg/kmol
Specific Enthalpy (Overall Liquid)	0	kJ/kg
Specific Entropy (Overall Liquid)	0	kJ/[kg.K]
Molar Enthalpy (Overall Liquid)	NaN	kJ/kmol

Property	Value	Unit
Molar Entropy (Overall Liquid)	NaN	kJ/[kmol.K]
Thermal Conductivity (Overall Liquid)	0	W/[m.K]
Kinematic Viscosity (Overall Liquid)	0	m2/s
Dynamic Viscosity (Overall Liquid)	0	Pa.s
Heat Capacity (Overall Liquid)	0	kJ/[kg.K]
Heat Capacity Ratio (Overall Liquid)	NaN	
Mass Flow (Overall Liquid)	0	kg/h
Molar Flow (Overall Liquid)	0	kmol/h
Volumetric Flow (Overall Liquid)	0	m3/h
Compressibility Factor (Overall Liquid)	0	
Molar Fraction (Overall Liquid)	0	
Mass Fraction (Overall Liquid)	0	
Volumetric Fraction (Overall Liquid)	0	
Molar Fraction (Liquid 1)		
Methane	1.06587E-19	
Ethane	3.7602E-26	
Propane	4.55852E-21	
N-butane	0	
Isobutane	0	
Nitrogen	5.3647E-49	
Carbon dioxide	3.95165E-14	
Water	1	
Oxygen	4.96775E-36	
Argon	1.79426E-39	
Hydrogen sulfide	0	
Hydrogen	1.83477E-210	
Carbonyl sulfide	0	
Carbon monoxide	5.86504E-38	
Ammonia	0	
Molar Fraction (Liquid 2)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	

Property	Value	Unit
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Molar Fraction (Aqueous)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Molar Fraction (Solid)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	

17 (Material Stream)

Property	Value	Unit
Temperature	961.749	C
Pressure	36.5	bar
Mass Flow	-7.20666E-11	kg/h
Molar Flow	-∞	kmol/h
Volumetric Flow	0	m3/h
Mixture Molar Fraction		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Density (Mixture)	0	kg/m3
Molecular Weight (Mixture)	0	kg/kmol
Specific Enthalpy (Mixture)	2048	kJ/kg
Specific Entropy (Mixture)	0	kJ/[kg.K]
Molar Enthalpy (Mixture)	0	kJ/kmol
Molar Entropy (Mixture)	0	kJ/[kmol.K]
Thermal Conductivity (Mixture)	0	W/[m.K]
Molar Fraction (Vapor)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	

Property	Value	Unit
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Density (Vapor)	0	kg/m3
Molecular Weight (Vapor)	0	kg/kmol
Specific Enthalpy (Vapor)	0	kJ/kg
Specific Entropy (Vapor)	0	kJ/[kg.K]
Molar Enthalpy (Vapor)	0	kJ/kmol
Molar Entropy (Vapor)	0	kJ/[kmol.K]
Thermal Conductivity (Vapor)	0	W/[m.K]
Kinematic Viscosity (Vapor)	0	m2/s
Dynamic Viscosity (Vapor)	0	Pa.s
Heat Capacity (Vapor)	0	kJ/[kg.K]
Heat Capacity Ratio (Vapor)	NaN	
Mass Flow (Vapor)	0	kg/h
Molar Flow (Vapor)	0	kmol/h
Volumetric Flow (Vapor)	0	m3/h
Compressibility Factor (Vapor)	0	
Molar Fraction (Vapor)	0	
Mass Fraction (Vapor)	0	
Volumetric Fraction (Vapor)	NaN	
Molar Fraction (Overall Liquid)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Density (Overall Liquid)	0	kg/m3
Molecular Weight (Overall Liquid)	0	kg/kmol
Specific Enthalpy (Overall Liquid)	0	kJ/kg
Specific Entropy (Overall Liquid)	0	kJ/[kg.K]
Molar Enthalpy (Overall Liquid)	0	kJ/kmol

Property	Value	Unit
Molar Entropy (Overall Liquid)	0	kJ/[kmol.K]
Thermal Conductivity (Overall Liquid)	0	W/[m.K]
Kinematic Viscosity (Overall Liquid)	0	m2/s
Dynamic Viscosity (Overall Liquid)	0	Pa.s
Heat Capacity (Overall Liquid)	0	kJ/[kg.K]
Heat Capacity Ratio (Overall Liquid)	NaN	
Mass Flow (Overall Liquid)	0	kg/h
Molar Flow (Overall Liquid)	0	kmol/h
Volumetric Flow (Overall Liquid)	0	m3/h
Compressibility Factor (Overall Liquid)	0	
Molar Fraction (Overall Liquid)	0	
Mass Fraction (Overall Liquid)	0	
Volumetric Fraction (Overall Liquid)	NaN	
Molar Fraction (Liquid 1)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Molar Fraction (Liquid 2)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	

Property	Value	Unit
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Molar Fraction (Aqueous)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Molar Fraction (Solid)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	

STM-502 (Material Stream)

Property	Value	Unit
Temperature	380	C
Pressure	41	bar
Mass Flow	50000	kg/h
Molar Flow	2775.42	kmol/h
Volumetric Flow	3415.92	m3/h
Mixture Molar Fraction		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	1	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Density (Mixture)	14.6373	kg/m3
Molecular Weight (Mixture)	18.0153	kg/kmol
Specific Enthalpy (Mixture)	618.344	kJ/kg
Specific Entropy (Mixture)	-0.270283	kJ/[kg.K]
Molar Enthalpy (Mixture)	11139.6	kJ/kmol
Molar Entropy (Mixture)	-4.86922	kJ/[kmol.K]
Thermal Conductivity (Mixture)	0.0594012	W/[m.K]
Molar Fraction (Vapor)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	1	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	

Property	Value	Unit
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Density (Vapor)	14.6373	kg/m3
Molecular Weight (Vapor)	18.0153	kg/kmol
Specific Enthalpy (Vapor)	618.344	kJ/kg
Specific Entropy (Vapor)	-0.270283	kJ/[kg.K]
Molar Enthalpy (Vapor)	11139.6	kJ/kmol
Molar Entropy (Vapor)	-4.86922	kJ/[kmol.K]
Thermal Conductivity (Vapor)	0.0594012	W/[m.K]
Kinematic Viscosity (Vapor)	1.66364E-06	m2/s
Dynamic Viscosity (Vapor)	2.43512E-05	Pa.s
Heat Capacity (Vapor)	2.23428	kJ/[kg.K]
Heat Capacity Ratio (Vapor)	1.40357	
Mass Flow (Vapor)	50000	kg/h
Molar Flow (Vapor)	2775.42	kmol/h
Volumetric Flow (Vapor)	3415.92	m3/h
Compressibility Factor (Vapor)	0.929265	
Molar Fraction (Vapor)	1	
Mass Fraction (Vapor)	1	
Volumetric Fraction (Vapor)	1	
Molar Fraction (Overall Liquid)		
Methane	NaN	
Ethane	NaN	
Propane	NaN	
N-butane	NaN	
Isobutane	NaN	
Nitrogen	NaN	
Carbon dioxide	NaN	
Water	NaN	
Oxygen	NaN	
Argon	NaN	
Hydrogen sulfide	NaN	
Hydrogen	NaN	
Carbonyl sulfide	NaN	
Carbon monoxide	NaN	
Ammonia	NaN	
Density (Overall Liquid)	0	kg/m3
Molecular Weight (Overall Liquid)	NaN	kg/kmol
Specific Enthalpy (Overall Liquid)	0	kJ/kg
Specific Entropy (Overall Liquid)	0	kJ/[kg.K]
Molar Enthalpy (Overall Liquid)	NaN	kJ/kmol

Property	Value	Unit
Molar Entropy (Overall Liquid)	NaN	kJ/[kmol.K]
Thermal Conductivity (Overall Liquid)	0	W/[m.K]
Kinematic Viscosity (Overall Liquid)	0	m2/s
Dynamic Viscosity (Overall Liquid)	0	Pa.s
Heat Capacity (Overall Liquid)	0	kJ/[kg.K]
Heat Capacity Ratio (Overall Liquid)	NaN	
Mass Flow (Overall Liquid)	0	kg/h
Molar Flow (Overall Liquid)	0	kmol/h
Volumetric Flow (Overall Liquid)	0	m3/h
Compressibility Factor (Overall Liquid)	0	
Molar Fraction (Overall Liquid)	0	
Mass Fraction (Overall Liquid)	0	
Volumetric Fraction (Overall Liquid)	0	
Molar Fraction (Liquid 1)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	1	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Molar Fraction (Liquid 2)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	1	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	

Property	Value	Unit
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Molar Fraction (Aqueous)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Molar Fraction (Solid)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	1	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	

RFG-402 (Material Stream)

Property	Value	Unit
Temperature	616.53	C
Pressure	35.5	bar
Mass Flow	162279	kg/h
Molar Flow	11188.9	kmol/h
Volumetric Flow	23407.6	m3/h
Mixture Molar Fraction		
Methane	0.00831565	
Ethane	1.25739E-17	
Propane	2.85769E-19	
N-butane	0	
Isobutane	0	
Nitrogen	0.109333	
Carbon dioxide	0.0478962	
Water	0.293919	
Oxygen	1.61709E-08	
Argon	0.00129686	
Hydrogen sulfide	0	
Hydrogen	0.432844	
Carbonyl sulfide	0	
Carbon monoxide	0.106395	
Ammonia	0	
Density (Mixture)	6.93277	kg/m3
Molecular Weight (Mixture)	14.5036	kg/kmol
Specific Enthalpy (Mixture)	1326.75	kJ/kg
Specific Entropy (Mixture)	2.24181	kJ/[kg.K]
Molar Enthalpy (Mixture)	19242.6	kJ/kmol
Molar Entropy (Mixture)	32.5143	kJ/[kmol.K]
Thermal Conductivity (Mixture)	0.221234	W/[m.K]
Molar Fraction (Vapor)		
Methane	0.00831565	
Ethane	1.25739E-17	
Propane	2.85769E-19	
N-butane	0	
Isobutane	0	
Nitrogen	0.109333	
Carbon dioxide	0.0478962	
Water	0.293919	
Oxygen	1.61709E-08	
Argon	0.00129686	
Hydrogen sulfide	0	
Hydrogen	0.432844	

Property	Value	Unit
Carbonyl sulfide	0	
Carbon monoxide	0.106395	
Ammonia	0	
Density (Vapor)	6.93277	kg/m3
Molecular Weight (Vapor)	14.5036	kg/kmol
Specific Enthalpy (Vapor)	1326.75	kJ/kg
Specific Entropy (Vapor)	2.24181	kJ/[kg.K]
Molar Enthalpy (Vapor)	19242.6	kJ/kmol
Molar Entropy (Vapor)	32.5143	kJ/[kmol.K]
Thermal Conductivity (Vapor)	0.221234	W/[m.K]
Kinematic Viscosity (Vapor)	4.1064E-06	m2/s
Dynamic Viscosity (Vapor)	2.84687E-05	Pa.s
Heat Capacity (Vapor)	2.40717	kJ/[kg.K]
Heat Capacity Ratio (Vapor)	1.32176	
Mass Flow (Vapor)	162279	kg/h
Molar Flow (Vapor)	11188.9	kmol/h
Volumetric Flow (Vapor)	23407.6	m3/h
Compressibility Factor (Vapor)	1.00405	
Molar Fraction (Vapor)	1	
Mass Fraction (Vapor)	1	
Volumetric Fraction (Vapor)	1	
Molar Fraction (Overall Liquid)		
Methane	NaN	
Ethane	NaN	
Propane	NaN	
N-butane	NaN	
Isobutane	NaN	
Nitrogen	NaN	
Carbon dioxide	NaN	
Water	NaN	
Oxygen	NaN	
Argon	NaN	
Hydrogen sulfide	NaN	
Hydrogen	NaN	
Carbonyl sulfide	NaN	
Carbon monoxide	NaN	
Ammonia	NaN	
Density (Overall Liquid)	0	kg/m3
Molecular Weight (Overall Liquid)	NaN	kg/kmol
Specific Enthalpy (Overall Liquid)	0	kJ/kg
Specific Entropy (Overall Liquid)	0	kJ/[kg.K]
Molar Enthalpy (Overall Liquid)	NaN	kJ/kmol

Property	Value	Unit
Molar Entropy (Overall Liquid)	NaN	kJ/[kmol.K]
Thermal Conductivity (Overall Liquid)	0	W/[m.K]
Kinematic Viscosity (Overall Liquid)	0	m2/s
Dynamic Viscosity (Overall Liquid)	0	Pa.s
Heat Capacity (Overall Liquid)	0	kJ/[kg.K]
Heat Capacity Ratio (Overall Liquid)	NaN	
Mass Flow (Overall Liquid)	0	kg/h
Molar Flow (Overall Liquid)	0	kmol/h
Volumetric Flow (Overall Liquid)	0	m3/h
Compressibility Factor (Overall Liquid)	0	
Molar Fraction (Overall Liquid)	0	
Mass Fraction (Overall Liquid)	0	
Volumetric Fraction (Overall Liquid)	0	
Molar Fraction (Liquid 1)		
Methane	4.11109E-11	
Ethane	2.55799E-21	
Propane	7.95174E-20	
N-butane	0	
Isobutane	0	
Nitrogen	2.187E-25	
Carbon dioxide	3.38864E-07	
Water	1	
Oxygen	2.94717E-22	
Argon	2.05215E-21	
Hydrogen sulfide	0	
Hydrogen	2.92809E-110	
Carbonyl sulfide	0	
Carbon monoxide	8.5597E-20	
Ammonia	0	
Molar Fraction (Liquid 2)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	

Property	Value	Unit
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Molar Fraction (Aqueous)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Molar Fraction (Solid)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	

METH-901 (Material Stream)

Property	Value	Unit
Temperature	40	C
Pressure	34	bar
Mass Flow	43357.9	kg/h
Molar Flow	5123.79	kmol/h
Volumetric Flow	3949.5	m3/h
Mixture Molar Fraction		
Methane	0.0912208	
Ethane	2.2122E-17	
Propane	5.02772E-19	
N-butane	0	
Isobutane	0	
Nitrogen	0.192357	
Carbon dioxide	0.00103911	
Water	0.00223153	
Oxygen	2.84492E-08	
Argon	0.00228155	
Hydrogen sulfide	0	
Hydrogen	0.71087	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Density (Mixture)	10.9781	kg/m3
Molecular Weight (Mixture)	8.46208	kg/kmol
Specific Enthalpy (Mixture)	44.9108	kJ/kg
Specific Entropy (Mixture)	-2.08992	kJ/[kg.K]
Molar Enthalpy (Mixture)	380.039	kJ/kmol
Molar Entropy (Mixture)	-17.6851	kJ/[kmol.K]
Thermal Conductivity (Mixture)	0.141764	W/[m.K]
Molar Fraction (Vapor)		
Methane	0.0912208	
Ethane	2.2122E-17	
Propane	5.02772E-19	
N-butane	0	
Isobutane	0	
Nitrogen	0.192357	
Carbon dioxide	0.00103911	
Water	0.00223153	
Oxygen	2.84492E-08	
Argon	0.00228155	
Hydrogen sulfide	0	
Hydrogen	0.71087	

Property	Value	Unit
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Density (Vapor)	10.9781	kg/m3
Molecular Weight (Vapor)	8.46208	kg/kmol
Specific Enthalpy (Vapor)	44.9108	kJ/kg
Specific Entropy (Vapor)	-2.08992	kJ/[kg.K]
Molar Enthalpy (Vapor)	380.039	kJ/kmol
Molar Entropy (Vapor)	-17.6851	kJ/[kmol.K]
Thermal Conductivity (Vapor)	0.141764	W/[m.K]
Kinematic Viscosity (Vapor)	1.04633E-06	m2/s
Dynamic Viscosity (Vapor)	1.14867E-05	Pa.s
Heat Capacity (Vapor)	3.55293	kJ/[kg.K]
Heat Capacity Ratio (Vapor)	1.42364	
Mass Flow (Vapor)	43357.9	kg/h
Molar Flow (Vapor)	5123.79	kmol/h
Volumetric Flow (Vapor)	3949.5	m3/h
Compressibility Factor (Vapor)	1.00662	
Molar Fraction (Vapor)	1	
Mass Fraction (Vapor)	1	
Volumetric Fraction (Vapor)	1	
Molar Fraction (Overall Liquid)		
Methane	NaN	
Ethane	NaN	
Propane	NaN	
N-butane	NaN	
Isobutane	NaN	
Nitrogen	NaN	
Carbon dioxide	NaN	
Water	NaN	
Oxygen	NaN	
Argon	NaN	
Hydrogen sulfide	NaN	
Hydrogen	NaN	
Carbonyl sulfide	NaN	
Carbon monoxide	NaN	
Ammonia	NaN	
Density (Overall Liquid)	0	kg/m3
Molecular Weight (Overall Liquid)	NaN	kg/kmol
Specific Enthalpy (Overall Liquid)	0	kJ/kg
Specific Entropy (Overall Liquid)	0	kJ/[kg.K]
Molar Enthalpy (Overall Liquid)	NaN	kJ/kmol

Property	Value	Unit
Molar Entropy (Overall Liquid)	NaN	kJ/[kmol.K]
Thermal Conductivity (Overall Liquid)	0	W/[m.K]
Kinematic Viscosity (Overall Liquid)	0	m2/s
Dynamic Viscosity (Overall Liquid)	0	Pa.s
Heat Capacity (Overall Liquid)	0	kJ/[kg.K]
Heat Capacity Ratio (Overall Liquid)	NaN	
Mass Flow (Overall Liquid)	0	kg/h
Molar Flow (Overall Liquid)	0	kmol/h
Volumetric Flow (Overall Liquid)	0	m3/h
Compressibility Factor (Overall Liquid)	0	
Molar Fraction (Overall Liquid)	0	
Mass Fraction (Overall Liquid)	0	
Volumetric Fraction (Overall Liquid)	0	
Molar Fraction (Liquid 1)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Molar Fraction (Liquid 2)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	

Property	Value	Unit
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Molar Fraction (Aqueous)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Molar Fraction (Solid)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	

WATER-OUT-901 (Material Stream)

Property	Value	Unit
Temperature	40	C
Pressure	34	bar
Mass Flow	7246.88	kg/h
Molar Flow	402.258	kmol/h
Volumetric Flow	8.62317	m3/h
Mixture Molar Fraction		
Methane	2.91568E-08	
Ethane	1.86454E-25	
Propane	2.36866E-29	
N-butane	0	
Isobutane	0	
Nitrogen	1.24194E-06	
Carbon dioxide	1.51454E-05	
Water	0.999971	
Oxygen	3.31659E-12	
Argon	2.55208E-07	
Hydrogen sulfide	0	
Hydrogen	1.26365E-05	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Density (Mixture)	840.395	kg/m3
Molecular Weight (Mixture)	18.0155	kg/kmol
Specific Enthalpy (Mixture)	-2466.25	kJ/kg
Specific Entropy (Mixture)	-6.61468	kJ/[kg.K]
Molar Enthalpy (Mixture)	-44430.7	kJ/kmol
Molar Entropy (Mixture)	-119.167	kJ/[kmol.K]
Thermal Conductivity (Mixture)	0.63088	W/[m.K]
Molar Fraction (Vapor)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	

Property	Value	Unit
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Density (Vapor)	0	kg/m3
Molecular Weight (Vapor)	0	kg/kmol
Specific Enthalpy (Vapor)	0	kJ/kg
Specific Entropy (Vapor)	0	kJ/[kg.K]
Molar Enthalpy (Vapor)	0	kJ/kmol
Molar Entropy (Vapor)	0	kJ/[kmol.K]
Thermal Conductivity (Vapor)	0	W/[m.K]
Kinematic Viscosity (Vapor)	0	m2/s
Dynamic Viscosity (Vapor)	0	Pa.s
Heat Capacity (Vapor)	0	kJ/[kg.K]
Heat Capacity Ratio (Vapor)	NaN	
Mass Flow (Vapor)	0	kg/h
Molar Flow (Vapor)	0	kmol/h
Volumetric Flow (Vapor)	0	m3/h
Compressibility Factor (Vapor)	0	
Molar Fraction (Vapor)	0	
Mass Fraction (Vapor)	0	
Volumetric Fraction (Vapor)	0	
Molar Fraction (Overall Liquid)		
Methane	2.91568E-08	
Ethane	1.86454E-25	
Propane	2.36866E-29	
N-butane	0	
Isobutane	0	
Nitrogen	1.24194E-06	
Carbon dioxide	1.51454E-05	
Water	0.999971	
Oxygen	3.31659E-12	
Argon	2.55208E-07	
Hydrogen sulfide	0	
Hydrogen	1.26365E-05	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Density (Overall Liquid)	840.395	kg/m3
Molecular Weight (Overall Liquid)	18.0155	kg/kmol
Specific Enthalpy (Overall Liquid)	-2466.25	kJ/kg
Specific Entropy (Overall Liquid)	-6.61468	kJ/[kg.K]
Molar Enthalpy (Overall Liquid)	-44430.7	kJ/kmol

Property	Value	Unit
Molar Entropy (Overall Liquid)	-119.167	kJ/[kmol.K]
Thermal Conductivity (Overall Liquid)	0.63088	W/[m.K]
Kinematic Viscosity (Overall Liquid)	7.81696E-07	m ² /s
Dynamic Viscosity (Overall Liquid)	0.000656933	Pa.s
Heat Capacity (Overall Liquid)	3.91689	kJ/[kg.K]
Heat Capacity Ratio (Overall Liquid)	1.30788	
Mass Flow (Overall Liquid)	7246.88	kg/h
Molar Flow (Overall Liquid)	402.258	kmol/h
Volumetric Flow (Overall Liquid)	8.62317	m ³ /h
Compressibility Factor (Overall Liquid)	0.0279949	
Molar Fraction (Overall Liquid)	1	
Mass Fraction (Overall Liquid)	1	
Volumetric Fraction (Overall Liquid)	1	
Molar Fraction (Liquid 1)		
Methane	2.91568E-08	
Ethane	1.86454E-25	
Propane	2.36866E-29	
N-butane	0	
Isobutane	0	
Nitrogen	1.24194E-06	
Carbon dioxide	1.51454E-05	
Water	0.999971	
Oxygen	3.31659E-12	
Argon	2.55208E-07	
Hydrogen sulfide	0	
Hydrogen	1.26365E-05	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Molar Fraction (Liquid 2)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	

Property	Value	Unit
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Molar Fraction (Aqueous)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Molar Fraction (Solid)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	

CMP-101 (Material Stream)

Property	Value	Unit
Temperature	270.01	C
Pressure	55	bar
Mass Flow	43357.9	kg/h
Molar Flow	5123.79	kmol/h
Volumetric Flow	0.268088	m3/h
Mixture Molar Fraction		
Methane	0.0912208	
Ethane	2.2122E-17	
Propane	5.02772E-19	
N-butane	0	
Isobutane	0	
Nitrogen	0.192357	
Carbon dioxide	0.00103911	
Water	0.00223153	
Oxygen	2.84492E-08	
Argon	0.00228155	
Hydrogen sulfide	0	
Hydrogen	0.71087	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Density (Mixture)	161730	kg/m3
Molecular Weight (Mixture)	8.46208	kg/kmol
Specific Enthalpy (Mixture)	0	kJ/kg
Specific Entropy (Mixture)	0	kJ/[kg.K]
Molar Enthalpy (Mixture)	0	kJ/kmol
Molar Entropy (Mixture)	0	kJ/[kmol.K]
Thermal Conductivity (Mixture)	0.000195388	W/[m.K]
Molar Fraction (Vapor)		
Methane	0.0918574	
Ethane	2.25445E-17	
Propane	5.15041E-19	
N-butane	0	
Isobutane	0	
Nitrogen	0.191485	
Carbon dioxide	0.00106312	
Water	0.00235632	
Oxygen	2.86099E-08	
Argon	0.00229522	
Hydrogen sulfide	0	
Hydrogen	0.710943	

Property	Value	Unit
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Density (Vapor)	0	kg/m3
Molecular Weight (Vapor)	0	kg/kmol
Specific Enthalpy (Vapor)	0	kJ/kg
Specific Entropy (Vapor)	0	kJ/[kg.K]
Molar Enthalpy (Vapor)	0	kJ/kmol
Molar Entropy (Vapor)	0	kJ/[kmol.K]
Thermal Conductivity (Vapor)	0	W/[m.K]
Kinematic Viscosity (Vapor)	0	m2/s
Dynamic Viscosity (Vapor)	0	Pa.s
Heat Capacity (Vapor)	0	kJ/[kg.K]
Heat Capacity Ratio (Vapor)	NaN	
Mass Flow (Vapor)	0	kg/h
Molar Flow (Vapor)	0	kmol/h
Volumetric Flow (Vapor)	0	m3/h
Compressibility Factor (Vapor)	0	
Molar Fraction (Vapor)	0	
Mass Fraction (Vapor)	0	
Volumetric Fraction (Vapor)	0	
Molar Fraction (Overall Liquid)		
Methane	0.0912208	
Ethane	2.2122E-17	
Propane	5.02772E-19	
N-butane	0	
Isobutane	0	
Nitrogen	0.192357	
Carbon dioxide	0.00103911	
Water	0.00223153	
Oxygen	2.84492E-08	
Argon	0.00228155	
Hydrogen sulfide	0	
Hydrogen	0.71087	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Density (Overall Liquid)	161730	kg/m3
Molecular Weight (Overall Liquid)	8.46208	kg/kmol
Specific Enthalpy (Overall Liquid)	0	kJ/kg
Specific Entropy (Overall Liquid)	0	kJ/[kg.K]
Molar Enthalpy (Overall Liquid)	0	kJ/kmol

Property	Value	Unit
Molar Entropy (Overall Liquid)	0	kJ/[kmol.K]
Thermal Conductivity (Overall Liquid)	0.000195388	W/[m.K]
Kinematic Viscosity (Overall Liquid)	1.34594E-12	m2/s
Dynamic Viscosity (Overall Liquid)	2.17679E-07	Pa.s
Heat Capacity (Overall Liquid)	0	kJ/[kg.K]
Heat Capacity Ratio (Overall Liquid)	NaN	
Mass Flow (Overall Liquid)	43357.9	kg/h
Molar Flow (Overall Liquid)	5123.79	kmol/h
Volumetric Flow (Overall Liquid)	0.268088	m3/h
Compressibility Factor (Overall Liquid)	-0.0436563	
Molar Fraction (Overall Liquid)	1	
Mass Fraction (Overall Liquid)	1	
Volumetric Fraction (Overall Liquid)	1	
Molar Fraction (Liquid 1)		
Methane	0.0912208	
Ethane	2.2122E-17	
Propane	5.02772E-19	
N-butane	0	
Isobutane	0	
Nitrogen	0.192357	
Carbon dioxide	0.00103911	
Water	0.00223153	
Oxygen	2.84492E-08	
Argon	0.00228155	
Hydrogen sulfide	0	
Hydrogen	0.71087	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Molar Fraction (Liquid 2)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	

Property	Value	Unit
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Molar Fraction (Aqueous)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Molar Fraction (Solid)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	

IC-101 (Material Stream)

Property	Value	Unit
Temperature	40	C
Pressure	54	bar
Mass Flow	43357.9	kg/h
Molar Flow	5123.79	kmol/h
Volumetric Flow	2498.41	m3/h
Mixture Molar Fraction		
Methane	0.0912208	
Ethane	2.2122E-17	
Propane	5.02772E-19	
N-butane	0	
Isobutane	0	
Nitrogen	0.192357	
Carbon dioxide	0.00103911	
Water	0.00223153	
Oxygen	2.84492E-08	
Argon	0.00228155	
Hydrogen sulfide	0	
Hydrogen	0.71087	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Density (Mixture)	17.3542	kg/m3
Molecular Weight (Mixture)	8.46208	kg/kmol
Specific Enthalpy (Mixture)	37.6649	kJ/kg
Specific Entropy (Mixture)	-2.57104	kJ/[kg.K]
Molar Enthalpy (Mixture)	318.723	kJ/kmol
Molar Entropy (Mixture)	-21.7563	kJ/[kmol.K]
Thermal Conductivity (Mixture)	0.143257	W/[m.K]
Molar Fraction (Vapor)		
Methane	0.0912842	
Ethane	2.21373E-17	
Propane	5.03121E-19	
N-butane	0	
Isobutane	0	
Nitrogen	0.19249	
Carbon dioxide	0.00103982	
Water	0.00153774	
Oxygen	2.8469E-08	
Argon	0.00228313	
Hydrogen sulfide	0	
Hydrogen	0.711365	

Property	Value	Unit
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Density (Vapor)	17.329	kg/m3
Molecular Weight (Vapor)	8.45543	kg/kmol
Specific Enthalpy (Vapor)	41.3717	kJ/kg
Specific Entropy (Vapor)	-2.56504	kJ/[kg.K]
Molar Enthalpy (Vapor)	349.816	kJ/kmol
Molar Entropy (Vapor)	-21.6886	kJ/[kmol.K]
Thermal Conductivity (Vapor)	0.142918	W/[m.K]
Kinematic Viscosity (Vapor)	6.68598E-07	m2/s
Dynamic Viscosity (Vapor)	1.15862E-05	Pa.s
Heat Capacity (Vapor)	3.58558	kJ/[kg.K]
Heat Capacity Ratio (Vapor)	1.43932	
Mass Flow (Vapor)	43293.8	kg/h
Molar Flow (Vapor)	5120.23	kmol/h
Volumetric Flow (Vapor)	2498.34	m3/h
Compressibility Factor (Vapor)	1.01203	
Molar Fraction (Vapor)	0.999305	
Mass Fraction (Vapor)	0.998521	
Volumetric Fraction (Vapor)	0.999969	
Molar Fraction (Overall Liquid)		
Methane	4.43477E-08	
Ethane	2.71915E-25	
Propane	3.33157E-29	
N-butane	0	
Isobutane	0	
Nitrogen	1.92427E-06	
Carbon dioxide	2.24786E-05	
Water	0.999955	
Oxygen	5.1167E-12	
Argon	3.93818E-07	
Hydrogen sulfide	0	
Hydrogen	1.98728E-05	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Density (Overall Liquid)	840.628	kg/m3
Molecular Weight (Overall Liquid)	18.0156	kg/kmol
Specific Enthalpy (Overall Liquid)	-2464.32	kJ/kg
Specific Entropy (Overall Liquid)	-6.61591	kJ/[kg.K]
Molar Enthalpy (Overall Liquid)	-44396.2	kJ/kmol

Property	Value	Unit
Molar Entropy (Overall Liquid)	-119.189	kJ/[kmol.K]
Thermal Conductivity (Overall Liquid)	0.630855	W/[m.K]
Kinematic Viscosity (Overall Liquid)	7.84467E-07	m ² /s
Dynamic Viscosity (Overall Liquid)	0.000659444	Pa.s
Heat Capacity (Overall Liquid)	3.91415	kJ/[kg.K]
Heat Capacity Ratio (Overall Liquid)	1.30705	
Mass Flow (Overall Liquid)	64.1418	kg/h
Molar Flow (Overall Liquid)	3.56035	kmol/h
Volumetric Flow (Overall Liquid)	0.0763022	m ³ /h
Compressibility Factor (Overall Liquid)	0.0444504	
Molar Fraction (Overall Liquid)	0.000694866	
Mass Fraction (Overall Liquid)	0.00147936	
Volumetric Fraction (Overall Liquid)	3.05403E-05	
Molar Fraction (Liquid 1)		
Methane	4.43477E-08	
Ethane	2.71915E-25	
Propane	3.33157E-29	
N-butane	0	
Isobutane	0	
Nitrogen	1.92427E-06	
Carbon dioxide	2.24786E-05	
Water	0.999955	
Oxygen	5.1167E-12	
Argon	3.93818E-07	
Hydrogen sulfide	0	
Hydrogen	1.98728E-05	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Molar Fraction (Liquid 2)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	

Property	Value	Unit
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Molar Fraction (Aqueous)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Molar Fraction (Solid)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	

SYNGAS-1002 (Material Stream)

Property	Value	Unit
Temperature	40	C
Pressure	54	bar
Mass Flow	43293.8	kg/h
Molar Flow	5120.23	kmol/h
Volumetric Flow	2498.34	m3/h
Mixture Molar Fraction		
Methane	0.0912842	
Ethane	2.21373E-17	
Propane	5.03121E-19	
N-butane	0	
Isobutane	0	
Nitrogen	0.19249	
Carbon dioxide	0.00103982	
Water	0.00153774	
Oxygen	2.8469E-08	
Argon	0.00228313	
Hydrogen sulfide	0	
Hydrogen	0.711365	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Density (Mixture)	17.329	kg/m3
Molecular Weight (Mixture)	8.45543	kg/kmol
Specific Enthalpy (Mixture)	41.3717	kJ/kg
Specific Entropy (Mixture)	-2.56504	kJ/[kg.K]
Molar Enthalpy (Mixture)	349.816	kJ/kmol
Molar Entropy (Mixture)	-21.6886	kJ/[kmol.K]
Thermal Conductivity (Mixture)	0.142918	W/[m.K]
Molar Fraction (Vapor)		
Methane	0.0912842	
Ethane	2.21373E-17	
Propane	5.03121E-19	
N-butane	0	
Isobutane	0	
Nitrogen	0.19249	
Carbon dioxide	0.00103982	
Water	0.00153774	
Oxygen	2.8469E-08	
Argon	0.00228313	
Hydrogen sulfide	0	
Hydrogen	0.711365	

Property	Value	Unit
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Density (Vapor)	17.329	kg/m3
Molecular Weight (Vapor)	8.45543	kg/kmol
Specific Enthalpy (Vapor)	41.3717	kJ/kg
Specific Entropy (Vapor)	-2.56504	kJ/[kg.K]
Molar Enthalpy (Vapor)	349.816	kJ/kmol
Molar Entropy (Vapor)	-21.6886	kJ/[kmol.K]
Thermal Conductivity (Vapor)	0.142918	W/[m.K]
Kinematic Viscosity (Vapor)	6.68598E-07	m2/s
Dynamic Viscosity (Vapor)	1.15862E-05	Pa.s
Heat Capacity (Vapor)	3.58558	kJ/[kg.K]
Heat Capacity Ratio (Vapor)	1.43932	
Mass Flow (Vapor)	43293.8	kg/h
Molar Flow (Vapor)	5120.23	kmol/h
Volumetric Flow (Vapor)	2498.34	m3/h
Compressibility Factor (Vapor)	1.01203	
Molar Fraction (Vapor)	1	
Mass Fraction (Vapor)	1	
Volumetric Fraction (Vapor)	1	
Molar Fraction (Overall Liquid)		
Methane	NaN	
Ethane	NaN	
Propane	NaN	
N-butane	NaN	
Isobutane	NaN	
Nitrogen	NaN	
Carbon dioxide	NaN	
Water	NaN	
Oxygen	NaN	
Argon	NaN	
Hydrogen sulfide	NaN	
Hydrogen	NaN	
Carbonyl sulfide	NaN	
Carbon monoxide	NaN	
Ammonia	NaN	
Density (Overall Liquid)	0	kg/m3
Molecular Weight (Overall Liquid)	NaN	kg/kmol
Specific Enthalpy (Overall Liquid)	0	kJ/kg
Specific Entropy (Overall Liquid)	0	kJ/[kg.K]
Molar Enthalpy (Overall Liquid)	NaN	kJ/kmol

Property	Value	Unit
Molar Entropy (Overall Liquid)	NaN	kJ/[kmol.K]
Thermal Conductivity (Overall Liquid)	0	W/[m.K]
Kinematic Viscosity (Overall Liquid)	0	m2/s
Dynamic Viscosity (Overall Liquid)	0	Pa.s
Heat Capacity (Overall Liquid)	0	kJ/[kg.K]
Heat Capacity Ratio (Overall Liquid)	NaN	
Mass Flow (Overall Liquid)	0	kg/h
Molar Flow (Overall Liquid)	0	kmol/h
Volumetric Flow (Overall Liquid)	0	m3/h
Compressibility Factor (Overall Liquid)	0	
Molar Fraction (Overall Liquid)	0	
Mass Fraction (Overall Liquid)	0	
Volumetric Fraction (Overall Liquid)	0	
Molar Fraction (Liquid 1)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Molar Fraction (Liquid 2)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	

Property	Value	Unit
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Molar Fraction (Aqueous)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Molar Fraction (Solid)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	

WATER-OUT-1001 (Material Stream)

Property	Value	Unit
Temperature	40	C
Pressure	54	bar
Mass Flow	64.1418	kg/h
Molar Flow	3.56035	kmol/h
Volumetric Flow	0.0763022	m3/h
Mixture Molar Fraction		
Methane	4.43477E-08	
Ethane	2.71915E-25	
Propane	3.33157E-29	
N-butane	0	
Isobutane	0	
Nitrogen	1.92427E-06	
Carbon dioxide	2.24786E-05	
Water	0.999955	
Oxygen	5.1167E-12	
Argon	3.93818E-07	
Hydrogen sulfide	0	
Hydrogen	1.98728E-05	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Density (Mixture)	840.628	kg/m3
Molecular Weight (Mixture)	18.0156	kg/kmol
Specific Enthalpy (Mixture)	-2464.32	kJ/kg
Specific Entropy (Mixture)	-6.61591	kJ/[kg.K]
Molar Enthalpy (Mixture)	-44396.2	kJ/kmol
Molar Entropy (Mixture)	-119.189	kJ/[kmol.K]
Thermal Conductivity (Mixture)	0.630855	W/[m.K]
Molar Fraction (Vapor)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	

Property	Value	Unit
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Density (Vapor)	0	kg/m3
Molecular Weight (Vapor)	0	kg/kmol
Specific Enthalpy (Vapor)	0	kJ/kg
Specific Entropy (Vapor)	0	kJ/[kg.K]
Molar Enthalpy (Vapor)	0	kJ/kmol
Molar Entropy (Vapor)	0	kJ/[kmol.K]
Thermal Conductivity (Vapor)	0	W/[m.K]
Kinematic Viscosity (Vapor)	0	m2/s
Dynamic Viscosity (Vapor)	0	Pa.s
Heat Capacity (Vapor)	0	kJ/[kg.K]
Heat Capacity Ratio (Vapor)	NaN	
Mass Flow (Vapor)	0	kg/h
Molar Flow (Vapor)	0	kmol/h
Volumetric Flow (Vapor)	0	m3/h
Compressibility Factor (Vapor)	0	
Molar Fraction (Vapor)	0	
Mass Fraction (Vapor)	0	
Volumetric Fraction (Vapor)	0	
Molar Fraction (Overall Liquid)		
Methane	4.43477E-08	
Ethane	2.71915E-25	
Propane	3.33157E-29	
N-butane	0	
Isobutane	0	
Nitrogen	1.92427E-06	
Carbon dioxide	2.24786E-05	
Water	0.999955	
Oxygen	5.1167E-12	
Argon	3.93818E-07	
Hydrogen sulfide	0	
Hydrogen	1.98728E-05	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Density (Overall Liquid)	840.628	kg/m3
Molecular Weight (Overall Liquid)	18.0156	kg/kmol
Specific Enthalpy (Overall Liquid)	-2464.32	kJ/kg
Specific Entropy (Overall Liquid)	-6.61591	kJ/[kg.K]
Molar Enthalpy (Overall Liquid)	-44396.2	kJ/kmol

Property	Value	Unit
Molar Entropy (Overall Liquid)	-119.189	kJ/[kmol.K]
Thermal Conductivity (Overall Liquid)	0.630855	W/[m.K]
Kinematic Viscosity (Overall Liquid)	7.84467E-07	m ² /s
Dynamic Viscosity (Overall Liquid)	0.000659444	Pa.s
Heat Capacity (Overall Liquid)	3.91415	kJ/[kg.K]
Heat Capacity Ratio (Overall Liquid)	1.30705	
Mass Flow (Overall Liquid)	64.1418	kg/h
Molar Flow (Overall Liquid)	3.56035	kmol/h
Volumetric Flow (Overall Liquid)	0.0763022	m ³ /h
Compressibility Factor (Overall Liquid)	0.0444504	
Molar Fraction (Overall Liquid)	1	
Mass Fraction (Overall Liquid)	1	
Volumetric Fraction (Overall Liquid)	1	
Molar Fraction (Liquid 1)		
Methane	4.43477E-08	
Ethane	2.71915E-25	
Propane	3.33157E-29	
N-butane	0	
Isobutane	0	
Nitrogen	1.92427E-06	
Carbon dioxide	2.24786E-05	
Water	0.999955	
Oxygen	5.1167E-12	
Argon	3.93818E-07	
Hydrogen sulfide	0	
Hydrogen	1.98728E-05	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Molar Fraction (Liquid 2)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	

Property	Value	Unit
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Molar Fraction (Aqueous)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Molar Fraction (Solid)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	

CMP-102 (Material Stream)

Property	Value	Unit
Temperature	304.068	C
Pressure	91	bar
Mass Flow	43293.8	kg/h
Molar Flow	5120.23	kmol/h
Volumetric Flow	38.6961	m3/h
Mixture Molar Fraction		
Methane	0.0912842	
Ethane	2.21373E-17	
Propane	5.03121E-19	
N-butane	0	
Isobutane	0	
Nitrogen	0.19249	
Carbon dioxide	0.00103982	
Water	0.00153774	
Oxygen	2.8469E-08	
Argon	0.00228313	
Hydrogen sulfide	0	
Hydrogen	0.711365	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Density (Mixture)	1118.82	kg/m3
Molecular Weight (Mixture)	8.45543	kg/kmol
Specific Enthalpy (Mixture)	0	kJ/kg
Specific Entropy (Mixture)	0	kJ/[kg.K]
Molar Enthalpy (Mixture)	0	kJ/kmol
Molar Entropy (Mixture)	0	kJ/[kmol.K]
Thermal Conductivity (Mixture)	0.00018925	W/[m.K]
Molar Fraction (Vapor)		
Methane	0.0919616	
Ethane	2.25981E-17	
Propane	5.15372E-19	
N-butane	0	
Isobutane	0	
Nitrogen	0.190899	
Carbon dioxide	0.00106977	
Water	0.00165675	
Oxygen	2.86678E-08	
Argon	0.00230041	
Hydrogen sulfide	0	
Hydrogen	0.712112	

Property	Value	Unit
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Density (Vapor)	0	kg/m3
Molecular Weight (Vapor)	0	kg/kmol
Specific Enthalpy (Vapor)	0	kJ/kg
Specific Entropy (Vapor)	0	kJ/[kg.K]
Molar Enthalpy (Vapor)	0	kJ/kmol
Molar Entropy (Vapor)	0	kJ/[kmol.K]
Thermal Conductivity (Vapor)	0	W/[m.K]
Kinematic Viscosity (Vapor)	0	m2/s
Dynamic Viscosity (Vapor)	0	Pa.s
Heat Capacity (Vapor)	0	kJ/[kg.K]
Heat Capacity Ratio (Vapor)	NaN	
Mass Flow (Vapor)	0	kg/h
Molar Flow (Vapor)	0	kmol/h
Volumetric Flow (Vapor)	0	m3/h
Compressibility Factor (Vapor)	0	
Molar Fraction (Vapor)	0	
Mass Fraction (Vapor)	0	
Volumetric Fraction (Vapor)	0	
Molar Fraction (Overall Liquid)		
Methane	0.0912842	
Ethane	2.21373E-17	
Propane	5.03121E-19	
N-butane	0	
Isobutane	0	
Nitrogen	0.19249	
Carbon dioxide	0.00103982	
Water	0.00153774	
Oxygen	2.8469E-08	
Argon	0.00228313	
Hydrogen sulfide	0	
Hydrogen	0.711365	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Density (Overall Liquid)	1118.82	kg/m3
Molecular Weight (Overall Liquid)	8.45543	kg/kmol
Specific Enthalpy (Overall Liquid)	0	kJ/kg
Specific Entropy (Overall Liquid)	0	kJ/[kg.K]
Molar Enthalpy (Overall Liquid)	0	kJ/kmol

Property	Value	Unit
Molar Entropy (Overall Liquid)	0	kJ/[kmol.K]
Thermal Conductivity (Overall Liquid)	0.00018925	W/[m.K]
Kinematic Viscosity (Overall Liquid)	1.12087E-10	m2/s
Dynamic Viscosity (Overall Liquid)	1.25405E-07	Pa.s
Heat Capacity (Overall Liquid)	0	kJ/[kg.K]
Heat Capacity Ratio (Overall Liquid)	NaN	
Mass Flow (Overall Liquid)	43293.8	kg/h
Molar Flow (Overall Liquid)	5120.23	kmol/h
Volumetric Flow (Overall Liquid)	38.6961	m3/h
Compressibility Factor (Overall Liquid)	0.0143308	
Molar Fraction (Overall Liquid)	1	
Mass Fraction (Overall Liquid)	1	
Volumetric Fraction (Overall Liquid)	1	
Molar Fraction (Liquid 1)		
Methane	0.0912842	
Ethane	2.21373E-17	
Propane	5.03121E-19	
N-butane	0	
Isobutane	0	
Nitrogen	0.19249	
Carbon dioxide	0.00103982	
Water	0.00153774	
Oxygen	2.8469E-08	
Argon	0.00228313	
Hydrogen sulfide	0	
Hydrogen	0.711365	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Molar Fraction (Liquid 2)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	

Property	Value	Unit
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Molar Fraction (Aqueous)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Molar Fraction (Solid)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	

IC-102 (Material Stream)

Property	Value	Unit
Temperature	40	C
Pressure	91	bar
Mass Flow	43293.8	kg/h
Molar Flow	5120.23	kmol/h
Volumetric Flow	1499.93	m3/h
Mixture Molar Fraction		
Methane	0.0912842	
Ethane	2.21373E-17	
Propane	5.03121E-19	
N-butane	0	
Isobutane	0	
Nitrogen	0.19249	
Carbon dioxide	0.00103982	
Water	0.00153774	
Oxygen	2.8469E-08	
Argon	0.00228313	
Hydrogen sulfide	0	
Hydrogen	0.711365	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Density (Mixture)	28.8638	kg/m3
Molecular Weight (Mixture)	8.45543	kg/kmol
Specific Enthalpy (Mixture)	33.4766	kJ/kg
Specific Entropy (Mixture)	-3.11065	kJ/[kg.K]
Molar Enthalpy (Mixture)	283.059	kJ/kmol
Molar Entropy (Mixture)	-26.3019	kJ/[kmol.K]
Thermal Conductivity (Mixture)	0.145232	W/[m.K]
Molar Fraction (Vapor)		
Methane	0.0913271	
Ethane	2.21477E-17	
Propane	5.03358E-19	
N-butane	0	
Isobutane	0	
Nitrogen	0.192581	
Carbon dioxide	0.00104029	
Water	0.0010691	
Oxygen	2.84824E-08	
Argon	0.0022842	
Hydrogen sulfide	0	
Hydrogen	0.711699	

Property	Value	Unit
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Density (Vapor)	28.836	kg/m3
Molecular Weight (Vapor)	8.45094	kg/kmol
Specific Enthalpy (Vapor)	35.9724	kJ/kg
Specific Entropy (Vapor)	-3.10714	kJ/[kg.K]
Molar Enthalpy (Vapor)	304.001	kJ/kmol
Molar Entropy (Vapor)	-26.2582	kJ/[kmol.K]
Thermal Conductivity (Vapor)	0.145004	W/[m.K]
Kinematic Viscosity (Vapor)	4.09747E-07	m2/s
Dynamic Viscosity (Vapor)	1.18154E-05	Pa.s
Heat Capacity (Vapor)	3.636	kJ/[kg.K]
Heat Capacity Ratio (Vapor)	1.46301	
Mass Flow (Vapor)	43250.5	kg/h
Molar Flow (Vapor)	5117.83	kmol/h
Volumetric Flow (Vapor)	1499.88	m3/h
Compressibility Factor (Vapor)	1.02435	
Molar Fraction (Vapor)	0.999531	
Mass Fraction (Vapor)	0.999	
Volumetric Fraction (Vapor)	0.999966	
Molar Fraction (Overall Liquid)		
Methane	6.92306E-08	
Ethane	3.94628E-25	
Propane	4.54205E-29	
N-butane	0	
Isobutane	0	
Nitrogen	3.10255E-06	
Carbon dioxide	3.36201E-05	
Water	0.99993	
Oxygen	8.18602E-12	
Argon	6.30319E-07	
Hydrogen sulfide	0	
Hydrogen	3.28732E-05	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Density (Overall Liquid)	841.053	kg/m3
Molecular Weight (Overall Liquid)	18.0157	kg/kmol
Specific Enthalpy (Overall Liquid)	-2460.77	kJ/kg
Specific Entropy (Overall Liquid)	-6.61825	kJ/[kg.K]
Molar Enthalpy (Overall Liquid)	-44332.5	kJ/kmol

Property	Value	Unit
Molar Entropy (Overall Liquid)	-119.232	kJ/[kmol.K]
Thermal Conductivity (Overall Liquid)	0.630814	W/[m.K]
Kinematic Viscosity (Overall Liquid)	7.89499E-07	m2/s
Dynamic Viscosity (Overall Liquid)	0.00066401	Pa.s
Heat Capacity (Overall Liquid)	3.90915	kJ/[kg.K]
Heat Capacity Ratio (Overall Liquid)	1.30554	
Mass Flow (Overall Liquid)	43.2782	kg/h
Molar Flow (Overall Liquid)	2.40225	kmol/h
Volumetric Flow (Overall Liquid)	0.0514571	m3/h
Compressibility Factor (Overall Liquid)	0.0748697	
Molar Fraction (Overall Liquid)	0.000469168	
Mass Fraction (Overall Liquid)	0.00099964	
Volumetric Fraction (Overall Liquid)	3.43063E-05	
Molar Fraction (Liquid 1)		
Methane	6.92306E-08	
Ethane	3.94628E-25	
Propane	4.54205E-29	
N-butane	0	
Isobutane	0	
Nitrogen	3.10255E-06	
Carbon dioxide	3.36201E-05	
Water	0.99993	
Oxygen	8.18602E-12	
Argon	6.30319E-07	
Hydrogen sulfide	0	
Hydrogen	3.28732E-05	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Molar Fraction (Liquid 2)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	

Property	Value	Unit
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Molar Fraction (Aqueous)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Molar Fraction (Solid)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	

SYNGAS-1003 (Material Stream)

Property	Value	Unit
Temperature	40	C
Pressure	91	bar
Mass Flow	43250.5	kg/h
Molar Flow	5117.83	kmol/h
Volumetric Flow	1499.88	m3/h
Mixture Molar Fraction		
Methane	0.0913271	
Ethane	2.21477E-17	
Propane	5.03358E-19	
N-butane	0	
Isobutane	0	
Nitrogen	0.192581	
Carbon dioxide	0.00104029	
Water	0.0010691	
Oxygen	2.84824E-08	
Argon	0.0022842	
Hydrogen sulfide	0	
Hydrogen	0.711699	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Density (Mixture)	28.836	kg/m3
Molecular Weight (Mixture)	8.45094	kg/kmol
Specific Enthalpy (Mixture)	35.9724	kJ/kg
Specific Entropy (Mixture)	-3.10714	kJ/[kg.K]
Molar Enthalpy (Mixture)	304.001	kJ/kmol
Molar Entropy (Mixture)	-26.2582	kJ/[kmol.K]
Thermal Conductivity (Mixture)	0.145004	W/[m.K]
Molar Fraction (Vapor)		
Methane	0.0913271	
Ethane	2.21477E-17	
Propane	5.03358E-19	
N-butane	0	
Isobutane	0	
Nitrogen	0.192581	
Carbon dioxide	0.00104029	
Water	0.0010691	
Oxygen	2.84824E-08	
Argon	0.0022842	
Hydrogen sulfide	0	
Hydrogen	0.711699	

Property	Value	Unit
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Density (Vapor)	28.836	kg/m3
Molecular Weight (Vapor)	8.45094	kg/kmol
Specific Enthalpy (Vapor)	35.9724	kJ/kg
Specific Entropy (Vapor)	-3.10714	kJ/[kg.K]
Molar Enthalpy (Vapor)	304.001	kJ/kmol
Molar Entropy (Vapor)	-26.2582	kJ/[kmol.K]
Thermal Conductivity (Vapor)	0.145004	W/[m.K]
Kinematic Viscosity (Vapor)	4.09747E-07	m2/s
Dynamic Viscosity (Vapor)	1.18154E-05	Pa.s
Heat Capacity (Vapor)	3.636	kJ/[kg.K]
Heat Capacity Ratio (Vapor)	1.46301	
Mass Flow (Vapor)	43250.5	kg/h
Molar Flow (Vapor)	5117.83	kmol/h
Volumetric Flow (Vapor)	1499.88	m3/h
Compressibility Factor (Vapor)	1.02435	
Molar Fraction (Vapor)	1	
Mass Fraction (Vapor)	1	
Volumetric Fraction (Vapor)	1	
Molar Fraction (Overall Liquid)		
Methane	NaN	
Ethane	NaN	
Propane	NaN	
N-butane	NaN	
Isobutane	NaN	
Nitrogen	NaN	
Carbon dioxide	NaN	
Water	NaN	
Oxygen	NaN	
Argon	NaN	
Hydrogen sulfide	NaN	
Hydrogen	NaN	
Carbonyl sulfide	NaN	
Carbon monoxide	NaN	
Ammonia	NaN	
Density (Overall Liquid)	0	kg/m3
Molecular Weight (Overall Liquid)	NaN	kg/kmol
Specific Enthalpy (Overall Liquid)	0	kJ/kg
Specific Entropy (Overall Liquid)	0	kJ/[kg.K]
Molar Enthalpy (Overall Liquid)	NaN	kJ/kmol

Property	Value	Unit
Molar Entropy (Overall Liquid)	NaN	kJ/[kmol.K]
Thermal Conductivity (Overall Liquid)	0	W/[m.K]
Kinematic Viscosity (Overall Liquid)	0	m2/s
Dynamic Viscosity (Overall Liquid)	0	Pa.s
Heat Capacity (Overall Liquid)	0	kJ/[kg.K]
Heat Capacity Ratio (Overall Liquid)	NaN	
Mass Flow (Overall Liquid)	0	kg/h
Molar Flow (Overall Liquid)	0	kmol/h
Volumetric Flow (Overall Liquid)	0	m3/h
Compressibility Factor (Overall Liquid)	0	
Molar Fraction (Overall Liquid)	0	
Mass Fraction (Overall Liquid)	0	
Volumetric Fraction (Overall Liquid)	0	
Molar Fraction (Liquid 1)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Molar Fraction (Liquid 2)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	

Property	Value	Unit
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Molar Fraction (Aqueous)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Molar Fraction (Solid)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	

WATER-OUT-1002 (Material Stream)

Property	Value	Unit
Temperature	40	C
Pressure	91	bar
Mass Flow	43.2782	kg/h
Molar Flow	2.40225	kmol/h
Volumetric Flow	0.0514571	m3/h
Mixture Molar Fraction		
Methane	6.92306E-08	
Ethane	3.94628E-25	
Propane	4.54205E-29	
N-butane	0	
Isobutane	0	
Nitrogen	3.10255E-06	
Carbon dioxide	3.36201E-05	
Water	0.99993	
Oxygen	8.18602E-12	
Argon	6.30319E-07	
Hydrogen sulfide	0	
Hydrogen	3.28732E-05	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Density (Mixture)	841.053	kg/m3
Molecular Weight (Mixture)	18.0157	kg/kmol
Specific Enthalpy (Mixture)	-2460.77	kJ/kg
Specific Entropy (Mixture)	-6.61825	kJ/[kg.K]
Molar Enthalpy (Mixture)	-44332.5	kJ/kmol
Molar Entropy (Mixture)	-119.232	kJ/[kmol.K]
Thermal Conductivity (Mixture)	0.630814	W/[m.K]
Molar Fraction (Vapor)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	

Property	Value	Unit
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Density (Vapor)	0	kg/m3
Molecular Weight (Vapor)	0	kg/kmol
Specific Enthalpy (Vapor)	0	kJ/kg
Specific Entropy (Vapor)	0	kJ/[kg.K]
Molar Enthalpy (Vapor)	0	kJ/kmol
Molar Entropy (Vapor)	0	kJ/[kmol.K]
Thermal Conductivity (Vapor)	0	W/[m.K]
Kinematic Viscosity (Vapor)	0	m2/s
Dynamic Viscosity (Vapor)	0	Pa.s
Heat Capacity (Vapor)	0	kJ/[kg.K]
Heat Capacity Ratio (Vapor)	NaN	
Mass Flow (Vapor)	0	kg/h
Molar Flow (Vapor)	0	kmol/h
Volumetric Flow (Vapor)	0	m3/h
Compressibility Factor (Vapor)	0	
Molar Fraction (Vapor)	0	
Mass Fraction (Vapor)	0	
Volumetric Fraction (Vapor)	0	
Molar Fraction (Overall Liquid)		
Methane	6.92306E-08	
Ethane	3.94628E-25	
Propane	4.54205E-29	
N-butane	0	
Isobutane	0	
Nitrogen	3.10255E-06	
Carbon dioxide	3.36201E-05	
Water	0.99993	
Oxygen	8.18602E-12	
Argon	6.30319E-07	
Hydrogen sulfide	0	
Hydrogen	3.28732E-05	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Density (Overall Liquid)	841.053	kg/m3
Molecular Weight (Overall Liquid)	18.0157	kg/kmol
Specific Enthalpy (Overall Liquid)	-2460.77	kJ/kg
Specific Entropy (Overall Liquid)	-6.61825	kJ/[kg.K]
Molar Enthalpy (Overall Liquid)	-44332.5	kJ/kmol

Property	Value	Unit
Molar Entropy (Overall Liquid)	-119.232	kJ/[kmol.K]
Thermal Conductivity (Overall Liquid)	0.630814	W/[m.K]
Kinematic Viscosity (Overall Liquid)	7.89499E-07	m2/s
Dynamic Viscosity (Overall Liquid)	0.00066401	Pa.s
Heat Capacity (Overall Liquid)	3.90915	kJ/[kg.K]
Heat Capacity Ratio (Overall Liquid)	1.30554	
Mass Flow (Overall Liquid)	43.2782	kg/h
Molar Flow (Overall Liquid)	2.40225	kmol/h
Volumetric Flow (Overall Liquid)	0.0514571	m3/h
Compressibility Factor (Overall Liquid)	0.0748697	
Molar Fraction (Overall Liquid)	1	
Mass Fraction (Overall Liquid)	1	
Volumetric Fraction (Overall Liquid)	1	
Molar Fraction (Liquid 1)		
Methane	6.92306E-08	
Ethane	3.94628E-25	
Propane	4.54205E-29	
N-butane	0	
Isobutane	0	
Nitrogen	3.10255E-06	
Carbon dioxide	3.36201E-05	
Water	0.99993	
Oxygen	8.18602E-12	
Argon	6.30319E-07	
Hydrogen sulfide	0	
Hydrogen	3.28732E-05	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Molar Fraction (Liquid 2)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	

Property	Value	Unit
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Molar Fraction (Aqueous)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Molar Fraction (Solid)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	

SYNGAS-1004 (Material Stream)

Property	Value	Unit
Temperature	342.011	C
Pressure	150	bar
Mass Flow	43250.5	kg/h
Molar Flow	5117.83	kmol/h
Volumetric Flow	39.6985	m3/h
Mixture Molar Fraction		
Methane	0.0913271	
Ethane	2.21477E-17	
Propane	5.03358E-19	
N-butane	0	
Isobutane	0	
Nitrogen	0.192581	
Carbon dioxide	0.00104029	
Water	0.0010691	
Oxygen	2.84824E-08	
Argon	0.0022842	
Hydrogen sulfide	0	
Hydrogen	0.711699	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Density (Mixture)	1089.47	kg/m3
Molecular Weight (Mixture)	8.45094	kg/kmol
Specific Enthalpy (Mixture)	0	kJ/kg
Specific Entropy (Mixture)	0	kJ/[kg.K]
Molar Enthalpy (Mixture)	0	kJ/kmol
Molar Entropy (Mixture)	0	kJ/[kmol.K]
Thermal Conductivity (Mixture)	0.00018554	W/[m.K]
Molar Fraction (Vapor)		
Methane	0.0918865	
Ethane	2.25525E-17	
Propane	5.11605E-19	
N-butane	0	
Isobutane	0	
Nitrogen	0.189792	
Carbon dioxide	0.00107531	
Water	0.00118114	
Oxygen	2.87128E-08	
Argon	0.00230485	
Hydrogen sulfide	0	
Hydrogen	0.71376	

Property	Value	Unit
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Density (Vapor)	0	kg/m3
Molecular Weight (Vapor)	0	kg/kmol
Specific Enthalpy (Vapor)	0	kJ/kg
Specific Entropy (Vapor)	0	kJ/[kg.K]
Molar Enthalpy (Vapor)	0	kJ/kmol
Molar Entropy (Vapor)	0	kJ/[kmol.K]
Thermal Conductivity (Vapor)	0	W/[m.K]
Kinematic Viscosity (Vapor)	0	m2/s
Dynamic Viscosity (Vapor)	0	Pa.s
Heat Capacity (Vapor)	0	kJ/[kg.K]
Heat Capacity Ratio (Vapor)	NaN	
Mass Flow (Vapor)	0	kg/h
Molar Flow (Vapor)	0	kmol/h
Volumetric Flow (Vapor)	0	m3/h
Compressibility Factor (Vapor)	0	
Molar Fraction (Vapor)	0	
Mass Fraction (Vapor)	0	
Volumetric Fraction (Vapor)	0	
Molar Fraction (Overall Liquid)		
Methane	0.0913271	
Ethane	2.21477E-17	
Propane	5.03358E-19	
N-butane	0	
Isobutane	0	
Nitrogen	0.192581	
Carbon dioxide	0.00104029	
Water	0.0010691	
Oxygen	2.84824E-08	
Argon	0.0022842	
Hydrogen sulfide	0	
Hydrogen	0.711699	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Density (Overall Liquid)	1089.47	kg/m3
Molecular Weight (Overall Liquid)	8.45094	kg/kmol
Specific Enthalpy (Overall Liquid)	0	kJ/kg
Specific Entropy (Overall Liquid)	0	kJ/[kg.K]
Molar Enthalpy (Overall Liquid)	0	kJ/kmol

Property	Value	Unit
Molar Entropy (Overall Liquid)	0	kJ/[kmol.K]
Thermal Conductivity (Overall Liquid)	0.00018554	W/[m.K]
Kinematic Viscosity (Overall Liquid)	6.44392E-11	m2/s
Dynamic Viscosity (Overall Liquid)	7.02049E-08	Pa.s
Heat Capacity (Overall Liquid)	0	kJ/[kg.K]
Heat Capacity Ratio (Overall Liquid)	NaN	
Mass Flow (Overall Liquid)	43250.5	kg/h
Molar Flow (Overall Liquid)	5117.83	kmol/h
Volumetric Flow (Overall Liquid)	39.6985	m3/h
Compressibility Factor (Overall Liquid)	0.02275	
Molar Fraction (Overall Liquid)	1	
Mass Fraction (Overall Liquid)	1	
Volumetric Fraction (Overall Liquid)	1	
Molar Fraction (Liquid 1)		
Methane	0.0913271	
Ethane	2.21477E-17	
Propane	5.03358E-19	
N-butane	0	
Isobutane	0	
Nitrogen	0.192581	
Carbon dioxide	0.00104029	
Water	0.0010691	
Oxygen	2.84824E-08	
Argon	0.0022842	
Hydrogen sulfide	0	
Hydrogen	0.711699	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Molar Fraction (Liquid 2)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	

Property	Value	Unit
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Molar Fraction (Aqueous)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Molar Fraction (Solid)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	

IC-103 (Material Stream)

Property	Value	Unit
Temperature	40	C
Pressure	149	bar
Mass Flow	43250.5	kg/h
Molar Flow	5117.83	kmol/h
Volumetric Flow	937.689	m3/h
Mixture Molar Fraction		
Methane	0.0913271	
Ethane	2.21477E-17	
Propane	5.03358E-19	
N-butane	0	
Isobutane	0	
Nitrogen	0.192581	
Carbon dioxide	0.00104029	
Water	0.0010691	
Oxygen	2.84824E-08	
Argon	0.0022842	
Hydrogen sulfide	0	
Hydrogen	0.711699	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Density (Mixture)	46.1246	kg/m3
Molecular Weight (Mixture)	8.45094	kg/kmol
Specific Enthalpy (Mixture)	28.9351	kJ/kg
Specific Entropy (Mixture)	-3.62894	kJ/[kg.K]
Molar Enthalpy (Mixture)	244.529	kJ/kmol
Molar Entropy (Mixture)	-30.6679	kJ/[kmol.K]
Thermal Conductivity (Mixture)	0.148436	W/[m.K]
Molar Fraction (Vapor)		
Methane	0.09135	
Ethane	2.21533E-17	
Propane	5.03484E-19	
N-butane	0	
Isobutane	0	
Nitrogen	0.192629	
Carbon dioxide	0.00104054	
Water	0.0008189	
Oxygen	2.84895E-08	
Argon	0.00228478	
Hydrogen sulfide	0	
Hydrogen	0.711877	

Property	Value	Unit
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Density (Vapor)	46.1013	kg/m3
Molecular Weight (Vapor)	8.44855	kg/kmol
Specific Enthalpy (Vapor)	30.262	kJ/kg
Specific Entropy (Vapor)	-3.62734	kJ/[kg.K]
Molar Enthalpy (Vapor)	255.67	kJ/kmol
Molar Entropy (Vapor)	-30.6457	kJ/[kmol.K]
Thermal Conductivity (Vapor)	0.148315	W/[m.K]
Kinematic Viscosity (Vapor)	2.66542E-07	m2/s
Dynamic Viscosity (Vapor)	1.22879E-05	Pa.s
Heat Capacity (Vapor)	3.69654	kJ/[kg.K]
Heat Capacity Ratio (Vapor)	1.48904	
Mass Flow (Vapor)	43227.4	kg/h
Molar Flow (Vapor)	5116.55	kmol/h
Volumetric Flow (Vapor)	937.662	m3/h
Compressibility Factor (Vapor)	1.0488	
Molar Fraction (Vapor)	0.99975	
Mass Fraction (Vapor)	0.999466	
Volumetric Fraction (Vapor)	0.999971	
Molar Fraction (Overall Liquid)		
Methane	1.01558E-07	
Ethane	5.22429E-25	
Propane	5.50881E-29	
N-butane	0	
Isobutane	0	
Nitrogen	4.76525E-06	
Carbon dioxide	4.64156E-05	
Water	0.999895	
Oxygen	1.24295E-11	
Argon	9.57648E-07	
Hydrogen sulfide	0	
Hydrogen	5.23252E-05	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Density (Overall Liquid)	841.708	kg/m3
Molecular Weight (Overall Liquid)	18.0157	kg/kmol
Specific Enthalpy (Overall Liquid)	-2455.23	kJ/kg
Specific Entropy (Overall Liquid)	-6.62202	kJ/[kg.K]
Molar Enthalpy (Overall Liquid)	-44232.7	kJ/kmol

Property	Value	Unit
Molar Entropy (Overall Liquid)	-119.3	kJ/[kmol.K]
Thermal Conductivity (Overall Liquid)	0.63076	W/[m.K]
Kinematic Viscosity (Overall Liquid)	7.9721E-07	m2/s
Dynamic Viscosity (Overall Liquid)	0.000671018	Pa.s
Heat Capacity (Overall Liquid)	3.90149	kJ/[kg.K]
Heat Capacity Ratio (Overall Liquid)	1.30323	
Mass Flow (Overall Liquid)	23.0899	kg/h
Molar Flow (Overall Liquid)	1.28165	kmol/h
Volumetric Flow (Overall Liquid)	0.0274322	m3/h
Compressibility Factor (Overall Liquid)	0.122494	
Molar Fraction (Overall Liquid)	0.000250429	
Mass Fraction (Overall Liquid)	0.000533864	
Volumetric Fraction (Overall Liquid)	2.92551E-05	
Molar Fraction (Liquid 1)		
Methane	1.01558E-07	
Ethane	5.22429E-25	
Propane	5.50881E-29	
N-butane	0	
Isobutane	0	
Nitrogen	4.76525E-06	
Carbon dioxide	4.64156E-05	
Water	0.999895	
Oxygen	1.24295E-11	
Argon	9.57648E-07	
Hydrogen sulfide	0	
Hydrogen	5.23252E-05	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Molar Fraction (Liquid 2)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	

Property	Value	Unit
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Molar Fraction (Aqueous)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Molar Fraction (Solid)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	

RECYCLE-GAS-1201 (Material Stream)

Property	Value	Unit
Temperature	30.0015	C
Pressure	145	bar
Mass Flow	36640.4	kg/h
Molar Flow	3792.96	kmol/h
Volumetric Flow	642.948	m3/h
Mixture Molar Fraction		
Methane	0.118325	
Ethane	2.77702E-17	
Propane	6.06032E-19	
N-butane	0	
Isobutane	0	
Nitrogen	0.138891	
Carbon dioxide	0.00126157	
Water	4.02081E-06	
Oxygen	7.47993E-14	
Argon	0.00294551	
Hydrogen sulfide	0	
Hydrogen	0.591454	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0.14712	
Density (Mixture)	56.9881	kg/m3
Molecular Weight (Mixture)	9.66011	kg/kmol
Specific Enthalpy (Mixture)	-61.1639	kJ/kg
Specific Entropy (Mixture)	-2.78941	kJ/[kg.K]
Molar Enthalpy (Mixture)	-590.85	kJ/kmol
Molar Entropy (Mixture)	-26.946	kJ/[kmol.K]
Thermal Conductivity (Mixture)	0.130784	W/[m.K]
Molar Fraction (Vapor)		
Methane	0.118325	
Ethane	2.77702E-17	
Propane	6.06032E-19	
N-butane	0	
Isobutane	0	
Nitrogen	0.138891	
Carbon dioxide	0.00126157	
Water	3.9785E-06	
Oxygen	7.47995E-14	
Argon	0.00294552	
Hydrogen sulfide	0	
Hydrogen	0.591455	

Property	Value	Unit
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0.147117	
Density (Vapor)	56.9878	kg/m3
Molecular Weight (Vapor)	9.66008	kg/kmol
Specific Enthalpy (Vapor)	-61.1573	kJ/kg
Specific Entropy (Vapor)	-2.7894	kJ/[kg.K]
Molar Enthalpy (Vapor)	-590.785	kJ/kmol
Molar Entropy (Vapor)	-26.9459	kJ/[kmol.K]
Thermal Conductivity (Vapor)	0.130783	W/[m.K]
Kinematic Viscosity (Vapor)	2.17623E-07	m2/s
Dynamic Viscosity (Vapor)	1.24018E-05	Pa.s
Heat Capacity (Vapor)	3.6305	kJ/[kg.K]
Heat Capacity Ratio (Vapor)	1.58876	
Mass Flow (Vapor)	36640.2	kg/h
Molar Flow (Vapor)	3792.94	kmol/h
Volumetric Flow (Vapor)	642.948	m3/h
Compressibility Factor (Vapor)	0.975208	
Molar Fraction (Vapor)	0.999997	
Mass Fraction (Vapor)	0.999994	
Volumetric Fraction (Vapor)	0.999999	
Molar Fraction (Overall Liquid)		
Methane	0.0144063	
Ethane	1.47788E-17	
Propane	6.41408E-19	
N-butane	0	
Isobutane	0	
Nitrogen	0.00234315	
Carbon dioxide	0.00121146	
Water	0.0127787	
Oxygen	7.02883E-15	
Argon	0.000534263	
Hydrogen sulfide	0	
Hydrogen	0.0144582	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0.954268	
Density (Overall Liquid)	589.604	kg/m3
Molecular Weight (Overall Liquid)	16.8824	kg/kmol
Specific Enthalpy (Overall Liquid)	-1193.75	kJ/kg
Specific Entropy (Overall Liquid)	-4.8817	kJ/[kg.K]
Molar Enthalpy (Overall Liquid)	-20153.5	kJ/kmol

Property	Value	Unit
Molar Entropy (Overall Liquid)	-82.4151	kJ/[kmol.K]
Thermal Conductivity (Overall Liquid)	0.437291	W/[m.K]
Kinematic Viscosity (Overall Liquid)	2.34273E-07	m2/s
Dynamic Viscosity (Overall Liquid)	0.000138128	Pa.s
Heat Capacity (Overall Liquid)	4.48079	kJ/[kg.K]
Heat Capacity Ratio (Overall Liquid)	1.71934	
Mass Flow (Overall Liquid)	0.212151	kg/h
Molar Flow (Overall Liquid)	0.0125664	kmol/h
Volumetric Flow (Overall Liquid)	0.00035982	m3/h
Compressibility Factor (Overall Liquid)	0.18111	
Molar Fraction (Overall Liquid)	3.31308E-06	
Mass Fraction (Overall Liquid)	5.7901E-06	
Volumetric Fraction (Overall Liquid)	5.59641E-07	
Molar Fraction (Liquid 1)		
Methane	0.0144063	
Ethane	1.47788E-17	
Propane	6.41408E-19	
N-butane	0	
Isobutane	0	
Nitrogen	0.00234315	
Carbon dioxide	0.00121146	
Water	0.0127787	
Oxygen	7.02883E-15	
Argon	0.000534263	
Hydrogen sulfide	0	
Hydrogen	0.0144582	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0.954268	
Molar Fraction (Liquid 2)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	

Property	Value	Unit
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Molar Fraction (Aqueous)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Molar Fraction (Solid)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	

SYNGAS-1005 (Material Stream)

Property	Value	Unit
Temperature	40	C
Pressure	149	bar
Mass Flow	43227.4	kg/h
Molar Flow	5116.55	kmol/h
Volumetric Flow	937.662	m3/h
Mixture Molar Fraction		
Methane	0.09135	
Ethane	2.21533E-17	
Propane	5.03484E-19	
N-butane	0	
Isobutane	0	
Nitrogen	0.192629	
Carbon dioxide	0.00104054	
Water	0.0008189	
Oxygen	2.84895E-08	
Argon	0.00228478	
Hydrogen sulfide	0	
Hydrogen	0.711877	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Density (Mixture)	46.1013	kg/m3
Molecular Weight (Mixture)	8.44855	kg/kmol
Specific Enthalpy (Mixture)	30.262	kJ/kg
Specific Entropy (Mixture)	-3.62734	kJ/[kg.K]
Molar Enthalpy (Mixture)	255.67	kJ/kmol
Molar Entropy (Mixture)	-30.6457	kJ/[kmol.K]
Thermal Conductivity (Mixture)	0.148315	W/[m.K]
Molar Fraction (Vapor)		
Methane	0.09135	
Ethane	2.21533E-17	
Propane	5.03484E-19	
N-butane	0	
Isobutane	0	
Nitrogen	0.192629	
Carbon dioxide	0.00104054	
Water	0.0008189	
Oxygen	2.84895E-08	
Argon	0.00228478	
Hydrogen sulfide	0	
Hydrogen	0.711877	

Property	Value	Unit
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Density (Vapor)	46.1013	kg/m3
Molecular Weight (Vapor)	8.44855	kg/kmol
Specific Enthalpy (Vapor)	30.262	kJ/kg
Specific Entropy (Vapor)	-3.62734	kJ/[kg.K]
Molar Enthalpy (Vapor)	255.67	kJ/kmol
Molar Entropy (Vapor)	-30.6457	kJ/[kmol.K]
Thermal Conductivity (Vapor)	0.148315	W/[m.K]
Kinematic Viscosity (Vapor)	2.66542E-07	m2/s
Dynamic Viscosity (Vapor)	1.22879E-05	Pa.s
Heat Capacity (Vapor)	3.69654	kJ/[kg.K]
Heat Capacity Ratio (Vapor)	1.48904	
Mass Flow (Vapor)	43227.4	kg/h
Molar Flow (Vapor)	5116.55	kmol/h
Volumetric Flow (Vapor)	937.662	m3/h
Compressibility Factor (Vapor)	1.0488	
Molar Fraction (Vapor)	1	
Mass Fraction (Vapor)	1	
Volumetric Fraction (Vapor)	1	
Molar Fraction (Overall Liquid)		
Methane	NaN	
Ethane	NaN	
Propane	NaN	
N-butane	NaN	
Isobutane	NaN	
Nitrogen	NaN	
Carbon dioxide	NaN	
Water	NaN	
Oxygen	NaN	
Argon	NaN	
Hydrogen sulfide	NaN	
Hydrogen	NaN	
Carbonyl sulfide	NaN	
Carbon monoxide	NaN	
Ammonia	NaN	
Density (Overall Liquid)	0	kg/m3
Molecular Weight (Overall Liquid)	NaN	kg/kmol
Specific Enthalpy (Overall Liquid)	0	kJ/kg
Specific Entropy (Overall Liquid)	0	kJ/[kg.K]
Molar Enthalpy (Overall Liquid)	NaN	kJ/kmol

Property	Value	Unit
Molar Entropy (Overall Liquid)	NaN	kJ/[kmol.K]
Thermal Conductivity (Overall Liquid)	0	W/[m.K]
Kinematic Viscosity (Overall Liquid)	0	m2/s
Dynamic Viscosity (Overall Liquid)	0	Pa.s
Heat Capacity (Overall Liquid)	0	kJ/[kg.K]
Heat Capacity Ratio (Overall Liquid)	NaN	
Mass Flow (Overall Liquid)	0	kg/h
Molar Flow (Overall Liquid)	0	kmol/h
Volumetric Flow (Overall Liquid)	0	m3/h
Compressibility Factor (Overall Liquid)	0	
Molar Fraction (Overall Liquid)	0	
Mass Fraction (Overall Liquid)	0	
Volumetric Fraction (Overall Liquid)	0	
Molar Fraction (Liquid 1)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Molar Fraction (Liquid 2)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	

Property	Value	Unit
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Molar Fraction (Aqueous)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Molar Fraction (Solid)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	

WATER OUT -1003 (Material Stream)

Property	Value	Unit
Temperature	40	C
Pressure	149	bar
Mass Flow	23.0899	kg/h
Molar Flow	1.28165	kmol/h
Volumetric Flow	0.0274322	m3/h
Mixture Molar Fraction		
Methane	1.01558E-07	
Ethane	5.22429E-25	
Propane	5.50881E-29	
N-butane	0	
Isobutane	0	
Nitrogen	4.76525E-06	
Carbon dioxide	4.64156E-05	
Water	0.999895	
Oxygen	1.24295E-11	
Argon	9.57648E-07	
Hydrogen sulfide	0	
Hydrogen	5.23252E-05	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Density (Mixture)	841.708	kg/m3
Molecular Weight (Mixture)	18.0157	kg/kmol
Specific Enthalpy (Mixture)	-2455.23	kJ/kg
Specific Entropy (Mixture)	-6.62202	kJ/[kg.K]
Molar Enthalpy (Mixture)	-44232.7	kJ/kmol
Molar Entropy (Mixture)	-119.3	kJ/[kmol.K]
Thermal Conductivity (Mixture)	0.63076	W/[m.K]
Molar Fraction (Vapor)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	

Property	Value	Unit
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Density (Vapor)	0	kg/m3
Molecular Weight (Vapor)	0	kg/kmol
Specific Enthalpy (Vapor)	0	kJ/kg
Specific Entropy (Vapor)	0	kJ/[kg.K]
Molar Enthalpy (Vapor)	0	kJ/kmol
Molar Entropy (Vapor)	0	kJ/[kmol.K]
Thermal Conductivity (Vapor)	0	W/[m.K]
Kinematic Viscosity (Vapor)	0	m2/s
Dynamic Viscosity (Vapor)	0	Pa.s
Heat Capacity (Vapor)	0	kJ/[kg.K]
Heat Capacity Ratio (Vapor)	NaN	
Mass Flow (Vapor)	0	kg/h
Molar Flow (Vapor)	0	kmol/h
Volumetric Flow (Vapor)	0	m3/h
Compressibility Factor (Vapor)	0	
Molar Fraction (Vapor)	0	
Mass Fraction (Vapor)	0	
Volumetric Fraction (Vapor)	0	
Molar Fraction (Overall Liquid)		
Methane	1.01558E-07	
Ethane	5.22429E-25	
Propane	5.50881E-29	
N-butane	0	
Isobutane	0	
Nitrogen	4.76525E-06	
Carbon dioxide	4.64156E-05	
Water	0.999895	
Oxygen	1.24295E-11	
Argon	9.57648E-07	
Hydrogen sulfide	0	
Hydrogen	5.23252E-05	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Density (Overall Liquid)	841.708	kg/m3
Molecular Weight (Overall Liquid)	18.0157	kg/kmol
Specific Enthalpy (Overall Liquid)	-2455.23	kJ/kg
Specific Entropy (Overall Liquid)	-6.62202	kJ/[kg.K]
Molar Enthalpy (Overall Liquid)	-44232.7	kJ/kmol

Property	Value	Unit
Molar Entropy (Overall Liquid)	-119.3	kJ/[kmol.K]
Thermal Conductivity (Overall Liquid)	0.63076	W/[m.K]
Kinematic Viscosity (Overall Liquid)	7.9721E-07	m ² /s
Dynamic Viscosity (Overall Liquid)	0.000671018	Pa.s
Heat Capacity (Overall Liquid)	3.90149	kJ/[kg.K]
Heat Capacity Ratio (Overall Liquid)	1.30323	
Mass Flow (Overall Liquid)	23.0899	kg/h
Molar Flow (Overall Liquid)	1.28165	kmol/h
Volumetric Flow (Overall Liquid)	0.0274322	m ³ /h
Compressibility Factor (Overall Liquid)	0.122494	
Molar Fraction (Overall Liquid)	1	
Mass Fraction (Overall Liquid)	1	
Volumetric Fraction (Overall Liquid)	1	
Molar Fraction (Liquid 1)		
Methane	1.01558E-07	
Ethane	5.22429E-25	
Propane	5.50881E-29	
N-butane	0	
Isobutane	0	
Nitrogen	4.76525E-06	
Carbon dioxide	4.64156E-05	
Water	0.999895	
Oxygen	1.24295E-11	
Argon	9.57648E-07	
Hydrogen sulfide	0	
Hydrogen	5.23252E-05	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Molar Fraction (Liquid 2)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	

Property	Value	Unit
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Molar Fraction (Aqueous)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Molar Fraction (Solid)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	

RECYCLE-1201 (Material Stream)

Property	Value	Unit
Temperature	30.0015	C
Pressure	145	bar
Mass Flow	36640.4	kg/h
Molar Flow	3792.96	kmol/h
Volumetric Flow	642.948	m3/h
Mixture Molar Fraction		
Methane	0.118325	
Ethane	2.77702E-17	
Propane	6.06032E-19	
N-butane	0	
Isobutane	0	
Nitrogen	0.138891	
Carbon dioxide	0.00126157	
Water	4.02081E-06	
Oxygen	7.47993E-14	
Argon	0.00294551	
Hydrogen sulfide	0	
Hydrogen	0.591454	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0.14712	
Density (Mixture)	56.9881	kg/m3
Molecular Weight (Mixture)	9.66011	kg/kmol
Specific Enthalpy (Mixture)	-61.1639	kJ/kg
Specific Entropy (Mixture)	-2.78941	kJ/[kg.K]
Molar Enthalpy (Mixture)	-590.85	kJ/kmol
Molar Entropy (Mixture)	-26.946	kJ/[kmol.K]
Thermal Conductivity (Mixture)	0.130784	W/[m.K]
Molar Fraction (Vapor)		
Methane	0.118325	
Ethane	2.77702E-17	
Propane	6.06032E-19	
N-butane	0	
Isobutane	0	
Nitrogen	0.138891	
Carbon dioxide	0.00126157	
Water	3.9785E-06	
Oxygen	7.47995E-14	
Argon	0.00294552	
Hydrogen sulfide	0	
Hydrogen	0.591455	

Property	Value	Unit
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0.147117	
Density (Vapor)	56.9878	kg/m3
Molecular Weight (Vapor)	9.66008	kg/kmol
Specific Enthalpy (Vapor)	-61.1573	kJ/kg
Specific Entropy (Vapor)	-2.7894	kJ/[kg.K]
Molar Enthalpy (Vapor)	-590.785	kJ/kmol
Molar Entropy (Vapor)	-26.9459	kJ/[kmol.K]
Thermal Conductivity (Vapor)	0.130783	W/[m.K]
Kinematic Viscosity (Vapor)	2.17623E-07	m2/s
Dynamic Viscosity (Vapor)	1.24018E-05	Pa.s
Heat Capacity (Vapor)	3.6305	kJ/[kg.K]
Heat Capacity Ratio (Vapor)	1.58876	
Mass Flow (Vapor)	36640.2	kg/h
Molar Flow (Vapor)	3792.94	kmol/h
Volumetric Flow (Vapor)	642.948	m3/h
Compressibility Factor (Vapor)	0.975208	
Molar Fraction (Vapor)	0.999997	
Mass Fraction (Vapor)	0.999994	
Volumetric Fraction (Vapor)	0.999999	
Molar Fraction (Overall Liquid)		
Methane	0.0144063	
Ethane	1.47788E-17	
Propane	6.41408E-19	
N-butane	0	
Isobutane	0	
Nitrogen	0.00234315	
Carbon dioxide	0.00121146	
Water	0.0127787	
Oxygen	7.02883E-15	
Argon	0.000534263	
Hydrogen sulfide	0	
Hydrogen	0.0144582	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0.954268	
Density (Overall Liquid)	589.604	kg/m3
Molecular Weight (Overall Liquid)	16.8824	kg/kmol
Specific Enthalpy (Overall Liquid)	-1193.75	kJ/kg
Specific Entropy (Overall Liquid)	-4.8817	kJ/[kg.K]
Molar Enthalpy (Overall Liquid)	-20153.5	kJ/kmol

Property	Value	Unit
Molar Entropy (Overall Liquid)	-82.4151	kJ/[kmol.K]
Thermal Conductivity (Overall Liquid)	0.437291	W/[m.K]
Kinematic Viscosity (Overall Liquid)	2.34273E-07	m2/s
Dynamic Viscosity (Overall Liquid)	0.000138128	Pa.s
Heat Capacity (Overall Liquid)	4.48079	kJ/[kg.K]
Heat Capacity Ratio (Overall Liquid)	1.71934	
Mass Flow (Overall Liquid)	0.212151	kg/h
Molar Flow (Overall Liquid)	0.0125664	kmol/h
Volumetric Flow (Overall Liquid)	0.00035982	m3/h
Compressibility Factor (Overall Liquid)	0.18111	
Molar Fraction (Overall Liquid)	3.31308E-06	
Mass Fraction (Overall Liquid)	5.7901E-06	
Volumetric Fraction (Overall Liquid)	5.59641E-07	
Molar Fraction (Liquid 1)		
Methane	0.0144063	
Ethane	1.47788E-17	
Propane	6.41408E-19	
N-butane	0	
Isobutane	0	
Nitrogen	0.00234315	
Carbon dioxide	0.00121146	
Water	0.0127787	
Oxygen	7.02883E-15	
Argon	0.000534263	
Hydrogen sulfide	0	
Hydrogen	0.0144582	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0.954268	
Molar Fraction (Liquid 2)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	

Property	Value	Unit
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Molar Fraction (Aqueous)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Molar Fraction (Solid)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	

SYNGAS-MIX-1101 (Material Stream)

Property	Value	Unit
Temperature	40	C
Pressure	149	bar
Mass Flow	43227.4	kg/h
Molar Flow	5116.55	kmol/h
Volumetric Flow	937.662	m3/h
Mixture Molar Fraction		
Methane	0.09135	
Ethane	2.21533E-17	
Propane	5.03484E-19	
N-butane	0	
Isobutane	0	
Nitrogen	0.192629	
Carbon dioxide	0.00104054	
Water	0.0008189	
Oxygen	2.84895E-08	
Argon	0.00228478	
Hydrogen sulfide	0	
Hydrogen	0.711877	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Density (Mixture)	46.1013	kg/m3
Molecular Weight (Mixture)	8.44855	kg/kmol
Specific Enthalpy (Mixture)	30.2619	kJ/kg
Specific Entropy (Mixture)	-3.62734	kJ/[kg.K]
Molar Enthalpy (Mixture)	255.669	kJ/kmol
Molar Entropy (Mixture)	-30.6457	kJ/[kmol.K]
Thermal Conductivity (Mixture)	0.148315	W/[m.K]
Molar Fraction (Vapor)		
Methane	0.09135	
Ethane	2.21533E-17	
Propane	5.03484E-19	
N-butane	0	
Isobutane	0	
Nitrogen	0.192629	
Carbon dioxide	0.00104054	
Water	0.000818881	
Oxygen	2.84895E-08	
Argon	0.00228478	
Hydrogen sulfide	0	
Hydrogen	0.711877	

Property	Value	Unit
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Density (Vapor)	46.1013	kg/m ³
Molecular Weight (Vapor)	8.44855	kg/kmol
Specific Enthalpy (Vapor)	30.262	kJ/kg
Specific Entropy (Vapor)	-3.62734	kJ/[kg.K]
Molar Enthalpy (Vapor)	255.67	kJ/kmol
Molar Entropy (Vapor)	-30.6457	kJ/[kmol.K]
Thermal Conductivity (Vapor)	0.148315	W/[m.K]
Kinematic Viscosity (Vapor)	2.66542E-07	m ² /s
Dynamic Viscosity (Vapor)	1.22879E-05	Pa.s
Heat Capacity (Vapor)	3.69654	kJ/[kg.K]
Heat Capacity Ratio (Vapor)	1.48904	
Mass Flow (Vapor)	43227.4	kg/h
Molar Flow (Vapor)	5116.55	kmol/h
Volumetric Flow (Vapor)	937.662	m ³ /h
Compressibility Factor (Vapor)	1.0488	
Molar Fraction (Vapor)	1	
Mass Fraction (Vapor)	1	
Volumetric Fraction (Vapor)	1	
Molar Fraction (Overall Liquid)		
Methane	1.0156E-07	
Ethane	5.22441E-25	
Propane	5.50894E-29	
N-butane	0	
Isobutane	0	
Nitrogen	4.76536E-06	
Carbon dioxide	4.64167E-05	
Water	0.999895	
Oxygen	1.24298E-11	
Argon	9.5767E-07	
Hydrogen sulfide	0	
Hydrogen	5.23264E-05	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Density (Overall Liquid)	841.708	kg/m ³
Molecular Weight (Overall Liquid)	18.0157	kg/kmol
Specific Enthalpy (Overall Liquid)	-2455.23	kJ/kg
Specific Entropy (Overall Liquid)	-6.62202	kJ/[kg.K]
Molar Enthalpy (Overall Liquid)	-44232.7	kJ/kmol

Property	Value	Unit
Molar Entropy (Overall Liquid)	-119.3	kJ/[kmol.K]
Thermal Conductivity (Overall Liquid)	0.63076	W/[m.K]
Kinematic Viscosity (Overall Liquid)	7.9721E-07	m2/s
Dynamic Viscosity (Overall Liquid)	0.000671018	Pa.s
Heat Capacity (Overall Liquid)	3.90149	kJ/[kg.K]
Heat Capacity Ratio (Overall Liquid)	1.30323	
Mass Flow (Overall Liquid)	0.00174978	kg/h
Molar Flow (Overall Liquid)	9.7125E-05	kmol/h
Volumetric Flow (Overall Liquid)	2.07884E-06	m3/h
Compressibility Factor (Overall Liquid)	0.122494	
Molar Fraction (Overall Liquid)	1.89825E-08	
Mass Fraction (Overall Liquid)	4.04784E-08	
Volumetric Fraction (Overall Liquid)	2.21705E-09	
Molar Fraction (Liquid 1)		
Methane	1.0156E-07	
Ethane	5.22441E-25	
Propane	5.50894E-29	
N-butane	0	
Isobutane	0	
Nitrogen	4.76536E-06	
Carbon dioxide	4.64167E-05	
Water	0.999895	
Oxygen	1.24298E-11	
Argon	9.5767E-07	
Hydrogen sulfide	0	
Hydrogen	5.23264E-05	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Molar Fraction (Liquid 2)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	

Property	Value	Unit
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Molar Fraction (Aqueous)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Molar Fraction (Solid)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	

SYNGAS-1101 (Material Stream)

Property	Value	Unit
Temperature	40.7932	C
Pressure	150	bar
Mass Flow	43227.4	kg/h
Molar Flow	5116.55	kmol/h
Volumetric Flow	934.248	m3/h
Mixture Molar Fraction		
Methane	0.09135	
Ethane	2.21533E-17	
Propane	5.03484E-19	
N-butane	0	
Isobutane	0	
Nitrogen	0.192629	
Carbon dioxide	0.00104054	
Water	0.0008189	
Oxygen	2.84895E-08	
Argon	0.00228478	
Hydrogen sulfide	0	
Hydrogen	0.711877	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Density (Mixture)	46.2697	kg/m3
Molecular Weight (Mixture)	8.44855	kg/kmol
Specific Enthalpy (Mixture)	33.1273	kJ/kg
Specific Entropy (Mixture)	-3.62506	kJ/[kg.K]
Molar Enthalpy (Mixture)	279.878	kJ/kmol
Molar Entropy (Mixture)	-30.6265	kJ/[kmol.K]
Thermal Conductivity (Mixture)	0.148632	W/[m.K]
Molar Fraction (Vapor)		
Methane	0.09135	
Ethane	2.21533E-17	
Propane	5.03484E-19	
N-butane	0	
Isobutane	0	
Nitrogen	0.192629	
Carbon dioxide	0.00104054	
Water	0.0008189	
Oxygen	2.84895E-08	
Argon	0.00228478	
Hydrogen sulfide	0	
Hydrogen	0.711877	

Property	Value	Unit
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Density (Vapor)	46.2697	kg/m3
Molecular Weight (Vapor)	8.44855	kg/kmol
Specific Enthalpy (Vapor)	33.1273	kJ/kg
Specific Entropy (Vapor)	-3.62506	kJ/[kg.K]
Molar Enthalpy (Vapor)	279.878	kJ/kmol
Molar Entropy (Vapor)	-30.6265	kJ/[kmol.K]
Thermal Conductivity (Vapor)	0.148632	W/[m.K]
Kinematic Viscosity (Vapor)	2.66129E-07	m2/s
Dynamic Viscosity (Vapor)	1.23137E-05	Pa.s
Heat Capacity (Vapor)	3.69722	kJ/[kg.K]
Heat Capacity Ratio (Vapor)	1.48869	
Mass Flow (Vapor)	43227.4	kg/h
Molar Flow (Vapor)	5116.55	kmol/h
Volumetric Flow (Vapor)	934.248	m3/h
Compressibility Factor (Vapor)	1.04934	
Molar Fraction (Vapor)	1	
Mass Fraction (Vapor)	1	
Volumetric Fraction (Vapor)	1	
Molar Fraction (Overall Liquid)		
Methane	NaN	
Ethane	NaN	
Propane	NaN	
N-butane	NaN	
Isobutane	NaN	
Nitrogen	NaN	
Carbon dioxide	NaN	
Water	NaN	
Oxygen	NaN	
Argon	NaN	
Hydrogen sulfide	NaN	
Hydrogen	NaN	
Carbonyl sulfide	NaN	
Carbon monoxide	NaN	
Ammonia	NaN	
Density (Overall Liquid)	0	kg/m3
Molecular Weight (Overall Liquid)	NaN	kg/kmol
Specific Enthalpy (Overall Liquid)	0	kJ/kg
Specific Entropy (Overall Liquid)	0	kJ/[kg.K]
Molar Enthalpy (Overall Liquid)	NaN	kJ/kmol

Property	Value	Unit
Molar Entropy (Overall Liquid)	NaN	kJ/[kmol.K]
Thermal Conductivity (Overall Liquid)	0	W/[m.K]
Kinematic Viscosity (Overall Liquid)	0	m2/s
Dynamic Viscosity (Overall Liquid)	0	Pa.s
Heat Capacity (Overall Liquid)	0	kJ/[kg.K]
Heat Capacity Ratio (Overall Liquid)	NaN	
Mass Flow (Overall Liquid)	0	kg/h
Molar Flow (Overall Liquid)	0	kmol/h
Volumetric Flow (Overall Liquid)	0	m3/h
Compressibility Factor (Overall Liquid)	0	
Molar Fraction (Overall Liquid)	0	
Mass Fraction (Overall Liquid)	0	
Volumetric Fraction (Overall Liquid)	0	
Molar Fraction (Liquid 1)		
Methane	1.11026E-07	
Ethane	5.81398E-25	
Propane	6.28375E-29	
N-butane	0	
Isobutane	0	
Nitrogen	5.12742E-06	
Carbon dioxide	4.83156E-05	
Water	0.999889	
Oxygen	1.32203E-11	
Argon	1.01933E-06	
Hydrogen sulfide	0	
Hydrogen	5.62552E-05	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Molar Fraction (Liquid 2)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	

Property	Value	Unit
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Molar Fraction (Aqueous)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Molar Fraction (Solid)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	

FEED-1101 (Material Stream)

Property	Value	Unit
Temperature	430	C
Pressure	149	bar
Mass Flow	43227.4	kg/h
Molar Flow	5116.55	kmol/h
Volumetric Flow	2090.62	m3/h
Mixture Molar Fraction		
Methane	0.09135	
Ethane	2.21533E-17	
Propane	5.03484E-19	
N-butane	0	
Isobutane	0	
Nitrogen	0.192629	
Carbon dioxide	0.00104054	
Water	0.0008189	
Oxygen	2.84895E-08	
Argon	0.00228478	
Hydrogen sulfide	0	
Hydrogen	0.711877	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Density (Mixture)	20.6768	kg/m3
Molecular Weight (Mixture)	8.44855	kg/kmol
Specific Enthalpy (Mixture)	1498.99	kJ/kg
Specific Entropy (Mixture)	-0.591273	kJ/[kg.K]
Molar Enthalpy (Mixture)	12664.3	kJ/kmol
Molar Entropy (Mixture)	-4.99539	kJ/[kmol.K]
Thermal Conductivity (Mixture)	0.268734	W/[m.K]
Molar Fraction (Vapor)		
Methane	0.09135	
Ethane	2.21533E-17	
Propane	5.03484E-19	
N-butane	0	
Isobutane	0	
Nitrogen	0.192629	
Carbon dioxide	0.00104054	
Water	0.0008189	
Oxygen	2.84895E-08	
Argon	0.00228478	
Hydrogen sulfide	0	
Hydrogen	0.711877	

Property	Value	Unit
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Density (Vapor)	20.6768	kg/m3
Molecular Weight (Vapor)	8.44855	kg/kmol
Specific Enthalpy (Vapor)	1498.99	kJ/kg
Specific Entropy (Vapor)	-0.591273	kJ/[kg.K]
Molar Enthalpy (Vapor)	12664.3	kJ/kmol
Molar Entropy (Vapor)	-4.99539	kJ/[kmol.K]
Thermal Conductivity (Vapor)	0.268734	W/[m.K]
Kinematic Viscosity (Vapor)	9.757E-07	m2/s
Dynamic Viscosity (Vapor)	2.01743E-05	Pa.s
Heat Capacity (Vapor)	3.87868	kJ/[kg.K]
Heat Capacity Ratio (Vapor)	1.35673	
Mass Flow (Vapor)	43227.4	kg/h
Molar Flow (Vapor)	5116.55	kmol/h
Volumetric Flow (Vapor)	2090.62	m3/h
Compressibility Factor (Vapor)	1.04142	
Molar Fraction (Vapor)	1	
Mass Fraction (Vapor)	1	
Volumetric Fraction (Vapor)	1	
Molar Fraction (Overall Liquid)		
Methane	NaN	
Ethane	NaN	
Propane	NaN	
N-butane	NaN	
Isobutane	NaN	
Nitrogen	NaN	
Carbon dioxide	NaN	
Water	NaN	
Oxygen	NaN	
Argon	NaN	
Hydrogen sulfide	NaN	
Hydrogen	NaN	
Carbonyl sulfide	NaN	
Carbon monoxide	NaN	
Ammonia	NaN	
Density (Overall Liquid)	0	kg/m3
Molecular Weight (Overall Liquid)	NaN	kg/kmol
Specific Enthalpy (Overall Liquid)	0	kJ/kg
Specific Entropy (Overall Liquid)	0	kJ/[kg.K]
Molar Enthalpy (Overall Liquid)	NaN	kJ/kmol

Property	Value	Unit
Molar Entropy (Overall Liquid)	NaN	kJ/[kmol.K]
Thermal Conductivity (Overall Liquid)	0	W/[m.K]
Kinematic Viscosity (Overall Liquid)	0	m2/s
Dynamic Viscosity (Overall Liquid)	0	Pa.s
Heat Capacity (Overall Liquid)	0	kJ/[kg.K]
Heat Capacity Ratio (Overall Liquid)	NaN	
Mass Flow (Overall Liquid)	0	kg/h
Molar Flow (Overall Liquid)	0	kmol/h
Volumetric Flow (Overall Liquid)	0	m3/h
Compressibility Factor (Overall Liquid)	0	
Molar Fraction (Overall Liquid)	0	
Mass Fraction (Overall Liquid)	0	
Volumetric Fraction (Overall Liquid)	0	
Molar Fraction (Liquid 1)		
Methane	0.000261377	
Ethane	7.29693E-17	
Propane	1.0105E-16	
N-butane	0	
Isobutane	0	
Nitrogen	2.36487E-13	
Carbon dioxide	0.00060933	
Water	0.999129	
Oxygen	3.58284E-14	
Argon	1.25314E-11	
Hydrogen sulfide	0	
Hydrogen	4.69209E-67	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Molar Fraction (Liquid 2)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	

Property	Value	Unit
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Molar Fraction (Aqueous)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Molar Fraction (Solid)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	

NH3-HOT-1101 (Material Stream)

Property	Value	Unit
Temperature	430	C
Pressure	147	bar
Mass Flow	43227.4	kg/h
Molar Flow	4233.14	kmol/h
Volumetric Flow	1736.47	m3/h
Mixture Molar Fraction		
Methane	0.110414	
Ethane	2.67764E-17	
Propane	6.08556E-19	
N-butane	0	
Isobutane	0	
Nitrogen	0.128484	
Carbon dioxide	0.00125769	
Water	0.000989796	
Oxygen	3.4435E-08	
Argon	0.00276159	
Hydrogen sulfide	0	
Hydrogen	0.547403	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0.20869	
Density (Mixture)	24.8938	kg/m3
Molecular Weight (Mixture)	10.2117	kg/kmol
Specific Enthalpy (Mixture)	1335.44	kJ/kg
Specific Entropy (Mixture)	0.459074	kJ/[kg.K]
Molar Enthalpy (Mixture)	13637.1	kJ/kmol
Molar Entropy (Mixture)	4.68791	kJ/[kmol.K]
Thermal Conductivity (Mixture)	0.229675	W/[m.K]
Molar Fraction (Vapor)		
Methane	0.110414	
Ethane	2.67764E-17	
Propane	6.08556E-19	
N-butane	0	
Isobutane	0	
Nitrogen	0.128484	
Carbon dioxide	0.00125769	
Water	0.000989796	
Oxygen	3.4435E-08	
Argon	0.00276159	
Hydrogen sulfide	0	
Hydrogen	0.547403	

Property	Value	Unit
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0.20869	
Density (Vapor)	24.8938	kg/m3
Molecular Weight (Vapor)	10.2117	kg/kmol
Specific Enthalpy (Vapor)	1335.44	kJ/kg
Specific Entropy (Vapor)	0.459074	kJ/[kg.K]
Molar Enthalpy (Vapor)	13637.1	kJ/kmol
Molar Entropy (Vapor)	4.68791	kJ/[kmol.K]
Thermal Conductivity (Vapor)	0.229675	W/[m.K]
Kinematic Viscosity (Vapor)	8.55799E-07	m2/s
Dynamic Viscosity (Vapor)	2.13041E-05	Pa.s
Heat Capacity (Vapor)	3.68857	kJ/[kg.K]
Heat Capacity Ratio (Vapor)	1.3134	
Mass Flow (Vapor)	43227.4	kg/h
Molar Flow (Vapor)	4233.14	kmol/h
Volumetric Flow (Vapor)	1736.47	m3/h
Compressibility Factor (Vapor)	1.03149	
Molar Fraction (Vapor)	1	
Mass Fraction (Vapor)	1	
Volumetric Fraction (Vapor)	1	
Molar Fraction (Overall Liquid)		
Methane	NaN	
Ethane	NaN	
Propane	NaN	
N-butane	NaN	
Isobutane	NaN	
Nitrogen	NaN	
Carbon dioxide	NaN	
Water	NaN	
Oxygen	NaN	
Argon	NaN	
Hydrogen sulfide	NaN	
Hydrogen	NaN	
Carbonyl sulfide	NaN	
Carbon monoxide	NaN	
Ammonia	NaN	
Density (Overall Liquid)	0	kg/m3
Molecular Weight (Overall Liquid)	NaN	kg/kmol
Specific Enthalpy (Overall Liquid)	0	kJ/kg
Specific Entropy (Overall Liquid)	0	kJ/[kg.K]
Molar Enthalpy (Overall Liquid)	NaN	kJ/kmol

Property	Value	Unit
Molar Entropy (Overall Liquid)	NaN	kJ/[kmol.K]
Thermal Conductivity (Overall Liquid)	0	W/[m.K]
Kinematic Viscosity (Overall Liquid)	0	m2/s
Dynamic Viscosity (Overall Liquid)	0	Pa.s
Heat Capacity (Overall Liquid)	0	kJ/[kg.K]
Heat Capacity Ratio (Overall Liquid)	NaN	
Mass Flow (Overall Liquid)	0	kg/h
Molar Flow (Overall Liquid)	0	kmol/h
Volumetric Flow (Overall Liquid)	0	m3/h
Compressibility Factor (Overall Liquid)	0	
Molar Fraction (Overall Liquid)	0	
Mass Fraction (Overall Liquid)	0	
Volumetric Fraction (Overall Liquid)	0	
Molar Fraction (Liquid 1)		
Methane	1.57989E-05	
Ethane	4.41062E-18	
Propane	6.10799E-18	
N-butane	0	
Isobutane	0	
Nitrogen	7.8882E-15	
Carbon dioxide	3.68309E-05	
Water	0.0603923	
Oxygen	2.16564E-15	
Argon	7.57462E-13	
Hydrogen sulfide	0	
Hydrogen	1.80432E-68	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0.939555	
Molar Fraction (Liquid 2)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	

Property	Value	Unit
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Molar Fraction (Aqueous)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Molar Fraction (Solid)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	

55 (Material Stream)

Property	Value	Unit
Temperature	430	C
Pressure	147	bar
Mass Flow	0	kg/h
Molar Flow	0	kmol/h
Volumetric Flow	0	m3/h
Mixture Molar Fraction		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Density (Mixture)	0	kg/m3
Molecular Weight (Mixture)	0	kg/kmol
Specific Enthalpy (Mixture)	NaN	kJ/kg
Specific Entropy (Mixture)	0	kJ/[kg.K]
Molar Enthalpy (Mixture)	0	kJ/kmol
Molar Entropy (Mixture)	0	kJ/[kmol.K]
Thermal Conductivity (Mixture)	0	W/[m.K]
Molar Fraction (Vapor)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	

Property	Value	Unit
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Density (Vapor)	0	kg/m3
Molecular Weight (Vapor)	0	kg/kmol
Specific Enthalpy (Vapor)	0	kJ/kg
Specific Entropy (Vapor)	0	kJ/[kg.K]
Molar Enthalpy (Vapor)	0	kJ/kmol
Molar Entropy (Vapor)	0	kJ/[kmol.K]
Thermal Conductivity (Vapor)	0	W/[m.K]
Kinematic Viscosity (Vapor)	0	m2/s
Dynamic Viscosity (Vapor)	0	Pa.s
Heat Capacity (Vapor)	0	kJ/[kg.K]
Heat Capacity Ratio (Vapor)	NaN	
Mass Flow (Vapor)	0	kg/h
Molar Flow (Vapor)	0	kmol/h
Volumetric Flow (Vapor)	0	m3/h
Compressibility Factor (Vapor)	0	
Molar Fraction (Vapor)	0	
Mass Fraction (Vapor)	0	
Volumetric Fraction (Vapor)	NaN	
Molar Fraction (Overall Liquid)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Density (Overall Liquid)	0	kg/m3
Molecular Weight (Overall Liquid)	0	kg/kmol
Specific Enthalpy (Overall Liquid)	0	kJ/kg
Specific Entropy (Overall Liquid)	0	kJ/[kg.K]
Molar Enthalpy (Overall Liquid)	0	kJ/kmol

Property	Value	Unit
Molar Entropy (Overall Liquid)	0	kJ/[kmol.K]
Thermal Conductivity (Overall Liquid)	0	W/[m.K]
Kinematic Viscosity (Overall Liquid)	0	m2/s
Dynamic Viscosity (Overall Liquid)	0	Pa.s
Heat Capacity (Overall Liquid)	0	kJ/[kg.K]
Heat Capacity Ratio (Overall Liquid)	NaN	
Mass Flow (Overall Liquid)	0	kg/h
Molar Flow (Overall Liquid)	0	kmol/h
Volumetric Flow (Overall Liquid)	0	m3/h
Compressibility Factor (Overall Liquid)	0	
Molar Fraction (Overall Liquid)	0	
Mass Fraction (Overall Liquid)	0	
Volumetric Fraction (Overall Liquid)	NaN	
Molar Fraction (Liquid 1)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Molar Fraction (Liquid 2)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	

Property	Value	Unit
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Molar Fraction (Aqueous)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Molar Fraction (Solid)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	

NH3-SYN-1101 (Material Stream)

Property	Value	Unit
Temperature	200	C
Pressure	146	bar
Mass Flow	43226.5	kg/h
Molar Flow	4233.26	kmol/h
Volumetric Flow	0.0867835	m3/h
Mixture Molar Fraction		
Methane	0.110394	
Ethane	2.67754E-17	
Propane	6.08532E-19	
N-butane	0	
Isobutane	0	
Nitrogen	0.128472	
Carbon dioxide	0.00125765	
Water	0.000989796	
Oxygen	6.96279E-14	
Argon	0.00276148	
Hydrogen sulfide	0	
Hydrogen	0.54743	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0.208695	
Density (Mixture)	498097	kg/m3
Molecular Weight (Mixture)	10.2112	kg/kmol
Specific Enthalpy (Mixture)	0	kJ/kg
Specific Entropy (Mixture)	0	kJ/[kg.K]
Molar Enthalpy (Mixture)	0	kJ/kmol
Molar Entropy (Mixture)	0	kJ/[kmol.K]
Thermal Conductivity (Mixture)	0.00502224	W/[m.K]
Molar Fraction (Vapor)		
Methane	0.112716	
Ethane	3.04008E-17	
Propane	7.44034E-19	
N-butane	0	
Isobutane	0	
Nitrogen	0.120201	
Carbon dioxide	0.00139602	
Water	0.0013991	
Oxygen	6.88729E-14	
Argon	0.0027732	
Hydrogen sulfide	0	
Hydrogen	0.514917	

Property	Value	Unit
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0.246597	
Density (Vapor)	0	kg/m3
Molecular Weight (Vapor)	0	kg/kmol
Specific Enthalpy (Vapor)	0	kJ/kg
Specific Entropy (Vapor)	0	kJ/[kg.K]
Molar Enthalpy (Vapor)	0	kJ/kmol
Molar Entropy (Vapor)	0	kJ/[kmol.K]
Thermal Conductivity (Vapor)	0	W/[m.K]
Kinematic Viscosity (Vapor)	0	m2/s
Dynamic Viscosity (Vapor)	0	Pa.s
Heat Capacity (Vapor)	0	kJ/[kg.K]
Heat Capacity Ratio (Vapor)	NaN	
Mass Flow (Vapor)	0	kg/h
Molar Flow (Vapor)	0	kmol/h
Volumetric Flow (Vapor)	0	m3/h
Compressibility Factor (Vapor)	0	
Molar Fraction (Vapor)	0	
Mass Fraction (Vapor)	0	
Volumetric Fraction (Vapor)	0	
Molar Fraction (Overall Liquid)		
Methane	0.110394	
Ethane	2.67754E-17	
Propane	6.08532E-19	
N-butane	0	
Isobutane	0	
Nitrogen	0.128472	
Carbon dioxide	0.00125765	
Water	0.000989796	
Oxygen	6.96279E-14	
Argon	0.00276148	
Hydrogen sulfide	0	
Hydrogen	0.54743	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0.208695	
Density (Overall Liquid)	498097	kg/m3
Molecular Weight (Overall Liquid)	10.2112	kg/kmol
Specific Enthalpy (Overall Liquid)	0	kJ/kg
Specific Entropy (Overall Liquid)	0	kJ/[kg.K]
Molar Enthalpy (Overall Liquid)	0	kJ/kmol

Property	Value	Unit
Molar Entropy (Overall Liquid)	0	kJ/[kmol.K]
Thermal Conductivity (Overall Liquid)	0.00502224	W/[m.K]
Kinematic Viscosity (Overall Liquid)	1.35181E-11	m2/s
Dynamic Viscosity (Overall Liquid)	6.7333E-06	Pa.s
Heat Capacity (Overall Liquid)	0	kJ/[kg.K]
Heat Capacity Ratio (Overall Liquid)	NaN	
Mass Flow (Overall Liquid)	43226.5	kg/h
Molar Flow (Overall Liquid)	4233.26	kmol/h
Volumetric Flow (Overall Liquid)	0.0867835	m3/h
Compressibility Factor (Overall Liquid)	-0.0845481	
Molar Fraction (Overall Liquid)	1	
Mass Fraction (Overall Liquid)	1	
Volumetric Fraction (Overall Liquid)	1	
Molar Fraction (Liquid 1)		
Methane	0.110394	
Ethane	2.67754E-17	
Propane	6.08532E-19	
N-butane	0	
Isobutane	0	
Nitrogen	0.128472	
Carbon dioxide	0.00125765	
Water	0.000989796	
Oxygen	6.96279E-14	
Argon	0.00276148	
Hydrogen sulfide	0	
Hydrogen	0.54743	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0.208695	
Molar Fraction (Liquid 2)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	

Property	Value	Unit
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Molar Fraction (Aqueous)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Molar Fraction (Solid)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	

SYNGAS-PRE-1101 (Material Stream)

Property	Value	Unit
Temperature	396.305	C
Pressure	149	bar
Mass Flow	43227.4	kg/h
Molar Flow	5116.55	kmol/h
Volumetric Flow	1992.83	m3/h
Mixture Molar Fraction		
Methane	0.09135	
Ethane	2.21533E-17	
Propane	5.03484E-19	
N-butane	0	
Isobutane	0	
Nitrogen	0.192629	
Carbon dioxide	0.00104054	
Water	0.0008189	
Oxygen	2.84895E-08	
Argon	0.00228478	
Hydrogen sulfide	0	
Hydrogen	0.711877	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Density (Mixture)	21.6915	kg/m3
Molecular Weight (Mixture)	8.44855	kg/kmol
Specific Enthalpy (Mixture)	1368.6	kJ/kg
Specific Entropy (Mixture)	-0.781298	kJ/[kg.K]
Molar Enthalpy (Mixture)	11562.7	kJ/kmol
Molar Entropy (Mixture)	-6.60083	kJ/[kmol.K]
Thermal Conductivity (Mixture)	0.258879	W/[m.K]
Molar Fraction (Vapor)		
Methane	0.09135	
Ethane	2.21533E-17	
Propane	5.03484E-19	
N-butane	0	
Isobutane	0	
Nitrogen	0.192629	
Carbon dioxide	0.00104054	
Water	0.0008189	
Oxygen	2.84895E-08	
Argon	0.00228478	
Hydrogen sulfide	0	
Hydrogen	0.711877	

Property	Value	Unit
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Density (Vapor)	21.6915	kg/m3
Molecular Weight (Vapor)	8.44855	kg/kmol
Specific Enthalpy (Vapor)	1368.6	kJ/kg
Specific Entropy (Vapor)	-0.781298	kJ/[kg.K]
Molar Enthalpy (Vapor)	11562.7	kJ/kmol
Molar Entropy (Vapor)	-6.60083	kJ/[kmol.K]
Thermal Conductivity (Vapor)	0.258879	W/[m.K]
Kinematic Viscosity (Vapor)	9.00749E-07	m2/s
Dynamic Viscosity (Vapor)	1.95386E-05	Pa.s
Heat Capacity (Vapor)	3.85474	kJ/[kg.K]
Heat Capacity Ratio (Vapor)	1.36202	
Mass Flow (Vapor)	43227.4	kg/h
Molar Flow (Vapor)	5116.55	kmol/h
Volumetric Flow (Vapor)	1992.83	m3/h
Compressibility Factor (Vapor)	1.04267	
Molar Fraction (Vapor)	1	
Mass Fraction (Vapor)	1	
Volumetric Fraction (Vapor)	1	
Molar Fraction (Overall Liquid)		
Methane	NaN	
Ethane	NaN	
Propane	NaN	
N-butane	NaN	
Isobutane	NaN	
Nitrogen	NaN	
Carbon dioxide	NaN	
Water	NaN	
Oxygen	NaN	
Argon	NaN	
Hydrogen sulfide	NaN	
Hydrogen	NaN	
Carbonyl sulfide	NaN	
Carbon monoxide	NaN	
Ammonia	NaN	
Density (Overall Liquid)	0	kg/m3
Molecular Weight (Overall Liquid)	NaN	kg/kmol
Specific Enthalpy (Overall Liquid)	0	kJ/kg
Specific Entropy (Overall Liquid)	0	kJ/[kg.K]
Molar Enthalpy (Overall Liquid)	NaN	kJ/kmol

Property	Value	Unit
Molar Entropy (Overall Liquid)	NaN	kJ/[kmol.K]
Thermal Conductivity (Overall Liquid)	0	W/[m.K]
Kinematic Viscosity (Overall Liquid)	0	m2/s
Dynamic Viscosity (Overall Liquid)	0	Pa.s
Heat Capacity (Overall Liquid)	0	kJ/[kg.K]
Heat Capacity Ratio (Overall Liquid)	NaN	
Mass Flow (Overall Liquid)	0	kg/h
Molar Flow (Overall Liquid)	0	kmol/h
Volumetric Flow (Overall Liquid)	0	m3/h
Compressibility Factor (Overall Liquid)	0	
Molar Fraction (Overall Liquid)	0	
Mass Fraction (Overall Liquid)	0	
Volumetric Fraction (Overall Liquid)	0	
Molar Fraction (Liquid 1)		
Methane	0.000744655	
Ethane	1.1881E-16	
Propane	1.04624E-16	
N-butane	0	
Isobutane	0	
Nitrogen	5.73296E-12	
Carbon dioxide	0.0012357	
Water	0.99802	
Oxygen	2.13159E-13	
Argon	1.35839E-10	
Hydrogen sulfide	0	
Hydrogen	8.55786E-61	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Molar Fraction (Liquid 2)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	

Property	Value	Unit
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Molar Fraction (Aqueous)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	
Molar Fraction (Solid)		
Methane	0	
Ethane	0	
Propane	0	
N-butane	0	
Isobutane	0	
Nitrogen	0	
Carbon dioxide	0	
Water	0	
Oxygen	0	
Argon	0	
Hydrogen sulfide	0	
Hydrogen	0	
Carbonyl sulfide	0	
Carbon monoxide	0	
Ammonia	0	

E26 (Energy Stream)

Property	Value	Unit
Energy Flow	2450.61	kW

E16 (Energy Stream)

Property	Value	Unit
Energy Flow	46200.6	kW

E15 (Energy Stream)

Property	Value	Unit
Energy Flow	0	kW

E14 (Energy Stream)

Property	Value	Unit
Energy Flow	23295.7	kW

E13 (Energy Stream)

Property	Value	Unit
Energy Flow	399.557	kW

E12 (Energy Stream)

Property	Value	Unit
Energy Flow	45714.1	kW

E11 (Energy Stream)

Property	Value	Unit
Energy Flow	0	kW

E10 (Energy Stream)

Property	Value	Unit
Energy Flow	13720.5	kW

E9 (Energy Stream)

Property	Value	Unit
Energy Flow	-10634.7	kW

E8 (Energy Stream)

Property	Value	Unit
Energy Flow	28294.9	kW

E6 (Energy Stream)

Property	Value	Unit
Energy Flow	9938.23	kW

E1 (Energy Stream)

Property	Value	Unit
Energy Flow	7348.72	kW

E4 (Energy Stream)

Property	Value	Unit
Energy Flow	41923.6	kW

E5 (Energy Stream)

Property	Value	Unit
Energy Flow	68968.9	kW

E3 (Energy Stream)

Property	Value	Unit
Energy Flow	0.00302567	kW

E7 (Energy Stream)

Property	Value	Unit
Energy Flow	53.8485	kW

E17 (Energy Stream)

Property	Value	Unit
Energy Flow	266.256	kW

E18 (Energy Stream)

Property	Value	Unit
Energy Flow	-453.631	kW

E19 (Energy Stream)

Property	Value	Unit
Energy Flow	194.773	kW

E20 (Energy Stream)

Property	Value	Unit
Energy Flow	-402.591	kW

E21 (Energy Stream)

Property	Value	Unit
Energy Flow	355.794	kW

E22 (Energy Stream)

Property	Value	Unit
Energy Flow	-347.627	kW

E23 (Energy Stream)

Property	Value	Unit
Energy Flow	34.4066	kW

E24 (Energy Stream)

Property	Value	Unit
Energy Flow	1565.74	kW

E25 (Energy Stream)

Property	Value	Unit
Energy Flow	-13226.9	kW

SP-1201 (Material Stream Splitter)

Property	Value	Unit
Stream 1 Flow Spec	0	
Stream 2 Flow Spec	0	
SR1	0.03	
SR2	0.97	

GLS-1201 (Vapor-Liquid Separator)

Property	Value	Unit
Vessel Orientation	0	
Operating Pressure	0	bar
Liquid Level	0	m
Get Volume from Dimensions	0	
Volume	1	m3
Get Height from Dimensions	0	
Height	2	m
Minimum Pressure	1.01325	bar
Initialize using Inlet Stream	-1	
Reset Content	0	
Liquid Outlet Nozzle Elevation	0	m
Gas Outlet Nozzle Elevation	0	m
Gas Outlet Transition Height	0.01	m
Gas Outlet Homogeneous	0	
Gas Outlet Liquid Fraction	0	
Liquid Outlet Transition Height	0.01	m
Liquid Outlet Gas Fraction	0	
Rigorous Energy Balance (UV)	0	
Split Wall (Wetted / Dry)	0	
Internal Heat Transfer Factor	1	
Fire Case (API 521)	0	
Fire Environment Factor	1	
Fire Adequate Drainage	-1	
Fire Dry Wall Heat Flux	0	
Vessel Bottom Elevation	0	m
Wetted Area	0	m2
Fire Heat Input	0	kW
Wetted Wall Heat Transfer Coefficient	0	W/[m2.K]
Dry Wall Heat Transfer Coefficient	0	W/[m2.K]
Wetted Wall Temperature	25	C
Dry Wall Temperature	25	C
Minimum Fluid Temperature	-273.15	C
Minimum Wetted Wall Temperature	-273.15	C
Minimum Dry Wall Temperature	-273.15	C
Maximum Dry Wall Temperature	-273.15	C
Separation Temperature (if overridden)	25	C
Separation Pressure (if overridden)	1.01325	bar

GLS-801 (Vapor-Liquid Separator)

Property	Value	Unit
Vessel Orientation	0	
Operating Pressure	0	bar
Liquid Level	0	m
Get Volume from Dimensions	0	
Volume	1	m3
Get Height from Dimensions	0	
Height	2	m
Minimum Pressure	1.01325	bar
Initialize using Inlet Stream	-1	
Reset Content	0	
Liquid Outlet Nozzle Elevation	0	m
Gas Outlet Nozzle Elevation	0	m
Gas Outlet Transition Height	0.01	m
Gas Outlet Homogeneous	0	
Gas Outlet Liquid Fraction	0	
Liquid Outlet Transition Height	0.01	m
Liquid Outlet Gas Fraction	0	
Rigorous Energy Balance (UV)	0	
Split Wall (Wetted / Dry)	0	
Internal Heat Transfer Factor	1	
Fire Case (API 521)	0	
Fire Environment Factor	1	
Fire Adequate Drainage	-1	
Fire Dry Wall Heat Flux	0	
Vessel Bottom Elevation	0	m
Wetted Area	0	m2
Fire Heat Input	0	kW
Wetted Wall Heat Transfer Coefficient	0	W/[m2.K]
Dry Wall Heat Transfer Coefficient	0	W/[m2.K]
Wetted Wall Temperature	25	C
Dry Wall Temperature	25	C
Minimum Fluid Temperature	-273.15	C
Minimum Wetted Wall Temperature	-273.15	C
Minimum Dry Wall Temperature	-273.15	C
Maximum Dry Wall Temperature	-273.15	C
Separation Temperature (if overridden)	25	C
Separation Pressure (if overridden)	1.01325	bar

GLS-901 (Vapor-Liquid Separator)

Property	Value	Unit
Vessel Orientation	0	
Operating Pressure	0	bar
Liquid Level	0	m
Get Volume from Dimensions	0	
Volume	1	m3
Get Height from Dimensions	0	
Height	2	m
Minimum Pressure	1.01325	bar
Initialize using Inlet Stream	-1	
Reset Content	0	
Liquid Outlet Nozzle Elevation	0	m
Gas Outlet Nozzle Elevation	0	m
Gas Outlet Transition Height	0.01	m
Gas Outlet Homogeneous	0	
Gas Outlet Liquid Fraction	0	
Liquid Outlet Transition Height	0.01	m
Liquid Outlet Gas Fraction	0	
Rigorous Energy Balance (UV)	0	
Split Wall (Wetted / Dry)	0	
Internal Heat Transfer Factor	1	
Fire Case (API 521)	0	
Fire Environment Factor	1	
Fire Adequate Drainage	-1	
Fire Dry Wall Heat Flux	0	
Vessel Bottom Elevation	0	m
Wetted Area	0	m2
Fire Heat Input	0	kW
Wetted Wall Heat Transfer Coefficient	0	W/[m2.K]
Dry Wall Heat Transfer Coefficient	0	W/[m2.K]
Wetted Wall Temperature	25	C
Dry Wall Temperature	25	C
Minimum Fluid Temperature	-273.15	C
Minimum Wetted Wall Temperature	-273.15	C
Minimum Dry Wall Temperature	-273.15	C
Maximum Dry Wall Temperature	-273.15	C
Separation Temperature (if overridden)	25	C
Separation Pressure (if overridden)	1.01325	bar

GLS-1001 (Vapor-Liquid Separator)

Property	Value	Unit
Vessel Orientation	0	
Operating Pressure	0	bar
Liquid Level	0	m
Get Volume from Dimensions	0	
Volume	1	m3
Get Height from Dimensions	0	
Height	2	m
Minimum Pressure	1.01325	bar
Initialize using Inlet Stream	-1	
Reset Content	0	
Liquid Outlet Nozzle Elevation	0	m
Gas Outlet Nozzle Elevation	0	m
Gas Outlet Transition Height	0.01	m
Gas Outlet Homogeneous	0	
Gas Outlet Liquid Fraction	0	
Liquid Outlet Transition Height	0.01	m
Liquid Outlet Gas Fraction	0	
Rigorous Energy Balance (UV)	0	
Split Wall (Wetted / Dry)	0	
Internal Heat Transfer Factor	1	
Fire Case (API 521)	0	
Fire Environment Factor	1	
Fire Adequate Drainage	-1	
Fire Dry Wall Heat Flux	0	
Vessel Bottom Elevation	0	m
Wetted Area	0	m2
Fire Heat Input	0	kW
Wetted Wall Heat Transfer Coefficient	0	W/[m2.K]
Dry Wall Heat Transfer Coefficient	0	W/[m2.K]
Wetted Wall Temperature	25	C
Dry Wall Temperature	25	C
Minimum Fluid Temperature	-273.15	C
Minimum Wetted Wall Temperature	-273.15	C
Minimum Dry Wall Temperature	-273.15	C
Maximum Dry Wall Temperature	-273.15	C
Separation Temperature (if overriden)	25	C
Separation Pressure (if overriden)	1.01325	bar

GLS-1002 (Vapor-Liquid Separator)

Property	Value	Unit
Vessel Orientation	0	
Operating Pressure	0	bar
Liquid Level	0	m
Get Volume from Dimensions	0	
Volume	1	m3
Get Height from Dimensions	0	
Height	2	m
Minimum Pressure	1.01325	bar
Initialize using Inlet Stream	-1	
Reset Content	0	
Liquid Outlet Nozzle Elevation	0	m
Gas Outlet Nozzle Elevation	0	m
Gas Outlet Transition Height	0.01	m
Gas Outlet Homogeneous	0	
Gas Outlet Liquid Fraction	0	
Liquid Outlet Transition Height	0.01	m
Liquid Outlet Gas Fraction	0	
Rigorous Energy Balance (UV)	0	
Split Wall (Wetted / Dry)	0	
Internal Heat Transfer Factor	1	
Fire Case (API 521)	0	
Fire Environment Factor	1	
Fire Adequate Drainage	-1	
Fire Dry Wall Heat Flux	0	
Vessel Bottom Elevation	0	m
Wetted Area	0	m2
Fire Heat Input	0	kW
Wetted Wall Heat Transfer Coefficient	0	W/[m2.K]
Dry Wall Heat Transfer Coefficient	0	W/[m2.K]
Wetted Wall Temperature	25	C
Dry Wall Temperature	25	C
Minimum Fluid Temperature	-273.15	C
Minimum Wetted Wall Temperature	-273.15	C
Minimum Dry Wall Temperature	-273.15	C
Maximum Dry Wall Temperature	-273.15	C
Separation Temperature (if overriden)	25	C
Separation Pressure (if overriden)	1.01325	bar

GLS-1003 (Vapor-Liquid Separator)

Property	Value	Unit
Vessel Orientation	0	
Operating Pressure	0	bar
Liquid Level	0	m
Get Volume from Dimensions	0	
Volume	1	m3
Get Height from Dimensions	0	
Height	2	m
Minimum Pressure	1.01325	bar
Initialize using Inlet Stream	-1	
Reset Content	0	
Liquid Outlet Nozzle Elevation	0	m
Gas Outlet Nozzle Elevation	0	m
Gas Outlet Transition Height	0.01	m
Gas Outlet Homogeneous	0	
Gas Outlet Liquid Fraction	0	
Liquid Outlet Transition Height	0.01	m
Liquid Outlet Gas Fraction	0	
Rigorous Energy Balance (UV)	0	
Split Wall (Wetted / Dry)	0	
Internal Heat Transfer Factor	1	
Fire Case (API 521)	0	
Fire Environment Factor	1	
Fire Adequate Drainage	-1	
Fire Dry Wall Heat Flux	0	
Vessel Bottom Elevation	0	m
Wetted Area	0	m2
Fire Heat Input	0	kW
Wetted Wall Heat Transfer Coefficient	0	W/[m2.K]
Dry Wall Heat Transfer Coefficient	0	W/[m2.K]
Wetted Wall Temperature	25	C
Dry Wall Temperature	25	C
Minimum Fluid Temperature	-273.15	C
Minimum Wetted Wall Temperature	-273.15	C
Minimum Dry Wall Temperature	-273.15	C
Maximum Dry Wall Temperature	-273.15	C
Separation Temperature (if overriden)	25	C
Separation Pressure (if overriden)	1.01325	bar

K-1001 (Adiabatic Compressor)

Property	Value	Unit
Flow Conductance	1	[kg/s]/[Pa^0.5]
Volume	0.01	m3
Minimum Pressure	1.01325	bar
Initialize using Inlet Stream	-1	
Reset Content	0	
Rotational Inertia	0	
Current Speed	3000	
Target Speed	3000	
Motor Torque	200	
Surge Flow Fraction	0	
Surge Alarm	0	
Pressure Increase	21	bar
Adiabatic Efficiency	75	%
Temperature Difference	230.01	C.
Power Required	266.256	kW
Outlet Pressure	55	bar
PolytropicEfficiency	99.3219	%
AdiabaticCoefficient	1.05396	
PolytropicCoefficient	-6.06065	
AdiabaticHead	20379.2	m
PolytropicHead	26988.1	m
RotationSpeed	1500	rpm
PressureRatio	1.61765	

K-1002 (Adiabatic Compressor)

Property	Value	Unit
Flow Conductance	1	[kg/s]/[Pa^0.5]
Volume	0.01	m3
Minimum Pressure	1.01325	bar
Initialize using Inlet Stream	-1	
Reset Content	0	
Rotational Inertia	0	
Current Speed	3000	
Target Speed	3000	
Motor Torque	200	
Surge Flow Fraction	0	
Surge Alarm	0	
Pressure Increase	37	bar
Adiabatic Efficiency	75	%
Temperature Difference	264.068	C.
Power Required	194.773	kW
Outlet Pressure	91	bar
PolytropicEfficiency	102.799	%
AdiabaticCoefficient	1.05761	
PolytropicCoefficient	-4.98913	
AdiabaticHead	23071.6	m
PolytropicHead	31623.3	m
RotationSpeed	1500	rpm
PressureRatio	1.68519	

K-1003 (Adiabatic Compressor)

Property	Value	Unit
Flow Conductance	1	[kg/s]/[Pa^0.5]
Volume	0.01	m3
Minimum Pressure	1.01325	bar
Initialize using Inlet Stream	-1	
Reset Content	0	
Rotational Inertia	0	
Current Speed	3000	
Target Speed	3000	
Motor Torque	200	
Surge Flow Fraction	0	
Surge Alarm	0	
Pressure Increase	59	bar
Adiabatic Efficiency	75	%
Temperature Difference	302.011	C.
Power Required	355.794	kW
Outlet Pressure	150	bar
PolytropicEfficiency	105.722	%
AdiabaticCoefficient	1.10738	
PolytropicCoefficient	-2.55739	
AdiabaticHead	23244.9	m
PolytropicHead	32766.7	m
RotationSpeed	1500	rpm
PressureRatio	1.64835	

K-1101 (Adiabatic Compressor)

Property	Value	Unit
Flow Conductance	1	[kg/s]/[Pa^0.5]
Volume	0.01	m3
Minimum Pressure	1.01325	bar
Initialize using Inlet Stream	-1	
Reset Content	0	
Rotational Inertia	0	
Current Speed	3000	
Target Speed	3000	
Motor Torque	200	
Surge Flow Fraction	0	
Surge Alarm	0	
Pressure Increase	1	bar
Adiabatic Efficiency	75	%
Temperature Difference	0.793214	C.
Power Required	34.4066	kW
Outlet Pressure	150	bar
PolytropicEfficiency	75.0237	%
AdiabaticCoefficient	1.56313	
PolytropicCoefficient	1.83417	
AdiabaticHead	220.937	m
PolytropicHead	221.007	m
RotationSpeed	1500	rpm
PressureRatio	1.00671	

H-901 (Material Stream Heater)

Property	Value	Unit
Flow Conductance	1	[kg/s]/[Pa^0.5]
Volume	1	m3
Minimum Pressure	1.01325	bar
Initialize using Inlet Stream	-1	
Reset Content	0	
Ambient Temperature	25	C
Ambient UA Product	0	W/[m2.K]
Wall Thermal Mass	0	
Wall Temperature	25	C
Pressure Drop	0	bar
Efficiency	100	
Outlet Temperature	420	C
Heat Added	23295.7	kW
Outlet Molar Vapor Fraction	1	
Temperature Difference	380	C.

H-301 (Material Stream Heater)

Property	Value	Unit
Flow Conductance	1	[kg/s]/[Pa^0.5]
Volume	1	m3
Minimum Pressure	1.01325	bar
Initialize using Inlet Stream	-1	
Reset Content	0	
Ambient Temperature	25	C
Ambient UA Product	0	W/[m2.K]
Wall Thermal Mass	0	
Wall Temperature	25	C
Pressure Drop	2	bar
Efficiency	100	
Outlet Temperature	850	C
Heat Added	41923.6	kW
Outlet Molar Vapor Fraction	1	
Temperature Difference	471.782	C.

H-101 (Material Stream Heater)

Property	Value	Unit
Flow Conductance	1	[kg/s]/[Pa^0.5]
Volume	1	m3
Minimum Pressure	1.01325	bar
Initialize using Inlet Stream	-1	
Reset Content	0	
Ambient Temperature	25	C
Ambient UA Product	0	W/[m2.K]
Wall Thermal Mass	0	
Wall Temperature	25	C
Pressure Drop	0.5	bar
Efficiency	100	
Outlet Temperature	350	C
Heat Added	7348.72	kW
Outlet Molar Vapor Fraction	1	
Temperature Difference	320	C.

H-1101 (Material Stream Heater)

Property	Value	Unit
Flow Conductance	1	[kg/s]/[Pa^0.5]
Volume	1	m3
Minimum Pressure	1.01325	bar
Initialize using Inlet Stream	-1	
Reset Content	0	
Ambient Temperature	25	C
Ambient UA Product	0	W/[m2.K]
Wall Thermal Mass	0	
Wall Temperature	25	C
Pressure Drop	0	bar
Efficiency	100	
Outlet Temperature	430	C
Heat Added	1565.74	kW
Outlet Molar Vapor Fraction	1	
Temperature Difference	33.6949	C.

C-1201 (Material Stream Cooler)

Property	Value	Unit
Flow Conductance	1	[kg/s]/[Pa^0.5]
Volume	1	m3
Minimum Pressure	1.01325	bar
Initialize using Inlet Stream	-1	
Reset Content	0	
Ambient Temperature	25	C
Ambient UA Product	0	W/[m2.K]
Wall Thermal Mass	0	
Wall Temperature	25	C
Pressure Drop	1	bar
Efficiency	100	
Outlet Temperature	30	C
Heat Removed	2450.61	kW
Outlet Molar Vapor Fraction	0.9237	
Temperature Difference	-170	C.

C-901 (Material Stream Cooler)

Property	Value	Unit
Flow Conductance	1	[kg/s]/[Pa^0.5]
Volume	1	m3
Minimum Pressure	1.01325	bar
Initialize using Inlet Stream	-1	
Reset Content	0	
Ambient Temperature	25	C
Ambient UA Product	0	W/[m2.K]
Wall Thermal Mass	0	
Wall Temperature	25	C
Pressure Drop	0	bar
Efficiency	100	
Outlet Temperature	40	C
Heat Removed	46200.6	kW
Outlet Molar Vapor Fraction	0.927207	
Temperature Difference	-812.093	C.

C-801 (Material Stream Cooler)

Property	Value	Unit
Flow Conductance	1	[kg/s]/[Pa^0.5]
Volume	1	m3
Minimum Pressure	1.01325	bar
Initialize using Inlet Stream	-1	
Reset Content	0	
Ambient Temperature	25	C
Ambient UA Product	0	W/[m2.K]
Wall Thermal Mass	0	
Wall Temperature	25	C
Pressure Drop	0	bar
Efficiency	100	
Outlet Temperature	40	C
Heat Removed	45714.1	kW
Outlet Molar Vapor Fraction	0.810177	
Temperature Difference	-195.019	C.

C-701 (Material Stream Cooler)

Property	Value	Unit
Flow Conductance	1	[kg/s]/[Pa^0.5]
Volume	1	m3
Minimum Pressure	1.01325	bar
Initialize using Inlet Stream	-1	
Reset Content	0	
Ambient Temperature	25	C
Ambient UA Product	0	W/[m2.K]
Wall Thermal Mass	0	
Wall Temperature	25	C
Pressure Drop	0	bar
Efficiency	100	
Outlet Temperature	220	C
Heat Removed	13720.5	kW
Outlet Molar Vapor Fraction	1	
Temperature Difference	-130	C.

C-601 (Material Stream Cooler)

Property	Value	Unit
Flow Conductance	1	[kg/s]/[Pa^0.5]
Volume	1	m3
Minimum Pressure	1.01325	bar
Initialize using Inlet Stream	-1	
Reset Content	0	
Ambient Temperature	25	C
Ambient UA Product	0	W/[m2.K]
Wall Thermal Mass	0	
Wall Temperature	25	C
Pressure Drop	0	bar
Efficiency	100	
Outlet Temperature	350	C
Heat Removed	28294.9	kW
Outlet Molar Vapor Fraction	1	
Temperature Difference	-266.53	C.

E-1001 (Material Stream Cooler)

Property	Value	Unit
Flow Conductance	1	[kg/s]/[Pa^0.5]
Volume	1	m3
Minimum Pressure	1.01325	bar
Initialize using Inlet Stream	-1	
Reset Content	0	
Ambient Temperature	25	C
Ambient UA Product	0	W/[m2.K]
Wall Thermal Mass	0	
Wall Temperature	25	C
Pressure Drop	1	bar
Efficiency	100	
Outlet Temperature	40	C
Heat Removed	-453.631	kW
Outlet Molar Vapor Fraction	0.999305	
Temperature Difference	-230.01	C.

E-1002 (Material Stream Cooler)

Property	Value	Unit
Flow Conductance	1	[kg/s]/[Pa^0.5]
Volume	1	m3
Minimum Pressure	1.01325	bar
Initialize using Inlet Stream	-1	
Reset Content	0	
Ambient Temperature	25	C
Ambient UA Product	0	W/[m2.K]
Wall Thermal Mass	0	
Wall Temperature	25	C
Pressure Drop	0	bar
Efficiency	100	
Outlet Temperature	40	C
Heat Removed	-402.591	kW
Outlet Molar Vapor Fraction	0.999531	
Temperature Difference	-264.068	C.

C-1003 (Material Stream Cooler)

Property	Value	Unit
Flow Conductance	1	[kg/s]/[Pa^0.5]
Volume	1	m3
Minimum Pressure	1.01325	bar
Initialize using Inlet Stream	-1	
Reset Content	0	
Ambient Temperature	25	C
Ambient UA Product	0	W/[m2.K]
Wall Thermal Mass	0	
Wall Temperature	25	C
Pressure Drop	1	bar
Efficiency	100	
Outlet Temperature	40	C
Heat Removed	-347.627	kW
Outlet Molar Vapor Fraction	0.99975	
Temperature Difference	-302.011	C.

R-901 (Equilibrium Reactor)

Property	Value	Unit
Operating Pressure (Dynamics)	0	bar
Liquid Level	0	m
Volume	1	m3
Height	2	m
Minimum Pressure	1.01325	bar
Initialize using Inlet Stream	0	
Reset Contents	0	
Pressure Drop	0.5	bar
Outlet Temperature	852.093	C
Calculation Mode	Adiabatic	
Initial Gibbs Energy	5348.84	kW
Final Gibbs Energy	-70593.2	kW
Methane: Conversion	-523.509	%
Water: Conversion	-3059.04	%
Hydrogen: Conversion	24.5553	%
Carbon monoxide: Conversion	1159.13	%
Carbon dioxide: Conversion	60.5733	%
CO Methanation: Extent	0	kmol/h
CO ₂ Methanation: Extent	0	kmol/h

R-701 (Equilibrium Reactor)

Property	Value	Unit
Operating Pressure (Dynamics)	0	bar
Liquid Level	0	m
Volume	1	m3
Height	2	m
Minimum Pressure	1.01325	bar
Initialize using Inlet Stream	0	
Reset Contents	0	
Pressure Drop	0.5	bar
Outlet Temperature	235.019	C
Calculation Mode	Adiabatic	
Initial Gibbs Energy	-321127	kW
Final Gibbs Energy	-322093	kW
Carbon dioxide: Conversion	-9.44881	%
Water: Conversion	6.36747	%
Hydrogen: Conversion	-2.48826	%
Carbon monoxide: Conversion	77.9537	%
HTS Reaction: Extent	0	kmol/h

R-601 (Equilibrium Reactor)

Property	Value	Unit
Operating Pressure (Dynamics)	0	bar
Liquid Level	0	m
Volume	1	m3
Height	2	m
Minimum Pressure	1.01325	bar
Initialize using Inlet Stream	0	
Reset Contents	0	
Pressure Drop	0.5	bar
Outlet Temperature	350	C
Calculation Mode	Isothermic	
Initial Gibbs Energy	-331318	kW
Final Gibbs Energy	-332007	kW
Carbon dioxide: Conversion	-187.311	%
Water: Conversion	30.5237	%
Hydrogen: Conversion	-20.7269	%
Carbon monoxide: Conversion	84.3226	%
HTS Reaction: Extent	1003.81	kmol/h

R-302 (Equilibrium Reactor)

Property	Value	Unit
Operating Pressure (Dynamics)	0	bar
Liquid Level	0	m
Volume	1	m3
Height	2	m
Minimum Pressure	1.01325	bar
Initialize using Inlet Stream	0	
Reset Contents	0	
Pressure Drop	0.5	bar
Outlet Temperature	850	C
Calculation Mode	Defined Temperature	
Initial Gibbs Energy	-356698	kW
Final Gibbs Energy	-357695	kW
Ethane: Conversion	100	%
Water: Conversion	6.03068	%
Hydrogen: Conversion	-11.7714	%
Carbon monoxide: Conversion	-28.2417	%
Propane: Conversion	100	%
N-butane: Conversion	100	%
ETHANE STEAM REFORMING: Extent	60.2259	kmol/h
PROPANE STEAM REFORMING: Extent	17.2074	kmol/h
n-BUTANE STEAM REFORMING: Extent	5.16222	kmol/h

R-301 (Equilibrium Reactor)

Property	Value	Unit
Operating Pressure (Dynamics)	0	bar
Liquid Level	0	m
Volume	1	m3
Height	2	m
Minimum Pressure	1.01325	bar
Initialize using Inlet Stream	0	
Reset Contents	0	
Pressure Drop	0.5	bar
Outlet Temperature	850	C
Calculation Mode	Defined Temperature	
Initial Gibbs Energy	-295158	kW
Final Gibbs Energy	-356383	kW
Methane: Conversion	72.7757	%
Water: Conversion	34.0076	%
Hydrogen: Conversion	-∞	%
Carbon monoxide: Conversion	-∞	%
Carbon dioxide: Conversion	-1868.25	%
METHANE STEAM REFORMING: Extent	1164.62	kmol/h
Water-Gas Shift: Extent	482.215	kmol/h

R-1101 (Equilibrium Reactor)

Property	Value	Unit
Operating Pressure (Dynamics)	0	bar
Liquid Level	0	m
Volume	1	m3
Height	2	m
Minimum Pressure	1.01325	bar
Initialize using Inlet Stream	0	
Reset Contents	0	
Pressure Drop	2	bar
Outlet Temperature	430	C
Calculation Mode	Defined Temperature	
Initial Gibbs Energy	11177.1	kW
Final Gibbs Energy	7208	kW
Nitrogen: Conversion	44.8162	%
Hydrogen: Conversion	36.3809	%
Ammonia: Conversion	-∞	%
AMMONIA SYNTHESIS: Extent	441.707	kmol/h

R-401 (Gibbs Reactor)

Property	Value	Unit
Operating Pressure (Dynamics)	0	bar
Liquid Level	0	m
Volume	1	m3
Height	2	m
Minimum Pressure	1.01325	bar
Initialize using Inlet Stream	0	
Reset Contents	0	
Pressure Drop	0	bar
Outlet Temperature	961.749	C
Calculation Mode	Adiabatic	
Initial Gibbs Energy	-420653	kW
Final Gibbs Energy	-455112	kW
Element Balance Residue	7.53218E-07	
Methane: Conversion	78.6436	%
Nitrogen: Conversion	0.00814904	%
Carbon dioxide: Conversion	-5.37182	%
Water: Conversion	-9.5119	%
Oxygen: Conversion	99.9999	%
Argon: Conversion	8.67524E-05	%
Hydrogen: Conversion	-8.9765	%
Carbon monoxide: Conversion	-36.0305	%

HX-1 (Heat Exchanger)

Property	Value	Unit
Cold Fluid Flow Conductance	1	[kg/s]/[Pa^0.5]
Hot Fluid Flow Conductance	1	[kg/s]/[Pa^0.5]
Volume for Cold Fluid	1	m3
Volume for Hot Fluid	1	m3
Cold Side Pressure	1.01325	bar
Hot Side Pressure	1.01325	bar
Minimum Pressure	1.01325	bar
Initialize using Inlet Streams	0	
Reset Contents	0	
Fouling Rate	0	
Current Fouling Resistance	0	
Wall Thermal Mass	0	
Wall Temperature	25	C
Number of Cells	10	
Substeps	1	
Global Heat Transfer Coefficient (U)	1000	W/[m2.K]
Heat Exchange Area (A)	70.9718	m2
Heat Load	38742.6	kW
Cold fluid outlet temperature	380	C
Hot fluid outlet temperature	616.53	C
[Shell and Tube] Internal Shell Diameter	500	mm
[Shell and Tube] Shell Fouling Factor	0	K.m2/W
[Shell and Tube] Baffle Cut	20	%
[Shell and Tube] Shells in Series	1	
[Shell and Tube] Baffle Spacing	250	mm
[Shell and Tube] Internal Tube Diameter	50	mm
[Shell and Tube] External Tube Diameter	60	mm
[Shell and Tube] Tube Length	5	m
[Shell and Tube] Tube Fouling factor	0	K.m2/W
[Shell and Tube] Tube Passes Per Shell	2	
[Shell and Tube] Number of Tubes	50	
[Shell and Tube] Tube Pitch	70	mm
[Shell and Tube] Fouling Factor (Design)	0	K.m2/W
[Shell and Tube] LMTD Correction Factor (F)	1	
Logarithmic mean temperature difference LMTD	545.887	C.
[Shell and Tube] Resistance heat transfer pipes	0	K.m2/W
[Shell and Tube] Resistance thermal conductivity pipes	0	K.m2/W
[Shell and Tube] Resistance heat transfer shell	0	K.m2/W
[Shell and Tube] Reynolds number shell	0	
[Shell and Tube] Reynolds number tubes	0	
Thermal Efficiency	67.2337	%

Property	Value	Unit
Maximum Theoretical Heat Exchange	57623.8	kW
Minimum Temperature Difference	0	C.
Heat Loss	0	kW
User-Defined LMTD Correction Factor	1	
Outlet Vapor Fraction (Stream 1)	0	
Outlet Vapor Fraction (Stream 2)	0	
Pressure Drop (Cold Side)	1	bar
Pressure Drop (Hot Side)	1	bar
Calculated MITA	0	C.
PROP_HX_35	1	
PROP_HX_36	1	
PROP_HX_37	10	bar
PROP_HX_38	10	bar
PROP_HX_39	200	C
PROP_HX_40	200	C
PROP_HX_41	15	bar
PROP_HX_42	15	bar
PROP_HX_43	3	mm
PROP_HX_44	10	mm
PROP_HX_45	0	mm
PROP_HX_46	0	mm
PROP_HX_47	0	mm
PROP_HX_48	0	mm
PROP_HX_49	0	mm
PROP_HX_50	0	m
PROP_HX_51	0	m2
PROP_HX_52	0	m2
PROP_HX_53	0	
PROP_HX_54	0	m2
PROP_HX_55	0	m2
PROP_HX_56	0	m/s
PROP_HX_57	0	m/s
PROP_HX_58	0	m/s
PROP_HX_59	0	m/s
PROP_HX_60	0	m/s
PROP_HX_61	0	m/s
PROP_HX_62	0	bar
PROP_HX_63	0	bar
PROP_HX_64	0	bar
PROP_HX_65	0	bar
PROP_HX_66	0	m3
PROP_HX_67	0	m3

Property	Value	Unit
PROP_HX_68	0	kg
PROP_HX_69	0	kg
PROP_HX_70	0	kg

HX-1101 (Heat Exchanger)

Property	Value	Unit
Cold Fluid Flow Conductance	1	[kg/s]/[Pa^0.5]
Hot Fluid Flow Conductance	1	[kg/s]/[Pa^0.5]
Volume for Cold Fluid	1	m3
Volume for Hot Fluid	1	m3
Cold Side Pressure	1.01325	bar
Hot Side Pressure	1.01325	bar
Minimum Pressure	1.01325	bar
Initialize using Inlet Streams	0	
Reset Contents	0	
Fouling Rate	0	
Current Fouling Resistance	0	
Wall Thermal Mass	0	
Wall Temperature	25	C
Number of Cells	10	
Substeps	1	
Global Heat Transfer Coefficient (U)	1000	W/[m2.K]
Heat Exchange Area (A)	198.398	m2
Heat Load	16035.8	kW
Cold fluid outlet temperature	396.305	C
Hot fluid outlet temperature	200	C
[Shell and Tube] Internal Shell Diameter	500	mm
[Shell and Tube] Shell Fouling Factor	0	K.m2/W
[Shell and Tube] Baffle Cut	20	%
[Shell and Tube] Shells in Series	1	
[Shell and Tube] Baffle Spacing	250	mm
[Shell and Tube] Internal Tube Diameter	50	mm
[Shell and Tube] External Tube Diameter	60	mm
[Shell and Tube] Tube Length	5	m
[Shell and Tube] Tube Fouling factor	0	K.m2/W
[Shell and Tube] Tube Passes Per Shell	2	
[Shell and Tube] Number of Tubes	50	
[Shell and Tube] Tube Pitch	70	mm
[Shell and Tube] Fouling Factor (Design)	0	K.m2/W
[Shell and Tube] LMTD Correction Factor (F)	1	
Logarithmic mean temperature difference LMTD	80.8264	C.
[Shell and Tube] Resistance heat transfer pipes	0	K.m2/W
[Shell and Tube] Resistance thermal conductivity pipes	0	K.m2/W
[Shell and Tube] Resistance heat transfer shell	0	K.m2/W
[Shell and Tube] Reynolds number shell	0	
[Shell and Tube] Reynolds number tubes	0	
Thermal Efficiency	93.5707	%

Property	Value	Unit
Maximum Theoretical Heat Exchange	17137.7	kW
Minimum Temperature Difference	0	C.
Heat Loss	0	kW
User-Defined LMTD Correction Factor	1	
Outlet Vapor Fraction (Stream 1)	0	
Outlet Vapor Fraction (Stream 2)	0	
Pressure Drop (Cold Side)	1	bar
Pressure Drop (Hot Side)	1	bar
Calculated MITA	0	C.
PROP_HX_35	1	
PROP_HX_36	1	
PROP_HX_37	10	bar
PROP_HX_38	10	bar
PROP_HX_39	200	C
PROP_HX_40	200	C
PROP_HX_41	15	bar
PROP_HX_42	15	bar
PROP_HX_43	3	mm
PROP_HX_44	10	mm
PROP_HX_45	0	mm
PROP_HX_46	0	mm
PROP_HX_47	0	mm
PROP_HX_48	0	mm
PROP_HX_49	0	mm
PROP_HX_50	0	m
PROP_HX_51	0	m2
PROP_HX_52	0	m2
PROP_HX_53	0	
PROP_HX_54	0	m2
PROP_HX_55	0	m2
PROP_HX_56	0	m/s
PROP_HX_57	0	m/s
PROP_HX_58	0	m/s
PROP_HX_59	0	m/s
PROP_HX_60	0	m/s
PROP_HX_61	0	m/s
PROP_HX_62	0	bar
PROP_HX_63	0	bar
PROP_HX_64	0	bar
PROP_HX_65	0	bar
PROP_HX_66	0	m3
PROP_HX_67	0	m3

Property	Value	Unit
PROP_HX_68	0	kg
PROP_HX_69	0	kg
PROP_HX_70	0	kg

S-801 (Compound Separator)

Property	Value	Unit
Volume	0.1	m3
Minimum Pressure	1.01325	bar
Initialize using Inlet Stream	-1	
Reset Content	0	
Energy Balance	399.557	kW
SpecifiedStreamIndex	0	
SepSpecValue_Methane	0	%
SepSpecValue_Ethane	0	%
SepSpecValue_Propane	0	%
SepSpecValue_N-butane	0	%
SepSpecValue_Isobutane	0	%
SepSpecValue_Nitrogen	0	%
SepSpecValue_Carbon dioxide	99	%
SepSpecValue_Water	0	%
SepSpecValue_Oxygen	0	%
SepSpecValue_Argon	0	%
SepSpecValue_Hydrogen sulfide	0	%
SepSpecValue_Hydrogen	0	%
SepSpecValue_Carbonyl sulfide	0	%
SepSpecValue_Carbon monoxide	0	%
SepSpecValue_Ammonia	0	%

SUL-101 (Compound Separator)

Property	Value	Unit
Volume	0.1	m3
Minimum Pressure	1.01325	bar
Initialize using Inlet Stream	-1	
Reset Content	0	
Energy Balance	0.00302567	kW
SpecifiedStreamIndex	0	
SepSpecValue_Methane	0	%
SepSpecValue_Ethane	0	%
SepSpecValue_Propane	0	%
SepSpecValue_N-butane	0	%
SepSpecValue_Isobutane	0	%
SepSpecValue_Nitrogen	0	%
SepSpecValue_Carbon dioxide	0	%
SepSpecValue_Water	0	%
SepSpecValue_Oxygen	0	%
SepSpecValue_Argon	0	%
SepSpecValue_Hydrogen sulfide	100	%
SepSpecValue_Hydrogen	0	%
SepSpecValue_Carbonyl sulfide	100	%
SepSpecValue_Carbon monoxide	0	%